

Midterm Evaluation of
Project for Accelerating Policy Change,
Translation and Implementation for Pneumonia
and Diarrhea Commodities

2016-2020

Pakistan

Midterm Evaluation Report

Contech International
December 2019

Acknowledgements

Contech International highly values the support extended by M/o NHR&C, Provincial Health Departments, health programmes, members of Child Survival Groups, Sindh Child Survival Programme, clinicians, Pakistan Pediatric Association, WHO and other development partners in enhancing the quality and scientific rigor of the evaluation as well as the report. We are also grateful to the pharmaceutical manufacturers who provided us with their invaluable insights about the industry and to the all those who were involved in and facilitated the data collection: Federal, Provincial and District Health Managers; data collection teams including field teams and the support staff. We would also like to extend our gratitude to facility and pharmacy in-charges, community health workers, GPs and parents/caregivers whose participation and freedom of expression allowed us to gather relevant and reliable information for this report. Last but not the least, we would like to express our gratitude to UNICEF's national and provincial project implementation teams and UNICEF Evaluation & Research Unit for their unstinted support and invaluable feedback during the process of this Midterm Evaluation. We hope that the evaluation findings and the frameworks hence developed will make a valuable contribution in improving child survival in Pakistan.

Title	MIDTERM EVALUATION OF PROJECT FOR ACCELERATING POLICY CHANGE, TRANSLATION AND IMPLEMENTATION FOR PNEUMONIA AND DIARRHEA COMMODITIES IN PAKISTAN
Geographic Location of the Project	Federal and Provincial (2 provinces, i.e. Punjab and Sindh)
Timeline of Evaluation	May 2019 – December 2019
Date of the Report	30 th December 2019
Country	Pakistan
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List of Acronyms and Abbreviations

Amox-DT	Amoxicillin Dispersible Tablets
ARI	Acute Respiratory Infection
BHUs	Basic Health Units
BMGF	Bill & Melinda Gates Foundation
CEO	Chief Executive Officer
CPSP	College of Physicians and Surgeons Pakistan
CSG	Child Survival Group
DCP	Disease Control Priorities
DFID	Department for International Development
DHIS	District Health Information System
DOC	Driver of Change
DT	Dispersible Tablets
EDO	Executive District Officer
EML	Essential Medicine List
EPI	Expanded Programme On Immunization
EPHS	Essential Package of Health Services
FGDs	Focus Group Discussions
FLCF	First Level Care Facility
GAPPD	Global Action Plan for Pneumonia & Diarrhea
GPs	General Practitioners
ICCM	Integrated Community Case Management
IMNCI	Integrated Management of Neonatal and Childhood Illnesses
IYCF	Infant and Young Child Feeding
JAF	Joint Accountability Framework
KII	Key Informant Interviews
LHW	Lady Health Worker
M&E	Monitoring & Evaluation
MoNHSR&C	Ministry of National Health Services Regulation & Coordination
MNCH	Maternal Neonatal Child Health
MSDP	Minimum Service Delivery Package
MSDS	Minimum Service Delivery Standards
NIH	National Institution of Health
OECD/DAC Criteria	Organisation for Economic Cooperation and Development (OECD)/ Development Assistance Committee (DAC) Criteria for Evaluation
ORS	Oral Rehydration Solution
PATS	Pakistan's Approach to Total Sanitation
PIEA	Political Institutional Economy Analysis
PMDC	Pakistan Medical and Dental Council
PMER	Planning Monitoring Reporting and Evaluation
PMRC	Pakistan Health (Medical) Research Council
RHCs	Rural Health Units
RMNCAH	Reproductive Maternal Neonatal Child & Adolescent Health
SOPs	Standard Operating Procedures
TOC	Theory of Change

TWG	Technical Working Group
U5	Under five years of age
UNICEF	United Nations Children's Fund
USAID	United States Agency for International Development
WHO	World Health Organization
Zinc-DT	Zinc Dispersible tablets

Contents

ACKNOWLEDGEMENTS	II
LIST OF ACRONYMS AND ABBREVIATIONS	IV
EXECUTIVE SUMMARY	VIII
EVALUATION PURPOSE & OBJECTIVES	VIII
METHODOLOGY	VIII
MAIN FINDINGS	VIII
CONCLUSION AND RECOMMENDATIONS	X
1. INTRODUCTION	11
1.1 OBJECT OF EVALUATION – PNEUMONIA AND DIARRHEA PROJECT	12
1.2 RATIONALE FOR COMMISSIONING MIDTERM EVALUATION	15
2. LITERATURE REVIEW	17
2.1 GLOBAL SCENARIO	17
2.2 PNEUMONIA AND DIARRHEA PAKISTAN CONTEXT	19
2.3 BMGF & PROGRAMME FOR CHILD SURVIVAL	21
3. EVALUATION METHODOLOGY	22
3.1 EVALUATION CRITERIA AND QUESTIONS	23
3.2 EVALUATION TECHNIQUE – MIXED METHOD	25
3.3 SECONDARY DATA	29
3.4 EVALUATION TEAM	30
3.5 PROCEDURAL STEPS AND QUALITY ASSURANCE MECHANISMS	31
3.6 DATA MANAGEMENT	35
3.7 LIMITATIONS OF THE EVALUATION	36
4. EVALUATION FINDINGS	37
4.1 STATUS OF PROJECT OUTCOMES	37
4.2.1 RELEVANCE	42
4.2.2 EFFECTIVENESS	49
4.2.3 EFFICIENCY	56
4.2.4 SUSTAINABILITY	61
5. CONCLUSIONS	65
5.1 PROPOSED THEORY OF CHANGE	67

6. RECOMMENDATIONS	69
7. DISSEMINATION	72
8. REFERENCES	73
9. BIBLIOGRAPHY	75
10. ANNEXES	76
ANNEX 1: EVALUATION MATRIX	76
ANNEX 2: EVALUATION TEAM COMPOSITION	83
ANNEX 3: ETHICAL REVIEW BOARD CERTIFICATE	84
ANNEX 4: CASE STUDIES	85
ANNEX 5: GUIDES FOR INTERVIEWS AND FOCUS GROUP DISCUSSIONS	89
ANNEX 6: RISK REGISTER	121
ANNEX 7: DESCRIPTION OF PROPOSED CHANGE PATHWAYS	124

LIST OF TABLES

TABLE 1: PROJECT IMPLEMENTATION AND SELECTION OF DISTRICTS	26
TABLE 2: DISTRICT SELECTION IN SINDH – HDI AND GEOGRAPHICAL VARIATIONS	26
TABLE 3: QUALITATIVE SAMPLING MATRIX FOR KEY INFORMANT INTERVIEWS	28
TABLE 4: QUALITATIVE SAMPLING MATRIX FOR FOCUS GROUP DISCUSSIONS	29
TABLE 5: QUANTITATIVE SAMPLING MATRIX FOR HEALTH FACILITIES	29
TABLE 6: TRAININGS UNDER P&D PROJECT	41
TABLE 7: RECOMMENDATIONS’ MATRIX	70

LIST OF FIGURES

FIGURE 1: PROJECT THEORY OF CHANGE	14
FIGURE 2: MIDTERM EVALUATION – CONCEPTUAL FRAMEWORK	22
FIGURE 3: SELECTED DISTRICTS FOR MIDTERM EVALUATION	25
FIGURE 4: ORGANOGRAM OF EVALUATION TEAM	30
FIGURE 5: REVISION OF IMNCI GUIDELINES	39
FIGURE 6: PERCENTAGE OF HEALTH FACILITIES HAVING UPDATED COMMODITIES	58
FIGURE 7: PERCENTAGE OF PNEUMONIA AND DIARRHEA PRESCRIPTIONS HAVING UPDATED COMMODITIES PRESCRIBED	59
FIGURE 8: UNDERLYING REASONS FOR NOT PRESCRIBING UPDATED COMMODITIES	59
FIGURE 9: PROPOSED THEORY OF CHANGE	68

Executive Summary

Evaluation Purpose & Objectives

UNICEF Pakistan, through financial assistance of Bill and Melinda Gates Foundation (BMGF) is implementing the project for 'Accelerating Policy Change, Translation and Implementation for Pneumonia and Diarrhea Commodities in Pakistan (hereinafter called the Project or P&D Project). Policy transformation is a complex and painstaking process that requires clear-cut, precise and well-timed interplay of a multitude of factors. These factors, both intrinsic like political will and commitment, and extrinsic like broader policy environment should all work coherently to complete the causal chain of policy change, policy translation and then its implementation and ultimately, knowledge management of the impact. Keeping this in view, the Project started off in 2016 to bring policies to build barriers between the children – girls and boys under five years of age – and two major contributors to child mortality: Pneumonia and Diarrhea. The scope of work of the Project mainly comprised of federal level support to Ministry of National Health Services Regulation and Coordination in Islamabad (MoNHSRC), Provincial Health Departments, Offices of District Health Officers (DHOs), health facilities and outreach workers in targeted districts. This midterm evaluation (MTE) was commissioned to assess the extent to which the Project was successful in achieving its intended results, and whether the stakeholders were productively involved in the causal chain of policy change, translation, implementation and knowledge management. It further explored opportunities and lessons learned. Specifically, MTE addressed the following objectives:

- To assess the extent to which the intended outcomes of the project are achieved by comparing it with results from the baseline studies / evaluability assessment of the project;
- To document the processes involved in achievement of the outcomes and identify gaps that has affected the project to ensure achieving the results;
- To review and assess proper utilization of supplies provided to beneficiaries at public facilities and identify the gaps in utilization;
- To assess the potential for replicability and scalability; and
- To provide guidance for improvement and course correction in all areas and programme strategies, and targets to ensure effective achievement of the results.

Methodology

A formative evaluation design with a mixed methods approach (quantitative and qualitative techniques) was adopted, and both primary and secondary data was collected with gender disaggregation done wherever possible. The evaluation team worked in close collaboration with all stakeholders during various stages of the MTE. Stakeholders included federal and provincial governments, along with health facilities, outreach workers, private sector care providers, donors and development partners and the ultimate beneficiaries. The end-users and other stakeholders were involved mainly as participants, i.e. interviewees and focus group participants. Gender mainstreaming was kept in consideration while devising the data collection processes and tools. Gender balance was incorporated in the approach of the evaluation along with exploring gender differentials and possible discriminatory practices against girl child. An evaluation framework was developed considering the evaluation objectives and evaluation questions as per OECD/DAC components of evaluation.

Main Findings

Findings of the evaluation are described under each component of the OECD/DAC Criteria for Evaluation.

Relevance was assessed to determine the extent to which the project suited the priorities and policies of the target group, recipient and donor. Findings revealed that there is high

relevance of the project as Pneumonia and Diarrhea contribute extensively to childhood deaths in Pakistan. Empirical evidence supported the use of updated commodities as per the GAPPD recommendations and guidelines of WHO for management of childhood Pneumonia and Diarrhea. Further, it was affirmed that the project objectives are highly consistent with Pakistan's national vision and priorities for child health.

Effectiveness was assessed to determine the extent to which the project was able to attain its objectives. Resultantly, the project updated national and provincial policies and guidelines in line with GAPPD recommendations, however implementation of these policies at district and community level needs to be strengthened. Effective coordination mechanisms between federal and provincial levels were established in the form of National RMNCAH&N TWG and provincial child survival TWGs. Reporting tools on facility (DHIS tools) and community level data (LHW MIS tools) were updated with inclusion of new commodities and indicators in alignment with GAPPD recommendations and WHO guidelines. The project engaged community level health workers – Lady Health Workers, and provision of ARI timers has empowered LHWs and improved their ability to manage ARI and timely refer Pneumonia.

Efficiency measured the outputs – qualitative and quantitative – in relation to the inputs and the project was able to achieve its planned milestones and outcomes within the stipulated timeline. There were numerous factors, mainly related to limited demand that hindered the availability of recommended commodities for management of childhood pneumonia and diarrhea in the open market. The project has added value by improving management practices of healthcare providers, mainly outreach workers and primary level facilities and majority of these facilities had adequate supply of P&D recommended commodities but prescription behavior of care providers needed improvement. It is pertinent to mention that project has made certain adjustments mid-way to increase effective implementation.

The project has contributed in building ownership of provincial governments of its interventions, which is further evident through inclusion of recommended commodities in MSDP, EML and procurement lists, DHIS and MIS tools, IMNCI guidelines and their incorporation in Maternal and Newborn Health Package of Services under Disease Control Priorities, Edition 3 (DCP3) of Universal Health Coverage Benefit Package for Pakistan.

Sustainability was assessed to measure whether the benefits of the project are likely to continue after donor funding is withdrawn. Sustainability and replicability of the project is to be ensured through institutionalization of key policies and guidelines within health system. Such challenges will be mitigated through enhanced government ownership and funding on P&D.

Lessons learnt during the course of MTE revealed that for **Policy Change**, political will and commitment was essential for policy transformation and reforms agenda. Joint accountability framework assisted the project in steering the progress and tracking as establishing oversight and coordination platforms at all levels were critical in building government ownership and steering the project activities. For **Policy Translation**, efforts should be strongly supported through empirical evidence for development of advocacy material. A key factor was building the acceptability of the care providers while introducing new/revised commodities and without financial commitment; public procurement of updated commodities is not possible. At the level of **Policy Implementation**, evidence of improved management of childhood illnesses through updated commodities accelerated the policy implementation phase. However, despite the resources and capacities, the local pharmaceutical manufacturing industry lacked interest in local production due to minimal demand of updated commodities in open market. In **Knowledge Management** of outcomes of the project interventions, updated federal and provincial policies and guidelines reflect the recognition of improved management of Pneumonia and Diarrhea as a shared goal and sustainability of P&D project will depend on achievement of results as perceived by the stakeholders including government, partner agencies, health facility staff and patients.

Conclusion and Recommendations

In the causal pathway of policy transformation and reforms, the stage of policy implementation is the most critical. In order to make the P&D Project interventions institutionalized in the existing system, the strengthening of health sector at individual, organizational and systems level is of utmost importance. A systems strengthening approach is recommended while moving forward so that the interventions under P&D become well entrenched in existing system for sustainability beyond the project life.

Enabling policy environment - Sustenance of federal-provincial-district linkages and coordination should be continued through existing platforms at federal level (RMNCAH Group), provincial level (CSG and TWG), and district level (District Health & Population Management Teams). The supply chain management (from DOH to health facilities and outreach) should be strengthened so as the government capacity to roll out LMIS and DHIS-02.

Private sector engagement – Being a service provider of more than two-thirds of population, the project should establish wider and proactive public-private partnership models. This includes engagement with pharma industry and Drugs Regulatory Authority, institutions regulating medical, nursing and paramedical education, and professional associations.

Advocacy and Communication - Concerted advocacy efforts with political leadership for sustained will and commitment is important in policy transformation and reforms. Side by side, continued networking and lobbying with pharmaceutical manufacturers for local production is very important. The project should engage relevant partners for leveraging resources and coordinated actions.

Scaling up and replication – Despite the devolution of health sector, which in a way made it easy to implement policy transformations in individual provinces, the project is encouraged to scale up the interventions to cover entire Pakistan. Now, after having experience of implementation in two of the big provinces of Pakistan, the provincial level interventions will be built upon the lessons learnt for replication through cross-sharing of achievements, challenges and lessons learnt among uncovered regions and provinces of Pakistan.

1. Introduction

The Sustainable Development Goals (SDGs) adopted by the United Nations in 2015 were established to promote healthy lives and well-being for all children. Third SDG is to end preventable deaths of newborns and under-5 children by 2030. Each day, together Diarrhea and Pneumonia kill more than 4,000 children under the age of five-year around the world (Bhutta et al., 2013). Contribution of these two diseases to under-five (U5) child deaths has remained stubbornly high. Together, Diarrhea and Pneumonia account for 29 (percent) of all deaths amongst U5 children and results in a loss of 2 million young lives each year (WHO and UNICEF, 2013). In 2015, both of these two diseases were responsible for one in four deaths that occurred in children under five years of age (Bhutta et al., 2013). However, the under-five mortality rate had been cut by more than half worldwide in the past quarter century (from 91 deaths per 1,000 live births in 1990 to 43 deaths per 1,000 live births in 2015). Similar trends were observed for Pakistan, where under-5 mortality has significantly declined from 112 deaths per 1,000 live births in 1990-91 to 74 deaths in 2017-18. Likewise, infant mortality rate also decreased from 86 to 62 deaths per 1,000 live births over the period of 1990-2018 (PDHS, 1991; PDHS 2018). Nonetheless, still many children do not survive to see their fifth birthday, especially in the poorest and deprived regions in the world. Pakistan is one of the fifteen developing countries in the world that constitute majority of the global burden of Pneumonia and Diarrhea, as 72 (percent) of the children who lose their lives to Pneumonia and Diarrhea before the age of five, are from these high-burden countries (Bhutta et al., 2013). Acute respiratory infections (ARIs), malaria, and dehydration caused by severe Diarrhea are major causes of childhood mortality in Pakistan. In total, Diarrhea, Pneumonia, and malaria collectively contribute to around half of all child deaths each year in Pakistan (Bhutta et al., 2013). The Global Action Plan for the Prevention and Control of Pneumonia and Diarrhea (GAPPD) ranks Pakistan as third highest in South Asia (with the score of 46), due to constantly high number of under-five deaths mainly caused by Pneumonia and Diarrhea.

Focusing further on the current scenario in Pakistan, the country has more than 27 million children under the age of 5 years. Among these, number of children with Diarrhea and ARI, including Pneumonia was estimated around 6.3 million and 4.4 million, respectively (Bhutta et al., 2013). Recent statistics also reveal that, approximately 74 children under the age of 5 years die in 1,000 live births every year in Pakistan (PDHS, 2018). Apart from the prevalence of these fatal childhood diseases, management of their illnesses is also a major challenge. It is critical that children suffering from Diarrhea and Pneumonia receive timely and appropriate treatment. Amoxicillin syrup remains the prescribed medicine for the treatment of Pneumonia at most health care facilities, while the prescribed treatment for Diarrhea is ORS and zinc syrup. A low proportion of children receive appropriate treatment. According to Pakistan Demographic and Health Survey (PDHS) 2017-2018, treatment from a health facility or provider was sought for 85% of children with ARI symptoms, almost equally for female and male child (84.4% and 84.6% respectively) and treatment was sought from a health facility or health provider for 71% of children with Diarrhea (70% girls and 72% boys). 37% of children with Diarrhea received a rehydration solution from an oral rehydration salt (ORS) packet Diarrhea (36.6 % girls and 38.2% boys); 13% of children with Diarrhea were given zinc supplements (12.2% girls and 12.7% boys), and 8% (7% girls and 8.8% boys) received both ORS and zinc supplements. Moreover, only 46% of the children suffering from acute respiratory infection received an appropriate antibiotic (PDHS, 2018). ORS requires accuracy in preparation and is therefore often not used correctly; meanwhile zinc syrup is not easily available at local pharmacies (UNICEF, 2018). Gender disaggregated data on child healthcare and survival is not quite readily available. However, to whatever extent data for U5 children is available, was incorporated in the literature of this report.

Estimates of infant mortality in Pakistan are available since the 1960s. These data show that IMR declined from 140 deaths per 1,000 live births in the 1960s to 105 in the mid-1980s,

and further to 85 deaths per 1,000 live births in the mid-1990s. However, the pace of infant mortality decline appears to have slowed down since then. As would be expected, the variations in infant and under-five mortality across provinces are wide. In 2017-18, the IMR ranged from a low of 53 in Khyber Pakhtunkhwa to a high of 73 in Punjab. The inter-provincial variations in U5 mortality rate are even greater – from a low of 64 in Khyber Pakhtunkhwa to a high of 85 in Punjab (PDHS, 2018).

A series of Multiple Indicators Cluster Survey (MICS) and Pakistan Demographic and Health Survey (PDHS) has revealed that the gap in mortality between rural and urban areas is consistent, with infant mortality in rural areas is 23 and 27 and points greater than in urban areas of Punjab and Sindh, respectively.

1.1 Object of Evaluation – Pneumonia and Diarrhea Project

With partnership of UNICEF and Bill and Melinda Gates Foundation, the project “Accelerating Policy Change, Translation and Implementation for Pneumonia and Diarrhea Commodities in Pakistan”, was initiated in 2016 with total investment of USD 12,516,903. The project was designed to improve the diagnosis, treatment as well as prevention of Diarrhea and Pneumonia in Pakistan by 2020. Working in close collaboration with Government of Pakistan and relevant stakeholders, the overall purpose was to implement a programme based on an integrated global action plan for the prevention and control of Pneumonia and Diarrhea and contribute to the efforts to increase child survival in Pakistan, particularly by strengthening policy solutions for better diagnosis and treatment of Pneumonia and Diarrhea in girls and boys under 5 year of age.

Aim of programme is to bring together evaluative interventions, warrant a healthy environment, promote practices that help decrease illnesses, and ensure that every girl and boy has access to proven preventive and treatment measures through evidence based updated medical commodities for Pneumonia and Diarrhea. The programme aims to link the most disadvantaged children and women with an integrated package of high impact and good quality healthcare commodities, with a special focus on systems strengthening and ensuring a continuum of healthcare. The project has made significant progress to meet its objectives through support in development and updating of national/provincial policies and guidelines in line with the global recommendations for the management of Diarrhea and Pneumonia among under five children in Pakistan. UNICEF provided technical support to update and revise child survival related policies and strategies aligned with Global Action Plan for Pneumonia and Diarrhea (GAPPD) and WHO updated recommendations, including updating Early Childhood Diseases (ECD) sensitive IMNCI guidelines, Amoxicillin DT, Zinc DT and Lo-ORS were included in provincial Essential Medicine Lists (EML), procurement lists, and in the Minimum Service Delivery Package (MSDP) in Punjab.

Promoting gender equality and child rights is fundamental to the fulfilment of UNICEF’s core mandate. Gender equality is not just an essential human right, but also a foundation for a peaceful, prosperous and sustainable world. Its principles apply to both development and humanitarian contexts. In the past, differential approach of curative health services was the main cause of greater survival of boys than girls (Booth and Verma 1992). Until recently, infant mortality was higher for females than for males, reflecting a trend common in the country. However, the recent Pakistan Demographic and Health Survey (PDHS, 2018), this trend has reversed; showing female infant and under five mortality to be considerably lower than male infant and under five mortality. Similarly, there was no difference in care seeking behavior in cases of ARI (84.1% for girls and 84.4 for boys) and Diarrhea (70.0% for girls and 71.5% for boys). Since this programme contributes to reducing childhood morbidity and mortality caused by Pneumonia and Diarrhea, it frames gender equality as one of the development objectives.

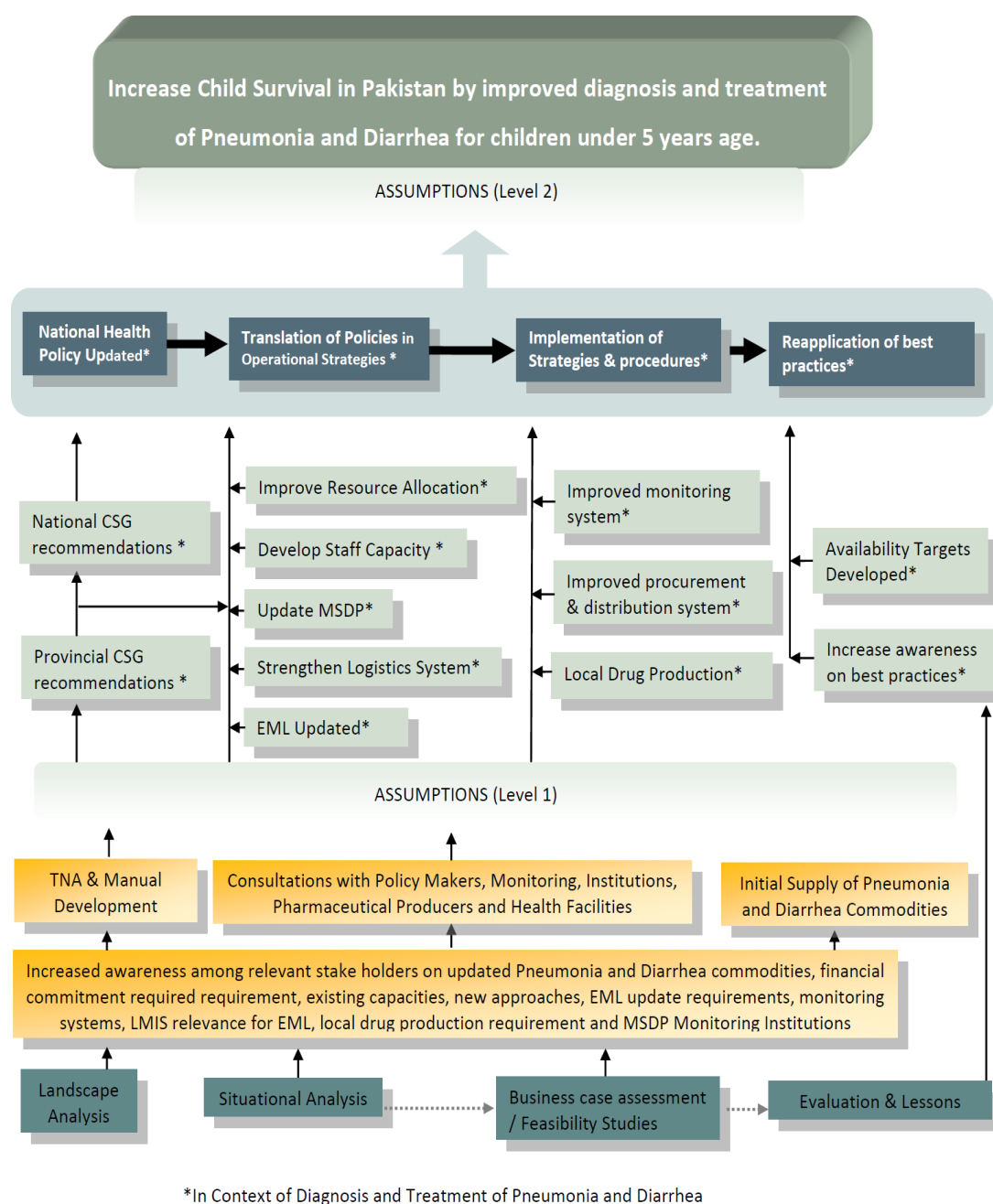
Health protection and services for poor families are crucial, given their pre-existing poor health and nutrition status to ensure equity. In 2017-18, the IMR of the poorest families was

almost 43 percent that of the richest families. Regarding U5MR, disparity is even wider with more than 78 percent increase in U5MR between poorest and the richest wealth index quintiles (PDHS, 2018). Considering a significant proportion of Pakistan's population are poor, ensuring health protection and services for this group remains a daunting challenge. Presence of a flush toilet is strongly associated with reduced risk of infant death, with the infant mortality rate in households having a flush toilet being 22 percent lower than in households without such a toilet (Bennet, 1999). Another dimension is education of the mothers and like rest of the world; educated mothers tend to have lower rates of infant and under-five mortality than illiterate mothers. A pattern is further observed that each incremental year of schooling is associated with significant gains in infant survival. The data indicate that even a few years of mother's schooling can help improve child survival and significantly lower the death rate of children in their early years of life. Social accessibility can be understood in terms of religious, tribal and cultural barriers. Social and cultural factors may contribute to delay in decision making for health seeking by a sick child. The utilization of appropriate healthcare services for children suffering from Pneumonia and Diarrhea is at large constrained by local customs, practices and prevailing norms.

This project attempts to increase child survival, equally for both the female and male child, thus adhering to the principles of gender equality and child rights. While the general project objective is to improve management of childhood Diarrhea and Pneumonia and increase child survival by the end of 2020, soon after initiation of project, UNICEF conducted an 'Evaluability Study' and 'Baseline Landscape Analysis', aimed to provide insight on barriers and facilitators to policy translation and commodity access in the public and private sectors at national and provincial level (Sindh and Punjab). As mentioned in the TORs of MTE, there was no Theory of Change (TOC) developed at the inception of the project. A retrospective TOC (Figure 1) was developed on the basis of '**Evaluability Study**' and '**Landscape Analysis**'.

As per the given TOC, the Project is expected to achieve the following outcomes, pertaining to policy change, translation and implementation of this project:

- **Outcome 1: Policy Change**
Existing national/provincial policies and guidelines are updated in line with global recommendations (GAPPD/WHO) for management of Diarrhea and Pneumonia among under five children in Pakistan by the end of 2020.
- **Outcome 2: Policy Translation**
Translation of revised and updated Pneumonia and Diarrhea treatment guidelines into gender-responsive relevant action plans by all provincial/areas health departments in Pakistan by the end of 2020.
- **Outcome 3: Policy Implementation**
Availability of essential commodities (amoxicillin DT, zinc DT, co-packaged ORS and zinc, oxygen, ARI timers, and pulse oximeters) for treatment of childhood Pneumonia and Diarrhea in Pakistan by the end of 2020.
- **Outcome 4: Knowledge Management**
Translation of lessons learnt from this investment to other settings/broader geographical areas within Pakistan.

Figure 1: Project Theory of Change

This project promotes gender equality and women empowerment at community level. The community outreach workers (LHS and LHWs) are females who work at the grass root level, to promote healthy behaviours and provide basic curative services for both female and male children, without any discrimination. Since the project involves provision of commodities including gadgets like ARI timers to LHWs, the project is equipping the females in the community and empowering them to perform better. The project also aims to improve Pneumonia and Diarrhea treatment equally for both female and male child, without any discrimination. Now after the implementation of the project, to ascertain the direction and progress made in this project in achieving the ultimate goal, UNICEF is getting a 'Midterm Evaluation' conducted as a part of the planned monitoring and evaluation activities for this

project. The table below demonstrates the involvement of stakeholders in this project at various tiers. These stakeholders were involved at all stages of the MTE.

P&D Project Stakeholders Roles and Responsibilities		
LEVEL	STAKEHOLDERS	ROLES
Federal	M/o National Health Services Regulations and Coordination, UNICEF	Knowledge Management, Steering, Monitoring, Policy Changing And Policy Translation Roles
Provincial	DOHs, LHW Programme, IRMNCH Programme, UNICEF Project Staff	Management And Implementation Roles
District	District Health Offices, Healthcare Providers, Lady Health Supervisors (LHSs) and Lady Health Workers (LHWs)	Coordination And Implementation Roles
Community	Concerned Population i.e. service users (mothers/caregivers of U5 children)	Recipients And Beneficiaries
Development Partners	BMGF	Funding Support
	UN Agencies	Technical Reviews And Implementation Support
	Aga Khan University	Implementation Partner for Community Mobilization in Sindh

1.2 Rationale for Commissioning Midterm Evaluation

UNICEF commissioned midterm evaluation (MTE) as part of the monitoring and evaluation activities planned for this project to ascertain that the direction and progress made were in the right direction in meeting its ultimate goals. It assessed the extent to which this project was successful in achieving its intended results and aided the beneficiaries, and whether the stakeholders were productively involved in the policy change, policy translation, and policy implementation and knowledge management. This evaluation determined systematically and objectively, the progress towards specified project objectives and the extent to which they were achieved and contributed to increase child survival for both girls and boys in Pakistan. This evaluation particularly made recommendations in improving the pathways of diagnosis and treatment of Pneumonia and Diarrhea in U5 children over the project period. Evaluation documented the overall progress and lessons learned from the project with recommendations that would be used in improving the project in accordance with the National Health Vision, and formulating a revised strategic plan for future planning. The evidences would also help in scaling up the project in other geographical areas in Pakistan and other settings.

Government health institutions, more specifically the MoNHSR&C (Ministry of National Health Services, Regulation and Coordination) and the Provincial/Area Departments of Health (DoH), Lady Health Worker Programme (LHWP), Integrated Reproductive Maternal Newborn, Child Health (IRMNCH) Programme, along with United Nations Children's Fund (UNICEF), and Bill & Melinda Gates Foundation (BMGF) are the primary intended users of this evaluation, and would benefit from the learning relevant to their work. The women, children, families, communities, health care providers, policy makers, opinion leaders and partners would be the secondary audience.

Primary aim of this midterm evaluation is to document the progress made so far and the lessons learnt from the project that will be used in improving the implementation of project interventions. In addition, secondary aim of this evaluation is knowledge management and reviewing potential for scalability in rest of the provinces. Therefore, the evidences generated will also help in scaling up the project in other geographical areas in Pakistan and other settings. In a nutshell, generated evaluation report would be used to inform project strategy and activity design. It would help identify and capitalize on project strengths, correct weaknesses, set

realistic goals, identify new areas of intervention, and provide guidance about best practices for replication and possible expansion.

Specific Objectives of this midterm evaluation are:

- To assess the extent to which the intended outcomes of the project are achieved by comparing it with the results from the baseline studies / evaluability assessment of the project;
- To document the processes involved in achievement of the outcomes and identify gaps that has affected the project to ensure achieving results;
- To review and assess proper utilization of supplies provided to beneficiaries at public facilities and identify the gaps in utilization;
- To assess the potential for replicability and scalability; and
- To provide guidance for improvement and course correction in all areas and programme strategies, and targets to ensure effective achievement of the results.

Since the implementation of Pneumonia and Diarrhea Project is midway, this is the opportune time to conduct the midterm evaluation. It will help to carve out the future course of action and course correction to achieve the goals of the project. Consequently, Theory of Change (TOC), prepared at the inception of the project and provided in the TORs is revisited and updated based on the findings of the midterm evaluation. The midterm evaluation also upholds gender mainstreaming as it gives a chance to mothers to raise issues about their child's health and the service provision. Mostly the mothers are not heard in their own households but this evaluation provides them with a platform to voice their concerns, which would ultimately contribute in policy change.

The midterm evaluation mainly focuses on the implementation and programmatic dimensions of the project and it did not include any impact assessment regarding childhood morbidity and mortality, which would require a household survey and was beyond the scope of this evaluation.

2. Literature Review

The literature review is conducted on the extant national and provincial level policies and management practices for Pneumonia and Diarrhea – including the shift from tablets and syrups to dispersible tablets; use of Low Osmolality ORS, Oxygen, ARI timers, Pulse Oximeters. It starts with a description of international context and best practices across the developing world, and current management practices for Pneumonia and Diarrhea. This section also gives a brief description of “Accelerating Policy Change, Translation and Implementation for Pneumonia and Diarrhea Commodities in Pakistan” being implemented through funding by Bill & Melinda Gates Foundation. It concludes with the gains if any that can be achieved at the midterm evaluation point keeping in view OECD/DAC criteria for evaluation. The literature included in this review was selected on the basis of their robustness of evidence as evident by the impact factor of the reviewed publications. Further secondary data for this midterm evaluation has been referred from sources like LHW-MIS, PSLM, PDHS, MICS and NNS. Desk review of project documents, existing child survival policies, strategies and guidelines related to Pneumonia and Diarrhea in relation to GAPPD have been taken into account but not limited to following.

- National IMNCI guidelines (2010)
- World Health Organization’s (WHO) 2014 IMCI recommendations
- Provincial and Areas Health Strategies (2012-2020)
- Pakistan’s National Drug Policy
- The National Infant and Young Child Feeding (IYCF) Strategy
- Pakistan’s Approach to Total Sanitation (PATS)
- Scale-up Plan for Essential Medicines for Child Health
- Ten Point Vision for RMNCAH & Nutrition 2015-2020
- Integrated Community Case Management (ICCM) guideline and Treatment Guidelines

This literature review further helped us identifying the list of indicators to address the evaluation objectives. The questions related to indicative areas (related to the OECD/DAC evaluation criteria, including relevance, effectiveness, efficiency, sustainability and cross-cutting areas of equity and gender equality), have been taken into account while selecting these indicators. This evaluation matrix formed the basis of data collection and analysis.

2.1 Global Scenario

Substantial developments in terms of child survival were observed throughout the globe during the past two decades making it possible for millions of children have a better chance of survival than earlier times. Estimated mortality rate of children under the age of 5 is 9 million per annum: nearly all occurring in low-and middle-income countries (Houweling & Kunst, 2009). Differences in equity of progress in reducing childhood mortality rates exist between and within countries. Dimensions of these inequities are complex and range from socioeconomic status, geographical location, urban and rural residence, gender and ethnic groups (Arregoces et al, 2015; UNICEF, 2018). Regions with socioeconomic disparities can experience under 5 mortality rates as high as 9 out of every 10 children (Arregoces et al, 2015; UNICEF, 2018). Global burden of child deaths especially in low- and middle-income countries is a call for urgent and intense action to further rally the survival chances of children around the world. In developing countries, health care often counts on both public and private sectors. Regrettably, these sectors are often poorly coordinated, regulated, and supported, resulting in gaps in services, lost opportunities, and unsustainable system (Alijanzadeh et al., 2016). Available public health facilities are exhausted and depersonalized with low personal accomplishments

due to massive disease burden and prevailing poverty and socio-economic constraints along with medical supply shortages (Rachiotis et al., 2014). While in private sector, patients have to bear out-of-pocket expenditure, which compels the underprivileged members of the society to fall in a vicious disease-poverty cycle, which leads to worsening of existing health condition and monetary constraints.

2.1.1 Global Initiatives for Pneumonia and Diarrhea

Since late 90s', the emphasis on child health has been seen, either through global goals (MDGs and SDGs), IMCI programmes or Pneumonia and Diarrhea programmes (Taylor, Schumacher & Davis, 2016).

Preventive and management initiatives around the globe are of significant consideration due to these high mortality rates. *The Global Strategies for Women's and Children's Health* developed by United Nations Secretary General in 2010 stresses on factors that can help control Pneumonia and Diarrhea by effective and proficient use of assets, that can in turn assist in establishment of synergies between allies (United Nations Secretary General, 2012).

In the same way, *The World Health Assembly Resolution 2010*, supported all countries to implement the GAPP action plans at national levels. GAPP, action plan calls for the establishment of evidence-based policies and national plans' for controlling Pneumonia (Resolution WHA63.24, 2010).

United Nations Commission on Life-Saving commodities for Women and Children, in 2012, targeted to improve marketing, supply and monitoring of neglected commodities while taking into consideration gap in demand and supply of commodities for Pneumonia and Diarrhea. Affordable commodities like Oral rehydration salt (ORS), Zinc DT and oral amoxicillin DT though inexpensive but proven to be effective measures for treating Pneumonia and Diarrhea were made part of commission's mandate.

To ensure the supply of commodities nationwide as well as to improve the market and quality of these commodities, the recommendation of involving potential stakeholders were made. As a result a complementary intervention The Global Vaccine Action Plan (GVAP) was launched by World Health Assembly, in 2012, to control the disease primarily Pneumonia and Diarrhea (WHO, 2012). Similarly, universal movement of UNICEF and USAID *committing to child survival: A Promise Renewed*, worked in 180 regimes and countries guaranteed scale up struggles to help curb the disease burden maternal, new-born and child mortality.

The Every New-born Action Plan was established to demonstrate consent on the activities that were necessary to be taken to enhance the improvement on survival of newborn. Political will and activities with dedication *were* of great need in order to deliver assets to help reduce mortality due to Pneumonia and Diarrhea for which a focused struggle at universal level was made in 2012, named as *Declaration on Scaling up Treatment of Diarrhea and Pneumonia*. Where in all donors, industry and non-governmental organizations calls on high burden countries and international community for commitment towards provision of resources, political will and focused action.

UNICEF and World health Organization tossed the *Integrated Global Action Plan for Pneumonia and Diarrhea* (GAPPD), suggesting a consistent methodology to address mortality rate due to Pneumonia and Diarrhea. GAPPD provides an opportunity for the global community to address these leading causes of child deaths in an integrated and coordinated fashion. By understanding the landscape of global child health initiatives and how they support and relate to the GAPPD, global donors and policymakers, alongside advocates and national decision-makers, can maximize their investments and help ensure every child has the opportunity to secure a prosperous future (WHO/UNICEF, 2013).

The use of ORS was successfully promoted in the era of mid 1980s; the use of ORS gained popularity in reducing Diarrhea related deaths in more than 100 states across the globe.

Apex management of UNICEF made substantial efforts to attract political will and support for its usage in Diarrhea related illnesses. Systematic reviews revealed the significance of ORS in reducing Diarrhea related mortality rates to a 93 (percent) (Munos, Walker & Black, 2010). Despite this, the use of ORS came to a downturn in the mid-2000s due to several reasons, among which the major reason is lack of awareness and practices.

For treatment of Pneumonia among children, Acute Respiratory Infection (ARI) programmes were initiated during 1990s, based on evidence from research studies focusing on childhood Pneumonia assessment and treatment with antibiotics in community settings. However, compared with the early years of Control of Diarrheal Diseases (CDD) programmes or immunization programmes, they were slower to be adopted or scaled up to a large extent.

GAVI funded the Pneumonia DIP and the HiB Initiative aimed at country adoption of the new vaccines. This combined effort provided new, strong momentum for Pneumonia and helped reactivate the network of Pneumonia proponents (Berlan, 2015). At the global level, increased advocacy for policy and resources by more formal groups such as the *Global Coalition against Child Pneumonia* were seen (Taylor, Schumacher & Davis, 2016). . In 2011, the Diarrhea Pneumonia Working Group (DPWG) accumulated efforts for both disease conditions and supports the expansion of programmes in 10 high-burden countries (Taylor, Schumacher & Davis, 2016).

UNICEF and WHO tossed the *Integrated Global Action Plan for Pneumonia and Diarrhea* (GAPPD), suggesting a consistent methodology to address mortality rate due to Pneumonia and Diarrhea. GAPPD provides an opportunity for the global community to address these leading causes of child deaths in an integrated and coordinated fashion. By understanding the landscape of global child health initiatives and how they support and relate to the GAPPD, global donors and policymakers, alongside advocates and national decision-makers, can maximize their investments and help ensure every child has the opportunity to secure a prosperous future (WHO/UNICEF, 2013).

2.2 Pneumonia and Diarrhea Pakistan Context

Among all the ailments, Diarrhea and Pneumonia are the prime causes of child mortality in Pakistan. Each year, approximately 91,000 deaths are attributed to Pneumonia and another 53,300 lose their lives because of Diarrhea (UNICEF, 2019). Two-thirds of the child mortality due to Pneumonia and Diarrhea occurs in countries, mostly within in Asia, with Pakistan ranking third highest amongst them (PSLM, 2014-15).

The policies working in Pakistan for child survival enhancement are; the National Health Vision 2016-2025, National and Provincial essential medicine lists, Integrated management system of Newborn and Childhood Illness, Global Action Plan for Pneumonia and Diarrhea, Child survival groups, current environment for updating commodities, Pakistan's approach to Total Sanitation (PATs).

2.2.1 Care Seeking and Referral

Low care seeking behavior for management of Diarrhea and Pneumonia has been observed in Pakistan. Disparities in care seeking behavior exist within different regions of Pakistan, such as in case of diarrhea, more than 75 percent and about 74 percent under-5 children pursue care from a facility or healthcare provider in provinces of Punjab and Sindh respectively, however, trends declined in Khyber Pakhtunkhwa (59.7 percent) and Baluchistan (63.1) (PDHS, 2018). Similar patterns have been noticed in Punjab and Sindh provinces for treatment of pneumonia amongst children under 5 years, who sought advice with 86.1 percent and 85.4 percent respectively. Nonetheless, improvement was seen for pneumonia treatment in Khyber Pakhtunkhwa (84.3%) and Baluchistan (62.2%) (PDHS, 2018). Use of Zinc supplements along with Low Osmolality ORS help reduce Diarrhea severity and duration (UNICEF National Nutrition Survey, 2018). Cost effective alternatives such as Amoxicillin

Dispersible tablets, Pulse Oximeter, ARI timers and Oxygen therapy can help diagnoses, manage, treat and prevent childhood Pneumonia (UNICEF, 2018). Gender disadvantage has pervasive effects across the lifetime, much of it mediated through poor care (Qadir et al, 2011). In Pakistan, male gender preference is deeply embedded in the culture. In Pakistan medical care is sought for children more often for sons than daughters (Qadir et al, 2011). Critically ill male children are twice as likely to be treated in hospitals as compared to their counterparts (Qadir et al, 2011).

2.2.2 Gender Equality

Gender discrimination in child rearing, nutrition, health care seeking and education make woman highly vulnerable members of the society (Shaikh & Hatcher, 2005). A core set of gender bottlenecks and barriers prevent the achievement of specific outcomes (UNICEF, 2014) and are discussed as follows:

a) Women's and girls' lack of safety and mobility:

Limited access to the outer world has been culturally entrenched in rural society, even if it is a matter of consulting a physician in emergency (Shaikh & Hatcher, 2005).

b) Women's and girls' lack of resources and decision making:

Men play a paramount role in determining the health needs of woman (Shaikh & Hatcher, 2005). The low status of women prevents them from recognizing and voicing their concerns about health care needs (Shaikh & Hatcher, 2005). Women despite being primary caregivers for families in most instances have low control or autonomy and are often not allowed to visit a healthcare facility alone (Shaikh & Hatcher, 2005). Lack of economic control and social dependence undermines women independence and decision making power (Shaikh & Hatcher, 2005). This certainly has repercussions on health particularly in the case that they are the primary caregivers (Shaikh & Hatcher, 2005).

c) Limited access to information, knowledge, and technology and decision-making:

Low or lack of Formal education amongst these women also accounts for poor understanding and recognition of seriousness of Diarrhea and Pneumonia resulting in unnecessary delay in care-seeking and low compliance (Bhutta & Hafeez). PDHS results on IMR clearly show that differences in equity of care exist between male and females gender disparities. Moreover, in the absence of mother as a primary caregiver, grandmothers are seen to be secondary caregivers (UNICEF, 2018). Knowledge of these elderly members is considered to be more reliable and acceptable on the basis of prior experience in raising children. However, limited exposure to new information and practices for treating Pneumonia and Diarrhea is seen as a disadvantage (Shaikh & Hatcher, 2005).

2.2.3 Cultural Perceptions

Across the developing world, females mostly suffer a great deal due to gender discrimination in child rearing, nutrition, healthcare seeking and education. A common practice in rural areas is to give more attention to a son when he is unwell as compared to a daughter and consequently taking the son to a formal facility while the daughter is given home remedies or taken to traditional healers. However, the extent and magnitude of gender discrimination varies by geographical, socio-economic, and demographic characteristics of the mother. Cultural perceptions and behavior around Pneumonia and Diarrheal diseases amongst caregivers of children under 5, impacts preventive practices, care seeking attitudes and management of these illnesses. Women are considered as primary caregivers for treating children with these illnesses. Low or lack of any formal education amongst a majority of these women results in delayed diagnosis which in turn leads to unnecessary delay in care seeking and low compliance for these illnesses (Bhutta & Hafeez, 2015). Women also face challenges with regards to financial

empowerment. Males are seen to play a key role in final decisions including when and where to seek treatment resulting in delayed care seeking (Fatimi & Van, 2002).

2.3 BMGF & Programme for Child Survival

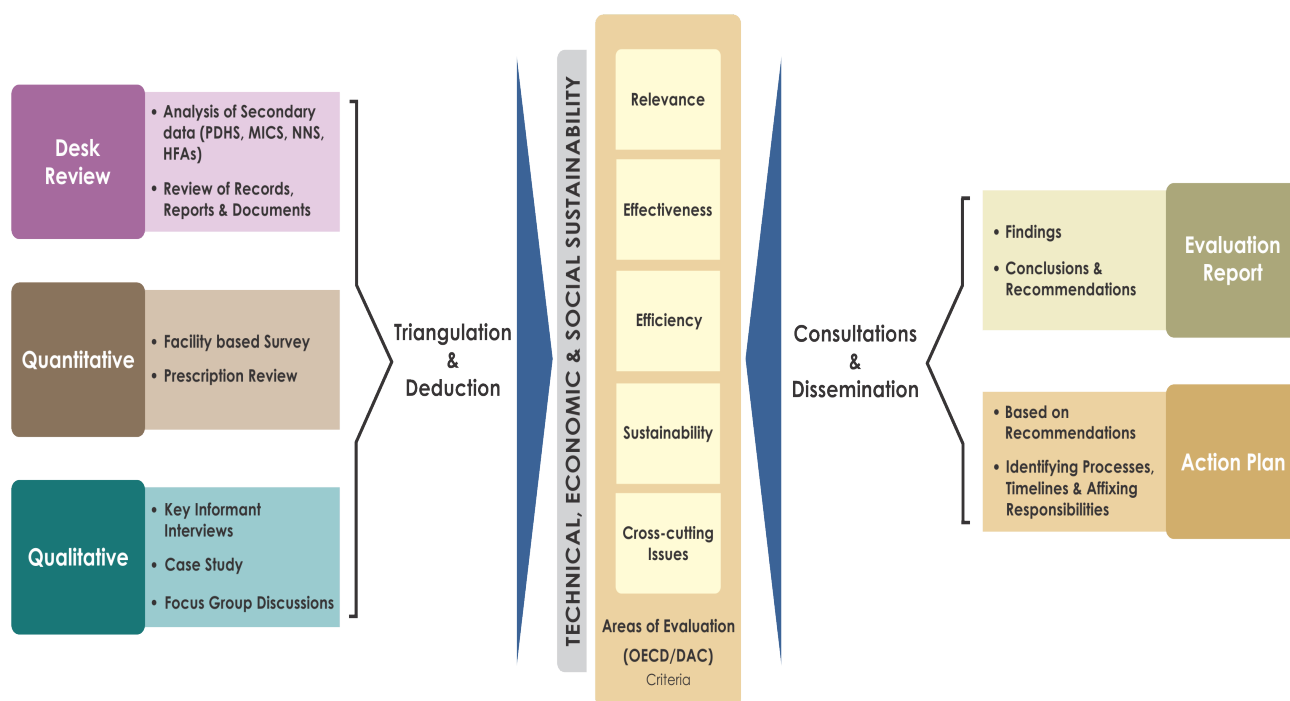
With the partnership of UNICEF and Bill and Melinda Gates Foundation (BMGF), the project “Accelerating Policy Change, Translation and Implementation for Pneumonia and Diarrhea Commodities in Pakistan”, has been established and is supposed to be implemented from 2016 to 2020. The project is designed to improve the diagnosis, treatment as well as prevention of Diarrhea and Pneumonia in Pakistan. Working in close collaboration of relevant stakeholders and Government of Pakistan, the overall purpose of this project is to contribute to the efforts to increase child survival in Pakistan, particularly by strengthening policy solutions for better diagnosis and treatment of Pneumonia and Diarrhea in girls and boys under 5 year of age.

3. Evaluation Methodology

It is formative evaluation (intended to improve performance), where a mixed methods approach (quantitative and qualitative techniques) was employed and both primary and secondary data was collected, in wherever cases possible, disaggregated by gender. Rationale for mixing methods was to obtain complementary data on the outcomes of the interventions, viz., availability of revised and updated commodities at the public health facilities and their prescription to the children suffering from Pneumonia and Diarrhea whereas effects on treatment outcomes and acceptability in the population was assessed through qualitative techniques. Qualitative component comprised of desk review, key informant interviews, focus group discussions and case studies whereas quantitative was relatively small in overall evaluation design and comprised of health facilities checklist and prescription reviews. Guides and tools developed for this purpose focused on the entire pathway of policy transformation processes (policy change, policy translation, policy implementation and knowledge management). Meetings were held with the stakeholders and implementers to elicit information related to achievements and impediments during the Project, which informed and strengthened the interview guides and quantitative tools. The evaluation team worked in close collaboration with all stakeholders. UNICEF project teams from federal and provinces were closely involved through provincial consultative and preparatory meetings for seeking their inputs and feedback on evaluation approach. Stakeholders included federal and provincial governments, health facilities, outreach workers, private sector, donors and development partners and the ultimate beneficiaries. Gender mainstreaming was kept in consideration while devising the data collection process and tools. Gender balance was incorporated in the approach of the evaluation methodology along with exploring gender differentials and possible discriminatory practices against girl child during data collection.

A conceptual framework for the midterm evaluation (Figure 2) was developed to describe the evaluation approach and techniques. Relevant methodologies were chosen for tracking effectiveness of different interventions and their effect on the beneficiaries.

Figure 2: Midterm Evaluation – Conceptual Framework



3.1 Evaluation Criteria and Questions

Following the TORs of MTE, Evaluation Matrix (Annex 1) was developed considering the evaluation objectives as per OECD/DAC criteria namely relevance, effectiveness, efficiency and sustainability. In addition to DAC Criteria, cross-cutting areas of equity and gender equality were taken into consideration while designing this evaluation as well as the Convention on the Rights of the Child (2 September 1990) and Convention on the Elimination of All Forms of Discrimination against Women (18 December 1979) for incorporation of the UN and UNICEF's commitment to a human rights-based approach to programming to gender equality and to equity. All major evaluation questions were considered for discussion with stakeholders to conclude the findings of evaluation. Certain aspects of the DAC Criteria, like efficiency in terms of cost per beneficiary and comparison of the project cost with other similar interventions was not included in the scope of this evaluation. The Reference Group designated to review the technical approach and methodology of the evaluation endorsed this during the inception phase. Being a midterm evaluation, the impact criterion was not included in this evaluation. The MTE mainly focused on performance evaluation to see if the project was on track to achieve its desired outcomes and based on the lessons learnt what course corrective measures should be taken by the project and thus there was no counterfactual design adopted for this evaluation,; however, some of the outcomes that could be assessed under this timeline were included under criteria of effectiveness and efficiency. Building on the objectives of the midterm evaluation, key evaluation questions were developed as part of the evaluation matrix, which addressed the following criteria:

3.2.1 Relevance

It is the extent to which the objectives of the development intervention suited to the priorities and policies of the target group, recipient and donor. In this MTE, relevance is evaluated through the following questions:

- How relevant and meaningful are the project objectives and activities in addressing the needs and priorities of the marginalized and vulnerable children in the project areas?
- To what extent the objectives of the project are consistent with the existing national/provincial policies and guidelines in line with global recommendations (WHO/ GAPPD) for management of Diarrhea and Pneumonia among under-five children in Pakistan and are sustainable?
- Are the strategies or approaches appropriate and adequate to achieve results?

3.1.2 Effectiveness

Effectiveness is measure of the extent to which the development intervention attained its objectives. It was evaluated by the following questions:

- To what extent the project has achieved its objectives/ outcomes and what were the major factors influencing the achievement or non-achievement of the objectives/ outcomes?
- To what extent the implementation of the project approaches worked as intended, particularly after the baseline in 2016 and subsequent adjustments?
- How effectively various Federal Ministry and Provincial Health Departments and Programmes coordinated among each other?
- Whether the mechanisms available to create awareness among communities are effectively linked to the project objectives?
- Whether the healthcare provider and community health workers have the required knowledge and skills to proper utilization of supplies as per standards?

- Whether the monitoring and reporting mechanisms exist and effectively implemented for effective tracking and improvement in system?
- How effective are the ‘innovative approaches like use of ARI timers and pulse oximeter, what results can be achieved, are replicable within the Sindh/Punjab context?
- To what extent the project has been able to contribute to ownership and leadership of the provincial/ area DOHs?
- What have been the major factors influencing the achievement or non-achievement of outcomes?

3.1.3 Efficiency

Efficiency measures the outputs – qualitative and quantitative – in relation to the inputs. It is an economic term and measures how economically resources/inputs (funds, expertise, time, etc.) are converted to results. For the MTE, efficiency was evaluated by the following questions:

- How well the resources, both human and financial, been managed to ensure timely, attainment of results?
- What are the implementation challenges from the perspective of both right holders and duty bearers especially for the under- five marginalized children of communities?
- What is the value added in terms of improved delivery of services for Pneumonia and Diarrhea?
- Whether the availability of P&D supplies is adequate, timely and whether they are prescribed?
- To what extent has the project achieved its goals in enhancing the health outcomes especially of U5 children in the catchment communities in Sindh and Punjab target districts?

3.1.5 Sustainability

Sustainability is concerned with measuring whether the benefits of an activity are likely to continue after donor funding has been withdrawn. In order to evaluate sustainability, following questions were framed:

- What evidences exists to see the likelihood of the project results are sustained and will be adopted by the Government to ensure that the ultimate goal of the project is achieved?
- What internal/external factors and drivers contribute to or constrain the sustainability of the project?
- What is required to ensure prospects of sustainability of the project outcomes and the potential for replication or scale up of good practices and/or innovative approaches?

3.1.6 Cross-cutting Areas

Equity across the socioeconomic strata, gender equality and human rights based approach are the crosscutting areas considered in this MTE and are, addressed through following questions. The findings regarding these questions were embedded across the OECD/DAC criteria.

- To what extent the crosscutting issues such as gender, equity and human rights were taken into account at various levels of planning and implementation?
- Are the services provided gender responsive and whether they take the gender and human rights-based approach into account during implementation?

- What have been the key lessons and experience of healthcare providers, Lady Health Workers and Lady Health Supervisors as agents of change and their contributions to other sectors?

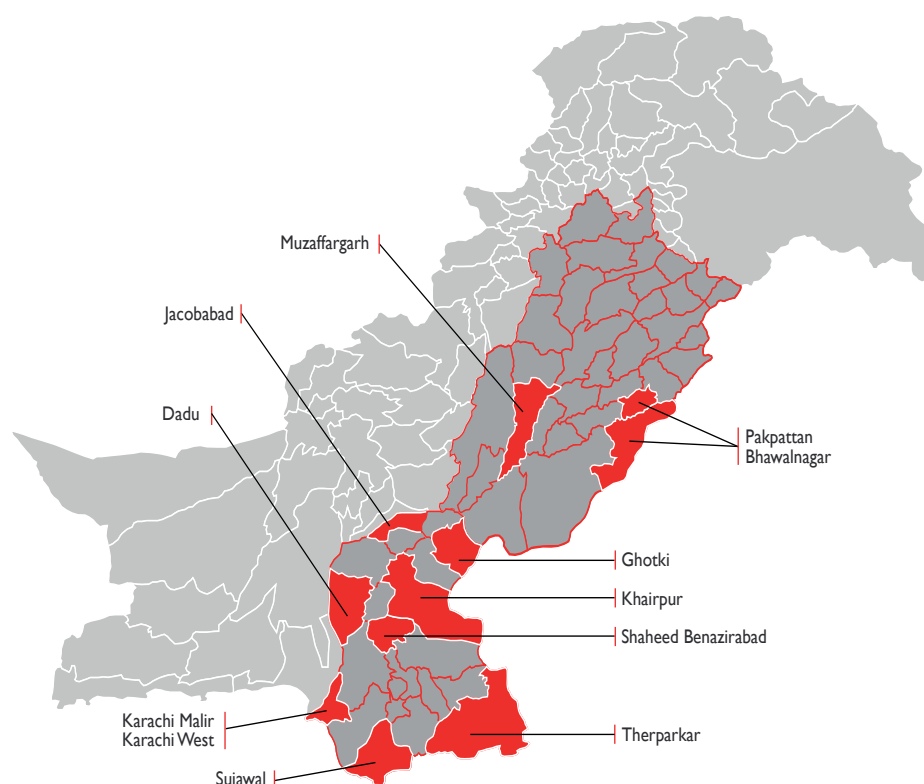
3.2 Evaluation Technique – Mixed Method

This MTE employed mixed method approach with both qualitative and quantitative components. Having a formative evaluation design, the MTE relied heavily on qualitative component. A quantitative component was required to answer certain research questions pertaining to efficiency criterion, whereas qualitative data mainly answered the rest of OECD/DAC criteria. Quantitative data was limited to availability of new/revised commodities and their prescription to children in the surveyed health facilities. Qualitative data was collected through key informant interviews and focus group discussions with the relevant stakeholders while quantitative data was gathered through facility checklists and prescription review from the surveyed health facilities. Both purposive and random sampling techniques were utilized respectively for sample selection. At the level of selection of districts, purposive sampling was done to cater the geographic, demographic and socio-economic factors in the districts selection. Within districts, health facilities were randomly selected. This approach proved useful in covering the vast variations across the districts of the same province and providing certain degree of randomness in selection of health facilities. This approach was finalized after consultation with members of MTE reference group during inception phase through individual meetings and inception workshop.

3.2.1 Geographical Scope

The geographical scope of this project included the 5 districts of Punjab and all the 29 districts of Sindh, where the Pneumonia and Diarrhea Project was being implemented.

Figure 3: Selected Districts for Midterm Evaluation



For sample selection of districts in Sindh, a divisional approach was adopted so that representation was uniform from within all 29 districts of Sindh. 7 districts were purposively selected from 6 divisions of Sindh while 2 districts from within 6 districts of Karachi were included. In total 9 districts of Sindh and 3 districts of Punjab (out of the five intervention district while catering to the geographical coverage of the project) were selected for the evaluation ensuring representation of all.

Table 1: Project Implementation and Selection of Districts	
PUNJAB	
Total intervention districts	05 (Bahawalnagar, Muzaffargarh, Rahim Yar Khan, Pakpattan, D G Khan)
Selected districts	Pakpattan Muzaffargarh Bahawalnagar
SINDH	
Total intervention districts	29 (entire province)
Selected districts	Karachi West Malir S. Benazirabad Ghotki Dadu Khairpur Jacobabad Tharparkar Sujaawal

Selection of districts in Sindh covered geographical, human development, and topographical variations. The following table shows the selected districts.

Table 2: District Selection in Sindh – HDI and Geographical Variations			
District	HDI Value	Selected Districts	Divisions of Selected Districts
Karachi (all 06)	0.789	Karachi West	Karachi Division
		Malir	
Ghotki	0.71 to 0.68	S. Benazirabad	S. Benazirabad Division
S. Benazirabad			
Sukkur			
TA Yar			
TM Khan		Ghotki	Sukkur Division
Matari			
Jamshoro			
Hyderabad	0.679 to 0.62	Dadu	Hyderabad Division
Mirpurkhas			

Dadu			
Khairpur			
Sanghar		Khairpur	Sukkur Division
Naushahro Feroze			
Badin			
Shikarpur	0.619 to 0.31	Sujawal	Larkana Division
Sujawal			
Thatta		Tharparkar	Mirpurkhas Division
Kambar			
Larkana		Jacobabad	Banbhore/Thatta Division
Jacobabad			
Tharparkar			

3.2.2 Development of Data Collection Tools

Data collection tools were developed in order to undertake a high quality; impartial, participatory, equity focused and gender responsive evaluation. Consultations and meetings were held with the relevant key stakeholders as a means to provide an opportunity for building consensus on evaluation methodology and tools. Tools were shared with government counterparts before the data collection began. A deductive approach was adopted in developing the data collection tools, which was guided by the Evaluation Matrix (Annex 1) along with desk review and documents and information received during consultations with the relevant stakeholders. All sets of tools were pre-tested. Feedback from pre-tests and review of relevant stakeholders were incorporated in finalizing the tools (See Annex 5 for data collection tools and guides).

3.2.3 Qualitative Data Collection

Stakeholders at primary departments, agencies and institutions at federal, provincial/regional level, selected districts level, facility in-charges; development partners and community level were engaged for qualitative data collection. Techniques employed for qualitative data collection included key informant interviews and focus group discussions. Limitations of both the techniques were kept in mind while designing the evaluation and identifying type of consultation for each stakeholder.

Stakeholders engaged at policy change and policy translation levels were contacted through key informant interviews whereas Focus Group Discussions mainly generated information about policy implementation and its challenges. Key informant interviews gathered information from multitude of stakeholders from different institutions and organizations, both within and outside government, whereas Focus Group Discussions were used for obtaining insights from groups of professional (like general practitioners, medical store keepers, community workers) and perspective of beneficiaries. Although key informant interviews are susceptible to limitation of selecting right interviewee and challenge of scheduling interviews with policy making level respondents. In addition to technical reviews, Reference Group formed under the Ministry facilitated in selection of the right key informants and scheduling appointments. List of key informants is given in the following table.

Table 3: Qualitative Sampling Matrix for Key Informant Interviews			
Sr. #	Key Stakeholders	Targeted Officials	No. of KIIs
Federal Level			
1.	Ministry of National Health Services Regulation and Coordination	Director & Deputy Director Programmes	2
2.	Neonatology Children hospital, PIMS	Associate Professor	1
Provincial Level			
3.	Department of Health Sindh	Representative	1
4.	Department of Primary and Secondary Healthcare Punjab	Representative	1
5.	Directorates General of Health Services Punjab and Sindh	DG Health	2
6.		Director DHIS	2
7.	Vertical Programmes (LHW Programme Sindh, MNCH Programme Sindh, IRMNCH & N Programme Punjab)	Representatives	3
8.	Implementing Partners of DOH Sindh (HANDS, MERF, PPHI, IHS)	Representative	4
9.	Pharmacists Associations / Pharmaceutical Manufacturers	Representatives	2
10.	Child Survival Groups Punjab and Sindh	Representatives	3
11.	Sindh Child Survival Programme	Representative	1
12.	Department of Pediatrics, Ganga Ram Hospital, Lahore	Head of Department	1
13.	Pakistan Pediatrics Association	Representative	1
UNICEF Programme Staff			
14.	Federal Staff	Focal Persons	2
15.	Provincial Staff	Focal Persons (Sindh & Punjab)	2
Development Partners			
16.	WHO	Focal Person	1
17.	Aga Khan University	Focal Person	1
District Level			
18.	District Health Department	CEO/DHO along with LHW, MNCH and DHIS Coordinators	12
19.	Health Facilities	In-charges	48

Focus Group Discussions (FGDs) were conducted with Local Service Providers, Pharmacists/medical storekeepers, Community health workers (LHWs, LHS) and service users/mothers/care-givers of children under 5 years (16 FGDs with 6-8 participants each).

Although FGDs face certain limitations but due to non-conflicting nature of childhood Pneumonia and Diarrhea, participants of the FGDs expressed their thoughts and discussed opportunities to save the children of their communities. Gender diversity was ensured among focus group members. The FGDs with LHWs and mothers (above 18 years) all had female participants. Numbers of FGDs are given in the following table.

Table 4: Qualitative Sampling Matrix for Focus Group Discussions		
Sr. No.	Types of Stakeholders	No. of FGDs
1.	Service Providers (Private Practitioners)	4
2.	Pharmacists and Medical Store Keeper	4
3.	Community Health Workers (LHWs, LHSs)	4
4.	Beneficiaries/Service Users (Over 18- Mothers/Caregivers of children under 5)	4

3.2.4 Quantitative Data Collection & Sampling

Facility based data on prescription review was collected from 48 health facilities (24 Basic Health Units and 24 Rural Health Centers) within 12 selected districts of both Punjab and Sindh. This sample of BHUs and RHCs was selected, considering the focus of P&D Project interventions at the district level. From each health facility total of 10 prescriptions were reviewed, making it a total of 360 prescriptions reviewed as part of the quantitative analysis, which was used to address the evaluation question under efficiency.

Table 5: Quantitative Sampling Matrix for Health Facilities	
BHUs per district	02
RHC per district	02
Prescription reviews per facility	10
PUNJAB	
Number of districts	03
Number of BHUs in 03 districts	06
Number of RHCs in 03 districts	06
Number of prescriptions reviewed	120
SINDH	
Number of districts	09
Number of BHUs in 09 districts	18
Number of RHCs in 09 districts	18
Number of prescriptions reviewed	360

3.3 Secondary Data

As reflected in the literature review section, desk review of all the relevant project data available at federal, provincial and district level was conducted to inform the scope of the project and to develop a better insight of the current scenario. A comprehensive list of documents, not limited to Pneumonia and Diarrhea Strategic Framework, Federal and Provincial Policies and strategies, Provincial PC-1s, federal and provincial programmes and operational

plans, Global mandates such as SDGs, UHC and National Health Vision, GAPDD Report, Evaluability study/baseline landscape and Pneumonia and Diarrhea project periodic monitoring and evaluation data, targets and monthly reporting data were taken into consideration to develop a thorough understanding of the project. Other documents reviewed included Project Documents for policy, systems, management and Supply chain management review as well as for Gender, Social Exclusion and Poverty (GSEP) analysis. Latest rounds of available datasets like PSLM, PDHS, MICS, HFA and NNS, Audit reports and financial documents with necessary information were taken into account. Routine information systems including DHIS, LHW-MIS and LMIS were also be a part of the review for their potential for updating and building linkages. Moreover, provincial and district offices were also visited to obtain relevant data as indicated in 'Evaluation Matrix'.

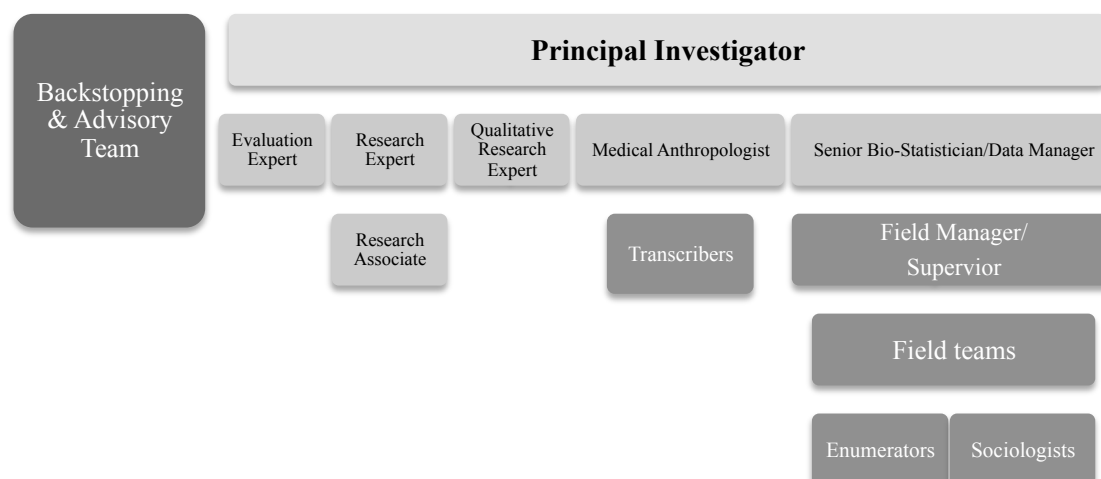
3.4 Evaluation Team

A team of experts having the required skill mix was organized for efficient and effective completion of the evaluation activities. The organogram of the midterm evaluation team is given in Figure 4 and details of individual team members are given in Annex 2.

3.4.1 Roles and Responsibilities

A Principal Investigator led the team that included evaluation and research experts, medical anthropologist, data manager and field data collection teams. Technical team was working in close association for developing tools and the collaboration continued in data collection and data analysis. Research associates as well as transcribers were engaged after the data collection phase. Field team comprised of field manager who supervised field teams. Technical team members participated in data collection at federal and provincial levels along with all interactions with the UNICEF counterparts. Field teams were responsible for gathering information at district level. Four field teams conducted facility-based survey, each team with 4 persons. 1 team was formed in Punjab with 1 male supervisor and 3 other team members (2 females and 1 male). In Sindh, 3 teams with similar composition were established. For District level KIIs, FGDs and case studies, 4 teams of Sociologists (1 Moderator and 1 Note taker in each team), 1 team in Punjab and 3 teams in Sindh were formed. Punjab's team covered all the 3 districts in Punjab while 3 teams in Sindh catered to the 9 districts of Sindh. It was ensured that all field team members were fluent in native language and well versed in local customs. Teams were identified, recruited, organized, trained and supervised. Guidance and support was sought throughout the evaluation process from backstopping team comprising of health policy and systems expert and procurement and supply chain expert.

Figure 4: Organogram of Evaluation Team



Through this qualified and experienced team, the following aspects of the evaluation were strengthened in addition to overall assurance of the quality of evaluation work.

- Developing and implementing a transparent system for evaluation
- Having adequate number of well trained and qualified staff
- Developing strong participation approach
- Basing decisions and making recommendations on quality information

3.5 Procedural Steps and Quality Assurance Mechanisms

3.5.1 Ethical Considerations

Ethical considerations for this evaluation are built on UNICEF Procedure for Ethical Standards in Research, Evaluation, Data Collection and Analysis and to the UNICEF Strategic Guidance Note on Institutionalizing Ethical Practice for UNICEF Research. The ethical review board (ERB) of UNICEF cleared this evaluation and their certificate is attached at Annex 3.

Both Field and technical teams maintained the highest standards of integrity, sensitivity, and confidentiality in dealing with informants, to ensure that the dignity, human, and civil rights of people involved, are respected. Overall, the ‘do no harm’ principle was applied throughout, especially when working in the field. UNICEF’s protocol on Ethical Standards in Research and Data Collection and UNEG’s ethical standards for data collection and evaluation was also incorporated in the trainings and ensured that it was strictly being observed. Additionally, following ethical considerations were imparted for data collection:

Informed Consent: Interviewers respected the rights of interviewed individuals. Every individual had the right to refuse to participate, or to refuse to answer specific questions. Verbal and written consent was sought after explaining the objective of evaluation, data collection procedures, along with risks and benefits. Field staff read out contents of consent forms and clarified any apprehensions of respondent/participants and interviews only continued after getting a formal approval.

Privacy: It is important that data collection process be conducted in a manner, comfortable for each respondent, and in which the individual is able to speak openly and honestly. The enumerators and facilitators made sure that the place of interview provided privacy for the interviews and FGDs.

Confidentiality: Interviewers may not discuss participants’ answers with anyone, except the supervisor when clarification was needed. Individuals’ names or other identifying information was not be linked to any responses.

Safety and security of data: Hard copies such as interview notes, prints of photographs and audio recordings was kept securely in a locked cabinet that can only be accessed by agreed members of the MTE team. Soft copies in the computers were encrypted / password protected. All data will be securely kept for up till three years and then safely disposed off.

Conflict of interest: Being a primary element of a staff member’s obligation to maintain integrity, independence and impartiality required, there should be no conflict of interest. No actual and potential conflicts of interests were identified for the evaluation team.

Addressing Gender and Child Rights Issues: Gender equality is a human rights issue and a prerequisite for sustainable development. According to human rights principles of equality and non-discrimination, everyone is entitled to equal enjoyment of their rights and the responsibilities and opportunities that come along, regardless of their gender. In Pakistani society gender is a major organizing principles. Local traditions and culture embody values pre-determining gender roles in the community. There is substantial diversity in the status of women as well as rights of the children across classes, regions, and rural/urban divide due to uneven

socioeconomic development and the impact of tribal, feudal, and social formations on women and children's lives. This evaluation kept in consideration gender, child rights and social exclusion barriers during varied phases of evaluation including activity designing, consultations and stakeholder engagement, staff hiring and training, monitoring and was ensured in data analysis.

Selection of field teams ensured gender balance. Further, during training of field teams, gender and child rights orientation was given as part of the evaluation design. Evaluation approaches and activities were revised to safeguard sensitivity in understanding gender values as well as child rights according to CRC. At large, all stages of the evaluation cycle were assessed through the lens of gender, social exclusion, child rights and equity, ensuring that it was reflected in the findings and recommendations.

3.5.2 Risks Management

Risk management plan was developed, identifying events in which the team foresees numerous potential risks and constraints, which may affect the evaluation adversely. Accordingly, mitigation measures were adopted to ensure a robust evaluation process and outcome like data quality, consistency and others. A risk register was prepared at the inception of the project and is attached as 6.

3.5.3 Backstopping

Backstopping and quality assurance mechanism was established and a consensus was built on what activities and outputs to be monitored, how to monitor and what information are required to monitor them. Backstopping and Advisory team carried out this task. Monitoring and regular progress updates were used as a way of assuring quality.

3.5.4 Continuous Liaison with UNICEF Team

Evaluation team worked in close collaboration and continuous liaison with UNICEF Evaluation and Research Unit, PMU as well as UNICEF's Gender Specialist and Gender Focal Point in Health Section. The data collected is the sole property of UNICEF.

3.5.5 Inception Phase

Preparatory meetings were held with relevant UNICEF staff at Punjab and Sindh to develop a deeper understanding of project component, implementation approaches, activities and guidance on evaluation framework. The aim of these initial meetings was to gain a consensus on evaluation design, methodology/implementation strategies and work plan. During these pre-inception meetings and informal consultations, many relevant documents and pieces of information were shared which have facilitated Contech in informing the scope of the evaluation. A Reference Group was established for quality assurance of all key deliverables. A formal Evaluation Management Team was also formed for this assessment, which comprises of evaluation teams from both UNICEF and Contech. This facilitated close coordination and collaboration with UNICEF since the commencement of the project.

The inception meeting took place on 4th July 2019 in Islamabad. Federal Ministry, provincial departments, donors, development partners and UNICEF project team participated in the meeting and gave comprehensive feedback on all sections of the inception report. Based on the feedback received during the meeting, the Inception Report was revised. The final inception report had the technical approach and methodology of the midterm evaluation among key stakeholders.

3.5.6 Field-testing and Finalization of Data Collection Tools

Approved tools after the inception meeting were further tested in the field, in same settings followed by adjustments, if needed, to determine:

- Any problem in language of questionnaire and skip patterns if any;
- Completion of interview in a given time; and
- Clarity of enumerators on questionnaire.

Field-testing was conducted systematically, with potential respondents by using the same method of administration. Considering the feedback of field-testing, any changes in tools were made and shared with UNICEF Evaluation Management Team.

3.5.7 Recruitment of Field Staff

Appropriate and qualified field staff members for evaluation were identified, short listed and then hired. Field teams was constituted striking a gender balance. Following were the structure and composition of the teams:

- 2 Teams with Technical team members conducted Federal and Provincial level KIIs in Punjab and Sindh
- Facility Based Survey (In total 12 persons) for 18 days
- Punjab: 1 team of 3 persons (2 females +1 male) for 18 days
- Sindh: 3 team of 3 persons each (6 females + 3 males) for 18 days
- District level KIIs, FGDs and Case studies (In total 8 persons) for 10 days
- District level KIIs were done by a team of Sociologists (male moderator and female note-taker)
- Punjab (3 districts): 1 team of 2 Sociologists (1 male and 1 female) for 10 days
- Sindh (9 districts): 3 team of 2 Sociologists (1 male and 1 female) each for 10 days

The technical team members conducted federal and provincial KIIs. The district data collection teams were responsible for conducting district specific KIIs, FGDs and districts' office visits. It was ensured that they must be graduates, fluent in native language and well versed with the local customs.

Teams were monitored by Field Manager, who shall act as coordinator and responsible for overall supervision and reporting to technical team. Field teams were identified, recruited, organized, trained and supervised by Contech, which was also be responsible for supervision, transportation and logistics of team movements, and their subsequent payments.

3.5.8 Training of Field Staff

The field training was conducted at the highest professional level and in a systematic manner that ensured timely and quality completion of data collection exercise within the stipulated time. This included finalization of field plans, availability of printed material, training guide and nametags etc. Specific attention was paid to ensure that training environment is conducive, without disturbance, with sufficient seating arrangements, keeping participants, trainers and field monitors in consideration. A team of trainers consisting of 2 males and 4 females conducted the three-day training sessions.

A detailed training manual for the field teams was prepared with clear instructions and guidelines, which was distributed among the data collectors. This training guide covered topics

including evaluation protocols, how to approach the concerned participants, how and when to probe, ethical requirements, filling of tools and recording necessary information while ensuring completeness and quality. This document was easily referred to in case of any confusion faced in the field. Ethical considerations including confidentiality, informed consent, privacy and anonymity were incorporated in the training sessions. Specific attention was paid to ensure that data collectors had a clear idea of the objectives of evaluation so that they were aware when to elicit information. Training topics included:

- Purpose of evaluation and its goals and objectives;
- Methodology and technique;
- Ethical requirements;
- Techniques of filling out tools; and
- Field simulation

Data collection teams were trained in the following two levels in order to achieve uniform standards while ensuring quality:

- One day training of technical team members for visits/consultations at federal/provincial level KIIs was conducted.
- A 3-day training of field team at a central level training workshop for enumerators and sociologists (moderators/note-takers) conducted in Lahore for district specific KIIs, FGDs, case studies and districts' offices visits.

3.5.9 Development of Field Micro-plans

Followed by the field simulation activity, detailed micro-plans were developed on the last day of training, which was shared with the monitoring and evaluation teams. The micro-plan entailed day-to-day data collection activities of each field team, ensuring that the provided trainings were put to practical use and that the data collection was being carried out in the required manner. Micro-plans facilitated the monitoring of data collection activities in the field and addressed any issues that arose during the process. The training agenda and field micro-plan were shared with UNICEF for the comments and feedback before finalization.

3.5.10 Monitoring of Field Data Collection

A thorough monitoring mechanism at both macro and micro levels was employed for assuring quality and gender disaggregation of collected data, by developing monitoring tools, their field-testing and modification. A monitoring and evaluation plan of the field activities were developed along with detail procedures and protocols that were shared with UNICEF. A participatory monitoring and evaluation approach was adopted, involving stakeholders and beneficiaries in the process of data collection. Specific measures were established to ensure proper accountability and transparency throughout the evaluation, specifically during the data collection phase. Senior Technical team was assigned task of assuring data quality and was responsible for:

- *Accompanying Calls:* observing some of the interviews to ensure that the interviewers were conducting the interview well, asking the questions in the right manner and interpreting the answers correctly.
- *Conducting Spot Checks:* to ensure that the questions were asked properly and recorded appropriately and that respondents are being identified correctly. UNICEF Evaluation and Research Unit also conducted the spot checks of the teams during the interviews and FGDs.
- *Questionnaire Completion Reviews:* reviewing a proportion of questionnaires to ensure that they were complete and internally consistent.

- *Micro-plan based data collection:* A day-wise field micro plan was prepared and shared with the UNICEF and the relevant stakeholders at Ministry and provincial departments for joint supervision.

3.5.11 Fieldwork and Data Collection

The course of data collection took place as per the detailed micro-plan. The entire evaluation fieldwork was completed within a span of 6 weeks. However, the technical team reconnected with certain key informants to elaborate further on their discussion during the report-writing phase.

3.6 Data Management

3.6.1 Data Handling and Record Keeping

In order to ensure the quality of data, field manager reviewed/checked tools for completeness and logical errors. Technical team reviewed collected data to assure quality standards. All completed tools were kept confidential after entry, analysis, and report writing. It was ensured that only authorized personnel have access to the filled tools. Electronic soft data was stored in a database while reports are entered whereas entered data was kept in folders and backup for safety purposes.

Data gathered through KIIs and FGDs was recorded and transcribed. After collection of field data each completed tool was reviewed before submission. Routine data transfer was done on a Periodic basis to the Team Leader by field team. To minimize human error, double data entry was done whereas transcription carried out carefully considering all field notes to ensure the proper flow and recording of valuable information. Physical validation of a sample of data was conducted during routine monitoring visits. Once data was collected, cleaned and secured, it was transcribed for analysis. Field teams transcribed data of KIIs and FGDs. To ensure secure and efficient data management during the fieldwork and data collection phase, following procedures was adopted:

- **Data capture** – FGDs and IDIs were recorded and transcripts were prepared; collection of survey data, semi-structured interviews were supervised by District Supervisors who were responsible for checking each completed questionnaire before submission to the Contech's Provincial/Regional Coordinator.
- **Routine data transfer** – Weekly reporting was done by Provincial/Regional Coordinator
- **To minimize human error** – Data entry was done after forms were checked by both District Supervisor and Provincial/Regional Coordinator. Desk editing team was placed at Head Office who edited the data of any errors.
- **Data verification** – Physical validation of data sample was conducted during routine monitoring visits and ensured that collected data was gender disaggregated.

3.6.2 Data Analysis

Qualitative analysis was characterized by reducing the overwhelming amount of data by identifying the content of more or less encompassing data segments. This analysis was conducted manually, exploring the complex phenomena hidden in data to manage, extract, compare, explore, and reassemble meaningful pieces from large amounts of data in creative, flexible, yet systematic ways. A "code" as abbreviation or name was attached to each segment. These codes were used as representatives of data segments or "units of meaning" in the data. A coding framework based on the themes given in the interview guides was designed. Transcripts were examined manually to identify codes using thematic content analysis and a system of constant comparison. They were read several times, choosing units of meaning, identifying

general themes, categorizing and classifying. This involved going through the data several times, making comparisons and connections until no further codes were identified and until data was saturated. Once codes were identified, they were categorized under each of the main themes used in the interview guides. The technical team carried out coding and thematic analysis and differences in opinion was resolved by mutual consensus. Quantitative data analysis was processed using SPSS. Initial task in analysis was to produce draft dummy tables and based on data, graphs and descriptions were included in the report. At the end, triangulation was applied to both qualitative and quantitative findings with secondary data while presenting evaluation findings and recommendations. The evaluation matrix provides a framework on the basis of which the collected data was analyzed and triangulated, as given in Annex 1.

3.6.3 Report Compilation

Initially, draft report entailing preliminary findings of data gathered was submitted to UNICEF on which feedback was sought. Draft report was developed, entailing details of evaluation scope and objectives, methodology, evaluation matrix, key findings and recommendations. The draft encompassed the comprehensive quantitative and qualitative analysis of strengths, weaknesses, achievements, and recommendations to improve performance in future. A 'Theory of Change' is an integral part of the final report that entailed pathway for course correction and continuity of these interventions beyond the project life. In order to effectively disseminate the findings, the evaluation team developed two case studies based on the achievements and successes of the project to bring sustainable improvement in management of childhood Pneumonia and Diarrhea by health facilities and outreach workers.

3.7 Limitations of the Evaluation

This midterm evaluation carries certain limitations that affected the interpretation of the findings. The technical team took appropriate measures and used certain techniques to minimize these limitations.

Attribution of the outcomes with project interventions was a key limitation. The evaluation team conducted the comparisons with the baselines studies and asking the respondents to compare the before and after comparisons of the Pneumonia and Diarrhea Project to estimate a fair level of attribution.

Generalizability of the quantitative findings will not be possible as the purposive sampling was done for selection of the districts and health facilities within districts.

Reliability and validity of findings could be a possible limitation as multiple teams collected data at different sites and they might have conducted the interview and discussion somewhat differently. Extensive efforts on development of questionnaires, pre-testing, actual practicing on the questionnaires and data collection tools during the training helped mitigating this limitation.

Limitations of OECD/DAC Criteria due to overlapping of information among criteria could result in repetitions of findings or analysis. Evaluation team remained cognizant of this limitation during interpretation of findings.

Knowledge Assessment of care providers on use of updated commodities and their level of competence was not assessed as it was beyond the scope of this evaluation.

4. Evaluation Findings

This section provides the findings of the midterm evaluation in terms of the achievement of project outcomes, involved processes and their conformance against the OECD/DAC evaluation criteria, including relevance, effectiveness, efficiency, sustainability and cross-cutting areas of equity and gender equality.

4.1 Status of Project Outcomes

Project outcomes pertaining to policy change, policy translation, policy implementation and knowledge management were identified at the inception of the project and later on were incorporated in the TOC, which was developed retrospectively. As part of this midterm evaluation, achievement against each outcome and areas for further improvement were assessed to propose actionable recommendations. Overall, objectives of the project were consistent with the national policies and strategic frameworks. The implementation of the project has paved way for building a dynamic policy environment for child survival. It further ensures that every girl and boy has access to proven preventive and treatment measures through evidence based updated medical commodities for management of Pneumonia and Diarrhea. With the help of this integrated package of high impact and good quality healthcare commodities, the project has contributed to uplift the health of marginalized children with a special focus ensuring a continuum of care from communities to health facilities. Various national and provincial policies and guidelines were updated through this project to align them with the global recommendations, including GAPPD, for the management of Pneumonia and Diarrhea. Status of the achievement of specific project targets and the underlying process is described for each outcome area.

4.1.1 Project Oversight and Coordination

A project accountability framework was developed as part of the baselines landscape analysis. This accountability framework identified the pathways of change in strategic areas along with timelines and assigned roles and responsibilities of all the stakeholders. At the federal level, a Project Steering Committee was established, which was later merged in National RMNCAH&N TWG due to similar TORs and membership. At provincial levels, initially Child Survival Groups were notified in both Sindh and Punjab provinces. However, as a course corrective measure, the CSG of Sindh was subsumed in the MNCH Technical Working Group to avoid duplication as both had the same mandate and stakeholders participation. Through these oversight and coordination forums, government stakeholders and key development partners had been meeting regularly over the project life to review implementation and progress against the planned child survival activities. UNICEF provided the essential support throughout the project in implementation of the decisions and recommendations of these committees/groups to achieve its overarching goal of child survival. A Senior Provincial Manager described the role of CSG as, *“...Before we had no platform to discuss the issues related to child survival, CSG are of immense significance in uniting all relevant stakeholders. CSG’s are a think tank to the government on child mortality.”*

A senior provincial manager described the CSG/TWG as “a think tank to the government on child mortality”.

The support in implementing the joint accountability framework in its strategic areas (policy change, updates to clinical management practices, strengthened logistic and procurement system, budgeting and role of media) ensued in practical implementation of child survival policies.

4.1.2 Revision of IMNCI Guidelines and Manuals

A key action proposed in the accountability framework of the project was to update and reduce the number of days for in-service abridged IMNCI training manuals to six days from 11 days, with an enhanced focus on management of Pneumonia and Diarrhea. Various concrete steps were taken by the relevant stakeholders under the stewardship of Federal Ministry and supported by Provincial Health Departments to update the IMNCI manuals. At MoNHSR&C, national consultant was hired in consultation with WHO to help draft the revised IMNCI training manuals so that they are aligned with Global Action Plan for Pneumonia and Diarrhea (GAPDD) and WHO updated recommendations. As a result, key development partners and stakeholders, including WHO, UNFPA, USAID, DFID, Agha Khan University, and other NGOs and academic institution worked closely to revise and update IMNCI guidelines. In this regard, a series of consultative meetings under the supervision of MoNHSR&C and respective Health Departments, development partners and key decision makers were organized for seeking technical inputs and consensus-building during the drafting of IMNCI manuals. This effort has been lead by the WHO and based on the findings from the sixteen controlled trials on updated IMNCI conducted in nineteen countries. Meanwhile, the pre-service and community IMNCI components were also drafted and being reviewed by UNICEF/ROSA and WHO/EMRO team at the time of MTE. They are now nearing finalization to be ready for use in the respective trainings. UNICEF provided technical assistance for the development of these new training guidelines; while all the stakeholders agreed to undergo trainings with the in-service IMNCI six-day abridged course. The abridged course underwent pilot trials and was finally endorsed by National Taskforce for IMNCI in 2017. The IMNCI package was disseminated after its notification by the MoNHSR&C in 2108. Federal Manager highlighting the significance of abridge version of IMNCI stated, *“IMNCI has been reduced from 11 days to six day abridge course with an enhanced focus on management of pneumonia and diarrhea.”*

A federal manager while discussing the revision of IMNCI explained, “This is a difficult task, which was achieved through coordinated efforts by all the partners, including WHO and UNICEF.”

Sindh government has trained six batches on in-service IMNCI guidelines with support from UNICEF and WHO and pre-service IMNCI guidelines are being made a part of curriculum and examination in medical universities. The reduced cost associated with trainings of shorter duration makes it cost effective and provides an opportunity to train more health workers in the health facilities.

4.1.3 Updating Essential Medicines and Procurement Lists

The project advocated extensively with federal and provincial governments for the revision of Essential Medicines and Procurement Lists to include the updated commodities for better management of childhood Pneumonia and Diarrhea for both intervention provinces.

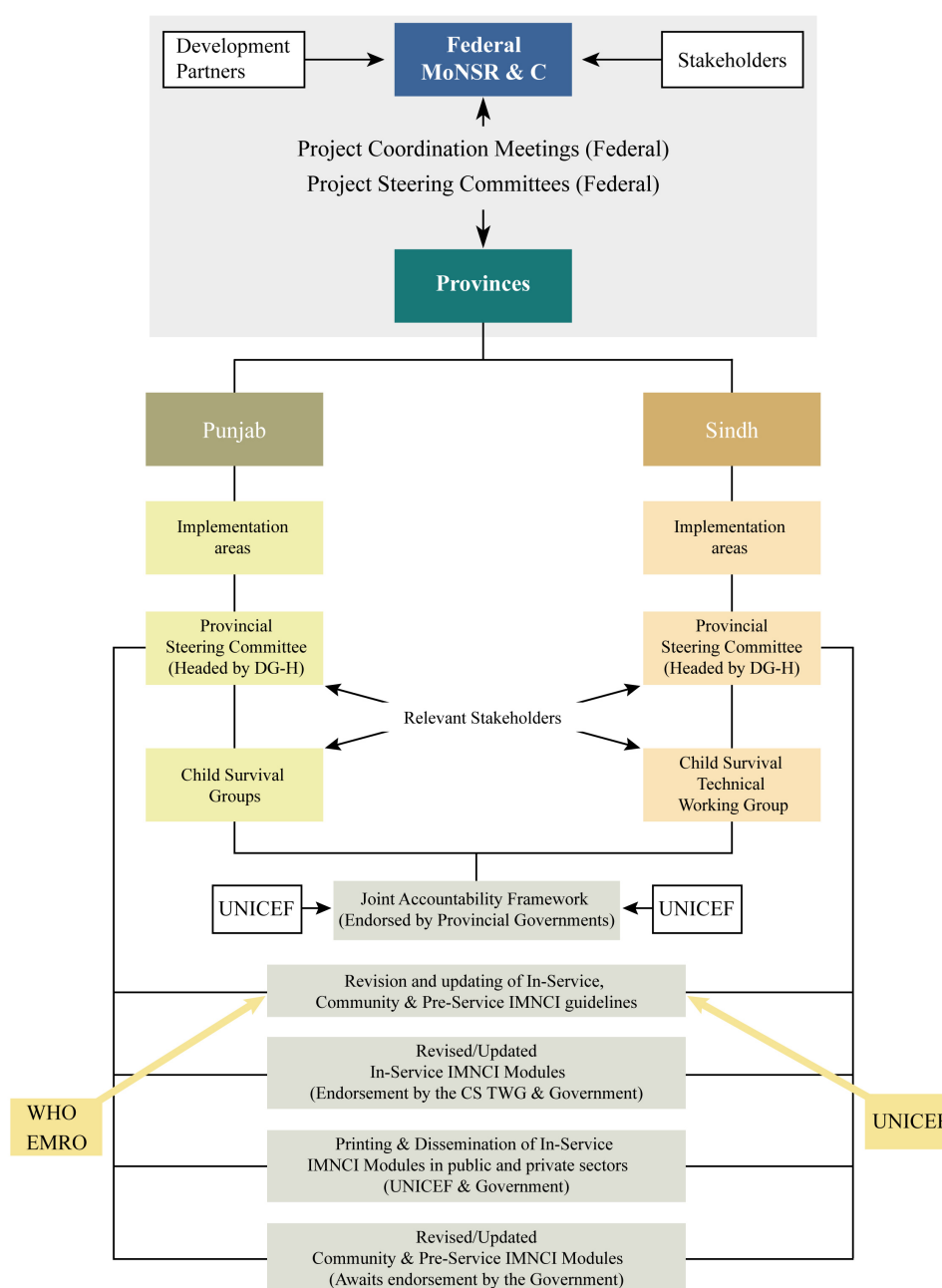
The project supported and facilitated widespread consultations (coordination committees, steering committees and programme management units) at national and provincial levels comprising of the relevant policy makers and partners to review the existing lists and update them with the inclusion of revised commodities for management of childhood Pneumonia and Diarrhea. Evidence-based policy briefs were prepared by the project on benefits of Amoxicillin DT, Zinc DT, co-packaged ORS and Zinc, Pulse Oximeters, ARI timer and oxygen in management of Pneumonia and Diarrhea, which were disseminated during the consultative meetings. Resultantly, updated lists included Amoxicillin DT, Lo-ORS, Zinc DT, ARI Timers, Pulse Oximeters and Oxygen for appropriate levels of service delivery. Now, the project is facilitating the operationalization of these lists for regular

“UNICEF Pneumonia and Diarrhea Project has popularised the DTs, now we all are aware of the benefits of switching to their use.”

Member of TWG/CSG

procurement by provincial procurement committees led by Health Secretary. The project further advocated for inclusion of the revised commodities in essential package of health services (EPHS) in Sindh, Minimum Service Delivery Package (MSDP) in Punjab, Minimum Service Delivery Standards (MSDS) and its endorsement from relevant CSG/TWG.

Figure 5: Revision of IMNCI Guidelines



4.1.4 Augmentation of Information Systems

The project supported the revision and enhancement of DHIS tools with inclusion of new indicators as well as the GAPPD recommendations, like updated commodities for management of Pneumonia and Diarrhea and indicators on safe drinking water, exclusive breast feeding and new vaccines. Furthermore, MIS of LHWs were updated to reflect the new commodities.

This change facilitated the report on stock outs of Amoxicillin DT and Zinc DT to the district and provincial managers. Sindh has upgraded and revised DHIS tools and procedure manual. New draft of DHIS tools, includes indicators about availability of Zinc DT, Amoxicillin DT and Low osmolarity ORS in the stock available at the facility. Section 12-A of DHIS includes Amoxicillin and Zinc DTs in the section of stock out reports. ARI Timers, Pulse Oximeter, ARI Timer and availability of Oxygen have been added in the section of stock out report. It also has a provision of entering OPD data for Diarrhea and dysentery separately for under 5 children (previously there was provision for entry of combined data only). Moreover, an age slot has been added to differentiate data of neonatal deaths from infant and U5 deaths. UNICEF has supported the printing of materials. All the GAPPD recommendations have been included in the updated DHIS tools. The revision of DHIS was not limited to the Pneumonia and Diarrhea commodities as the project leveraged the opportunity to update other MNCH related indicators including information on new vaccines and maternal health. A Provincial Manager, while highlighting the significance of updated DHIS tools, stated;

“Previously OPD data for diarrhea and dysentery were reported together. The updated DHIS has the provision of entering OPD data for Diarrhea and dysentery separately for under 5 children an age slot has been added to differentiate data of neonatal deaths from infant and U5 deaths.”

Considering the significant role of MIS system in combating mortality and morbidity due to Pneumonia and Diarrhea in Punjab, a new PC-1 is being developed for up-grading DHIS through technical support of UNICEF. In addition to development of PC-1, the project has committed support for upgrading the software (one time support). The updated information system – DHIS-2 – will have mechanism to collect and disseminate data of disease prevalence on daily basis and generate alert for any suspected disease outbreak/epidemic to ensure timely response. It will further have provision for data sharing at district, provincial, federal levels. Provincial Manager from Punjab stated,

“DHIS 2 would help Pakistan to move from no data to data visibility and would help improve all LMIS practices.”

“Technical experts from Oslo University have visited us and we are aiming at fast-track implementation with support of donors and development partners.”

Provincial Manager in Punjab

“The previous information system lacked collection and dissemination of data on a daily basis. The updated information system – DHIS-2 – will have mechanism to collect and disseminate data of disease prevalence on daily basis and generate alert for any suspected disease outbreak/epidemic to ensure timely response.”

Based on a recommendation made in the situation analysis of supply chain management to introduce a comprehensive HLMIS, the MoNHSR&C organized a think tank meeting to review and discuss the upgrading and linkages between LMIS and DHIS. UNICEF provided support to the government for establishment and strengthening of comprehensive supply chain management system including forecasting, procurement, distribution, and real time stock maintenance and warehousing. This involves different programmes to create a digitalized HLMIS, integration of the software with other available data management systems including vLMIS and cLMIS and to establish its linking with DHIS-2. The updated LMIS will not only contain information along these two illnesses but will also cover MNCH commodities listed in the EMLs. This project is about to be piloted in 5 districts of Sindh and 05 districts of Punjab, where rest of the project interventions are going on from December 2019.

4.1.5 Capacity building on updated commodities

A pool of master trainers was trained at the provincial level to roll out trainings for healthcare providers from the districts. These trainings included the orientation on the use of updated commodities in management of childhood Pneumonia and Diarrhea, and revised recording and reporting tools of DHIS. In Sindh, total of 9,742 doctors, including medical officers from district level healthcare faculties, private family physicians and paediatricians were trained on use of updated commodities. Out of these, 8,102 were also oriented on the use of revised DHIS tools for recording and reporting of information. Total of 22,700 Lady Health Workers (LHWs) were covered in these trainings from all 29 districts, covering ~100% of LHWs strength of Sindh province. In Punjab, in addition to doctors, LHWs and LHSs, Lady Health Visitors were also trained on use of updated commodities and MIS. Based on the project data, details of individual trainings are provided in the following table.

Table 6: Trainings under P&D Project <i>Cadre-wise number of staff receiving training under P&D Project</i>			
Cadre & Type of Trainings	Number of staff trainings conducted		
	Punjab	Sindh	Total
Doctors			
(Paediatricians and Medical Officers)			
DHIS	440	8,102	8,542
Use of updated commodities	440	9,742	10,182
IMNCI (In-service)	-	77	77
Lady Health Workers (LHWs)			
MIS	7,088	22,700	29,788
Use of updated commodities	7,088	22,700	29,788
Lady Health Visitors (LHVs)			
MIS	700	-	700
Use of updated commodities	700	-	700
Lady Health Supervisors (LHSs)			
MIS	280	725	1,005
Use of updated commodities	-	725	725

4.1.6 Knowledge Management

The good practices and lessons learnt during the project from its inception to its implementation in the selected provinces, the challenges faced and their mitigation strategies are altogether intangible assets that needs to be translated to broader settings for replication and up scaling within and outside Pakistan. UNICEF, starting from the inception phase, ensured the participation of government stakeholders for enhanced communication thereby helping in creating, storing, transferring and application of the knowledge to these provinces and regions. In this regard, the project engaged policy makers and planners from remaining two provinces (Balochistan and Khyber Pakhtunkhwa) and three regions (Gilgit Baltistan, Azad Jammu & Kashmir and the Federally Administered Tribal Areas – now merged districts of Khyber Pakhtunkhwa) and made them an essential part of National RMNCAH&N TWG. Building on the decisions taken during these meetings, the remaining provinces and regions has taken certain steps towards updating their existing systems, like inclusion of dispersible tablets in ‘very essential list of medicines’ of Khyber Pakhtunkhwa. During the pilot training on the abridged course of in-service IMNCI, participants from these provinces and regions were also trained to

further enhance capacities of the service providers in their respective areas. Moreover, lessons learned and recommendations given in this Midterm Evaluation (MTE) will be shared with all the provinces and areas for replication and scale up, through National dissemination of MTE report, followed by federal platforms of National RMNCAH TWG.

4.2 Findings against OECD/DAC Criteria

In line with the Organisation for Economic Cooperation and Development (OECD)/Development Assistance Committee (DAC) criteria, this evaluation assessed the findings against relevance, effectiveness, efficiency, sustainability and cross-cutting areas of equity and gender equality. A set of evaluation questions (EQs) were proposed in the TORs as per OECD/DAC criteria, against which, the findings of the evaluation report have been presented in the following section.

4.2.1 Relevance

Relevance has been assessed to determine the extent to which the project suited to the priorities and policies of the target group, recipient and donor.

EQ: How relevant and meaningful are the project objectives and activities in addressing the needs and priorities of the marginalized and vulnerable children in the project areas?

Finding: There is high relevance of the project as Pneumonia and Diarrhea contribute extensively to childhood deaths and morbidity in Pakistan.

Pakistan has one of the highest infant mortality rates, with 62 deaths per 1,000 live births, though declined over a period of time from 86 to 62 deaths per 1,000 live births (PDHS 2013; PDHS 2018). Keeping these above-mentioned statistics in view, Pakistan lags behind in Sustainable Development Goals (SDGs), adopted by the United Nations in 2015, to end preventable deaths of newborn and under-5 children by 2030. Similar is the case with GAPPD targets and timelines pledged by Pakistan. Among all the ailments, Diarrhea and Pneumonia are the prime causes of child mortality in Pakistan. Contribution of these two diseases to under-five (U5) child deaths has remained stubbornly high. In total, Diarrhea, Pneumonia, and malaria collectively contribute to around half of all child deaths each year in Pakistan (Bhutta et al., 2013).

The project was implemented to accelerate these efforts, in close coordination with the Government and relevant stakeholders to ensure sustainable improvements. The project had an overarching goal to ensure that relevant national policies are in place, understood, and adhered for prevention, promotion and treatment of childhood Pneumonia and Diarrhea. The project also focused on incorporating Pneumonia and Diarrhea treatment commodities into essential medicines lists and essential health service delivery packages. Based on key recommendations of landscape analysis and feasibility studies, the project has advocated for increased resource allocation for updated commodities and has catalysed the initial stages of the commodities procurement process with government authorities and pharmaceutical manufacturers.

Finding: There is empirical evidence supporting the use of updated commodities for management of childhood Pneumonia and Diarrhea at health facilities and in the communities.

Pneumonia and Diarrhea are fatal childhood illnesses requiring an integrated management approach. WHO and UNICEF initiated and conceptualized a cohesive approach for tackling Pneumonia and Diarrhea, resulting in launch of Global Action Plan for Pneumonia and Diarrhea in April 2013 (Qazi et al, 2015). GAPPD as a framework which emphasized on integrated approach to protect, prevent and treat both the diseases (WHO & UNICEF, 2013),

recommended the use of essential commodities including Dispersible Tablets of Amoxicillin & Zinc DT, Low Osmolarity ORS, ARI Timers, Pulse Oximeters and Oxygen.

To improve Pneumonia and Diarrhea management and services in Pakistan, UNICEF in alignment with GAPPD introduced these updated commodities through the project at appropriate levels of service delivery. UNICEF to advocate the use of and their inclusion in essential medicine list of the Government established the benefit and impact of these commodities on child survival. There is strong evidence that supports the use of these commodities for an improved management of both the childhood diseases, particularly the engagement of community level health workers for treatment of Diarrhea and Pneumonia, in terms of overcoming major barrier to access to health services (UNICEF, 2006).

Zinc plays a critical role in overall health and development of infants and young children, but among children of under developed and developing countries, its deficiencies are wide spread. During Diarrheal episodes zinc further depletes therefore, replacing this important micronutrient is essential for child recovery. WHO and UNICEF recommend a daily dosage of 20 mg of zinc supplements for 10 days for children suffering with Diarrhea and 10 mg for infants under six months (Bhutta et al, 2000). Similarly, WHO recommends a 250 mg amoxicillin DT, to be taken twice daily for 3-5 days for treating Pneumonia. Due to better results, UNICEF has been promoting DTs as they are more efficient than syrups. There are various characteristics of a dispersible tablet and suspension/syrup, which make them different in their usability. Their preparation, plasma concentration, dosage, palatability, packaging, storage, handling, logistics, cost etc. vary to a great degree. Suspension/ syrup is prepared in the ratio 20 mg/5 ml with prescribed amount of water to be placed in the suspension bottle, whereas dispersible tablet is to be dispersed in a small amount (5 to 10ml) of liquid; clean water or breast milk. Preparation of both commodities is also different as less water is required for DTs compared to suspension, since it is prepared per dose. Nonetheless, uniformity of content is not guaranteed in both forms. Use of clean and appropriate container to disperse the tablet is required. The liquid can be softly stirred to aid dispersion before swallowing. Dispersible tablets are easy to dispense, require minimal manipulation prior to use, which minimizes the risk of errors. Accuracy of dosage varies with Suspension/ Syrup. Dosing of liquids can be uncertain due to inaccuracy of spoon measurement/size. On the contrary dispersible tablet's dosage is more accurate as each dose requires the complete intake of one or half tablet. Therefore, dosing errors can be better avoided with dispersible tablets. Palatability of both commodities depended upon the formulation. Formulations can be composed according to the taste requirements. Taste masking has recently been adopted for dispersible tablets. Mostly suspension is packaged in 30 or 60 ml glass/ceramic bottle. On the other hand, dispersible tablets are usually packed in blisters (aluminium /PVC) or strips (aluminium), which are lighter in weight and packing is prone to breakage unlike suspension bottles. Manufacturer guarantees the stability of the dispersible tablet in this primary packaging, as they are less physical resistance than regular tablets, and more sensitive to moisture and humidity. Hence dispersible tablets have to be unpacked right before consumption because of its physical characteristics. Suspension form can be stored at room temperature but once reconstituted requires refrigeration at 2-8 °C for not more than 14 days while dispersible tablets are stored at room temperature, without any need for refrigeration, including lesser storage space. Transportation of dispersible tablets is more convenient due to its weight, easier and cheaper because of smaller volume compared to bottles of Suspension/ Syrup. Based on WHO price lists, dispersible tablets are more cost saving compared to suspensions i.e. dispersible tablet is \$ 0.22 and suspension is vs. \$ 0.8 (WHO 2010b).

WHO and UNICEF in 2004, jointly recommended zinc treatment for 10–14 days, in addition to low-osmolarity ORS, as an adjunct therapy that reduces the duration and severity of a Diarrhea episode and the likelihood of subsequent infections in the two to three months following treatment. The same is evident from PDHS 2017-18, where findings revealed that zinc was given to 13% of children under 5 years for treatment of diarrhea. However, some

provincial and urban/rural disparities were also observed regarding zinc for treatment of diarrhea, for instance 15.4% children were given zinc in Sindh (10% in urban and 22.1% in rural) and 13% children in Punjab (15.9% in urban and 11.7% in rural). UNICEF and WHO recommends low-osmolarity ORS for treatment of Diarrhea, as the earlier composition of ORS does not reduce stool output or duration of Diarrhea. The reduced osmolarity of ORS solution is to avoid possible adverse effects of hyper-tonicity on net fluid absorption. This was done by reducing the solution's glucose and salt (NaCl) concentrations (WHO, 2002).

Pulse Oximeter is a small device that stands as the standard for detection of hypoxemia in children at healthcare facilities, and in guiding whether children need oxygen support or not. However, the fragility and replacement of the probe is another determining factor. The delicate probes need to be replaced every year and sometimes, even before due to their easily damageable nature (Stop Pneumonia, 2016). On the contrary dispersible tablet's dosage is more accurate as each dose requires the complete intake of one or half tablet. Therefore, dosing errors can be better avoided with dispersible tablets. Hence UNICEF and WHO developed ARI Timer in 1990 to support health workers in determining how long they need to count a child's breath. These updated commodities and equipment have aided the healthcare providers and community health workers in timely and accurate diagnosis, and early referral of childhood Pneumonia.

Finding: The revised commodities are acceptable by both the care in public facilities and the community members for management of childhood Pneumonia & Diarrhea.

This finding is a discourse on the major overarching themes that emerged from the interviews and focus group discussions among relevant stakeholders at Provincial and District levels. It mainly highlights the availability and acceptance of new commodities along with prescribing behavior of healthcare providers at public sector health facilities for the management of childhood Pneumonia and Diarrhea.

Availability/functionality of commodities at public health facilities – Pulse Oximeter and Oxygen Therapy used in treatment of Pneumonia were present and used in majority of the health facilities. Oxygen cylinders and flow meters were found to be available at nearly all selected facilities of Punjab and Sindh. There were respondents from selected facilities in districts including Dadu, Ghotki, Karachi West, Khairpur and Thatta that reported neglect and disinterest on the part of health authorities when it came to availability of equipment. Facility in-charges in almost all facilities of Sindh and Punjab, seemed to be convinced about the benefits of using Pulse oximeters and Oxygen therapy to help treat Pneumonia. Facility in-charge from Sindh reported, ***“Oxygen plays an important role to save human lives particularly in severe condition of Pneumonia. Pulse Oximeter helps guide whether the child needs oxygen or not.”***

New forms of medication – Low osmolarity ORS, was available in most of the health facilities (80 percent) in implementing districts of Sindh and Punjab, with the in-charges being well aware of its benefits in comparison to the regular. Facilities In-charges in most instances had sound knowledge on how to administer L-ORS along with the added benefits of decrease in Diarrheal episodes. However, shortages of L-ORS at selected facilities in Muzaffargarh, Thatta and Karachi West were reported at the time of evaluation.

The facility in-charges in both Punjab and Sindh discussed DTs, in terms of their availability, acceptability and practicality as a medicine in reference to syrup. Dispersible Tablets have gained popularity in target districts especially in Sindh. Zinc Dispersible Tablets were present in majority of the facilities in Sindh and Punjab, and facility in-charges were aware of their benefits especially in terms of cost effectiveness and easy administration when compared to syrups. Facility in-charge from Sindh claimed about the benefits of DTs as compared to syrup, ***“DTs are more effective than syrup, exact dosage is administered through them which is essential for recovery.”***

Despite their benefits, majority of facility in-charges assumed that DTs were not available outside of public health sector facilities. Dispersible Tablets gathered varying views from the facility in-charges on its acceptability in the communities. Facility in-charges in both Sindh and Punjab stated that with proper counselling, a large number of caregivers have started using DTs, especially Zinc DT. Facility in-charges also assumed that DTs, specifically Amoxicillin DT has yet to win over complete community support over syrups. Without involving the Lady Health Workers and private health care providers its acceptance in communities is difficult. A facility in-charge from Sindh shared, ***“Community’s perception about DTs can be improved by counselling. Lady Health Workers can play a critical role...”***

Perceptions of Lady Health Workers on updated commodities – Lady Health Workers in the target districts of Sindh and Punjab had a certain level of awareness about the benefits of updated commodities especially of LoORS and ARI Timers but there wasn’t much awareness and acceptability about the benefits of DTs especially Amoxicillin. These LHWs seemed to have some shortage of supplies especially regarding DTs, however majority were being provided with LoORS and Zinc syrups. LHWs were aware of the benefits of ARI Timers in Acute Respiratory Tract Infections and have been provided with the same under the project. However, if an ARI Timer goes non-functional, LHWs showed their concern that it took up to six months before a new one was provided to them.

Perceptions of caregivers on updated commodities – Majority of parents/caregivers seemed to be aware of the benefits of LoORS, Amoxicillin and Zinc syrups, however they were mixed views about the presence of DTs for treating these illnesses. A few of them, who had access to dispersible tablets, were not clear on how to administer DTs accurately and therefore perceived them as a substance hard to swallow. Some of the caregivers from Karachi (Malir) viewed DTs as, ***“...before we were provided with syrup which was easy to use and not difficult to swallow... Tablets now given are hard to swallow.”***

Perceptions of paediatricians at federal and provincial levels – Perspective of the senior paediatricians at tertiary level health care facilities mentioned Diarrhea and Pneumonia as the most common diseases in their catchment areas. There was a general environment of awareness and acceptability of the new commodities introduced for treating these illnesses. Dispersible Tablets and Low Osmolarity ORS were not available and therefore not being prescribed for treating Diarrhea and Pneumonia at provincial level facilities.

Perceptions of GPs and Medical Store Keeper on updated commodities – There were mixed responses about the awareness of Dispersible Tablets and LoORS among General Practitioners and Pharmacy In-charges in treating Diarrhea and Pneumonia. Some GPs were aware of these modes of treatment. This awareness was relatively lower in Punjab. Among GPs, who were aware of low osmolarity ORS, some considered it better than the regular one, while others considered the opposite. Generally, GPs were unsure about its benefits. Similarly, most GPs were unaware of Dispersible Tablets (DTs) and amongst the few who were aware; they were concerned about their community acceptance. A GP from Punjab revealed, ***“oral Zinc has very important role in treatment...however I have never heard about Zinc DTs before.”*** Another GP from Sindh said, ***“...have heard of Zinc DTs for the first time. I think private sector should be involved in order to raise awareness.”*** Among the GPs who were aware about Zinc DT, some were doubtful of its practicality. A GP from Sindh justified syrup’s preference in these words, ***“Dispersible tablets dissolve in water, because the water is contaminated so syrup is better than dispersible tablets.”*** Majority of the Pharmacy in-charges interviewed were aware of the benefits of Dispersible Tablets in the treatment of Pneumonia and Diarrhea. However, the most commonly prescribed antibiotics for the treatment of respiratory tract infection and Diarrhea were Amoxicillin and Zinc in syrup forms. According to these Pharmacy in-charges, the current market for DTs is not very developed and there is negligible demand of DTs from the private sector and caregivers. Pharmacy In-charges from Tharparkar and Jacobabad commented, ***“There is no market for***

DTs. Private healthcare providers does not prescribe DTs; hence the DTs are not available in the market.”

The main suggestion to promote DTs was to create and increase demand through prescription from medical practitioners. Creating awareness about DTs in order to promote them was quite important according to pharmacy in-charges. They believed that medical representatives and doctors had a role to play in this. A pharmacy in-charge from Punjab emphasized this need; ***“... people should also have some awareness about usage and usefulness of such medicine...medical reps should suggest pharmaceutical companies to make such medicine in DTs given their usefulness in treatment of Pneumonia and Diarrhea.”*** Such awareness would also be beneficial for over the counter sales. A pharmacy in-charge from Punjab assumed that over counter sale of DTs would rise, if their prescriptions increase.

Finding: In remote areas of Pakistan, Lady Health Workers play an important role in management of Pneumonia and Diarrhea and are trusted, valued and carry influence on healthcare seeking behaviours.

Most of Facility In-charges in implementing districts of Punjab and Sindh agreed on the role and responsibilities of Lady Health Workers. They seemed satisfied with their performance, acknowledged their role in raising awareness and treating diseases as well as their contribution to the society. LHWs provide indispensable services to the country's poor and marginalized, particularly in the rural and remotes parts of the country. Their role was defined to be crucial in early diagnosis, initial management and referral of cases for Pneumonia and Diarrhea. Due to their door-to-door services, LHWs role was valued and believed to be critical in reducing the rate of mortality and morbidity caused due to childhood illnesses.

Provincial and district coordinators in Punjab and Sindh involved in monitoring LHWs performance seemed satisfied and acknowledged their role in raising awareness about illnesses and improved hygiene along with treating diseases. They were viewed as first point of contact between the community and healthcare system and agents of rural change. A district coordinator from Dadu highlighted their role as ***“LHWs are important workers in the community. They can reach and approach every corner of catchment area which are inaccessible for others.”***

Lady Health Workers enjoyed a good rapport with their respective caregivers. Caregivers were of the view that these Lady Health Workers bring about community participation through awareness creation, change of attitudes, and mobilization of support. They were seen to be geographically closer and more readily available than health care facilities. Caregivers from Karachi (Malir) district of Sindh were of the view, ***“.... Our LHWs provide care without cultural and linguistic barriers. Lady health workers are very polite. They have helped increase awareness for improved sanitation and hygiene.”***

LHWs contribution in preventing and treating ARI and Diarrheal illnesses was especially important in terms of awareness creation and timely management. Mothers from Jacobabad district of Sindh highlighted their role, and stated that ***“LHWs visit our house and provide awareness about benefits of ORS and Zinc syrup in case of Diarrheal illnesses.”***

EQ: To what extent the objectives of the project are consistent with the existing national/provincial policies and guidelines in line with global recommendations (WHO/GAPPD) for management of Diarrhea and Pneumonia among under-five children in Pakistan and are sustainable?

Finding: Project objectives are highly consistent with Pakistan's national vision and priorities for child health.

This finding highlights the policies and guidelines relevant to child health and how consistent are the project objectives with them according to the policy makers. There was an

agreement among the respondents at policy level that project objectives are highly consistent with Pakistan's National Health Vision (2016-2025) and priorities for child health. Federal and provincial policy makers and planners categorically mentioned that national/provincial policies/strategies and guidelines incorporated the global recommendations for Pneumonia and Diarrhea. Most of them believe that community case management for Pneumonia and Diarrhea is updated according to the global recommendations of WHO and GAPPD. Global Action Plan for Pneumonia and Diarrhea (GAPPD) was supported by WHO and UNICEF to help decrease in the rising mortality rates of children under 5 suffering from Diarrhea and Pneumonia.

Policy makers at MoNHSR&C and provincial Health Departments shared many details about how and where policy changes have been incorporated to improve the treatment services of these illnesses. This primary focus is to attain Sustainable Development Goals and fulfil its other global health responsibilities. SDG 3 ensures healthy lives and promotes wellbeing for all at all ages. By 2030, end preventable deaths of new-borns and children under 5 years of age, with all countries aiming to reduce neonatal mortality to at least as low as 12 per 1000 live births and under-5 mortality to at least as low as 25 per 1000 live births. It was also revealed that Pneumonia and Diarrhea has been included into the 12th 5-year plan from 2019-2023. A respondent shared that in accordance with Pakistan's National Health Vision, a 10-priority interventions document for RMNCAH was developed and translated into strategic action plans, keeping Pneumonia and Diarrhea in consideration. The ideology of National Health Vision 2016-2025 was to improve the health of all individuals, predominantly women and children through universal access to affordable quality essential health services, delivered through resilient and responsive health system. Officials from both provinces said that this project aligns with their provincial health sector strategies. The Government of Punjab, taking forward the intent of the newly elected political leadership to reform and restructure Punjab Health Sector towards better performance, has pursued the development of ten year Punjab Health Sector Strategy 2019-28. The defined priorities of the new leadership in the health sector include patient safety and quality of care, infection control, hospital waste management, environmental/one health, Health financing & public private partnership in the backdrop of three paradigms - biomedical, socio-environmental and lifestyle and behavioural contexts. Thematic strategic area of Punjab Health Sector Strategy 2019-28, no. 1 and no. 5 are relevant to the mid-term evaluation of this project. 1. Maternal and Child Health (MNCH) (to ensure timely free access to a quality MNCH services irrespective of ability to-pay, to all the people in Punjab; to institutionalise quality of care in MNCH services delivery system). 2. Health Management Information System (to enhance scope and contents of health data systems for policy and planning). The strategic framework for Sindh Health Sector Strategy 2012-2020 shall serve as an overarching umbrella to guide the operational plans of medium and long-term programmes and projects. It also provides estimates of resource envelopes, for the total budgetary outlays as well as costs of specific strategies requiring assistance. The key purpose of developing the strategic framework is to identify as to what is required for health systems strengthening in Sindh and to define a set of sub strategies for the major building blocks of the HSS including: service delivery; human resources; health management information; medical products, vaccines and technologies; financing; and leadership /governance/stewardship.

A provincial respondent elaborated that,

“In 2015 Global Financing Facility (GFF) established a Trust to fund 30 high priority districts and to support their unmet needs to achieve the SDG agenda. Further Neo-natal Units have also been established in the high priority Districts of Pakistan.”

It was also mentioned that national and provincial EMLs have been updated with the new Pneumonia and Diarrhea commodities in accordance with GAPPD. However, there is no pool procurement at national level, therefore procurement of drugs is managed at Provincial levels. There have been ongoing efforts for streamlining the LMIS in PHC facilities including the Pneumonia and Diarrhea commodities.

Support of development partners including WHO, UNFPA and UNICEF, to ministry of health and provincial department of health was appreciated by the policy makers. Technical as well as financial support of these international donors was acknowledged. A respondent stated,

“UN looks at health initiatives at global level, regional level and then facilitate to translate that at country level. Pneumonia & Diarrhea is included in this process.”

EQ: Are the strategies or approaches appropriate and adequate to achieve results?

Finding: Introduction of new commodities and management guidelines has contributed in improved management of P&D cases

The officials acknowledged the development of a separate module in IMNCI guidelines dedicated to Pneumonia and Diarrhea along with the introduction of new commodities for the management of these childhood illnesses. It was generally believed that these interventions contributed in improving Pneumonia and Diarrhea case management to a great extent, however some challenges were also spelled out.

The availability and acceptability of the new commodities is not widespread which impacts the management of Pneumonia and Diarrhea cases. However, the awareness of DTs was prevalent which shows improvement since the baseline was conducted. DTs are available at facility levels in both provinces and are well received by the community they cater to. However, the most GPs said that they don't prescribe them and majority pharmacy in-charges also said that they don't keep DTs due to low demand. There low demand and supply is like a vicious circle. A paediatrician was of the view that,

“There is a dire need to convince health care professionals to use DT and to promote their acceptability, they should be prescribed at tertiary level which in turn will set a precedent for the other levels.”

Acceptability of Low Osmolarity ORS was relatively more than the DTs. In both provinces and districts, majority of the respondents were aware that it is more effective in treating Diarrhea than regular ORS. Community was aware of its benefit to some extent. There was no denying the usefulness of ARI timers and pulse oximeters in timely diagnosing ARI/Pneumonia and Diarrhea and referral where required. Health care providers were aware that pulse oximeters could not be used without oxygen. Facility in-charges in Punjab complained of lack of requisite equipment and the need for proper maintenance. Majority of the LHWs claimed to have ARI timers, only a few from Sindh reported that it was out of order and hadn't been replaced. Front line workers in resource limited setting acknowledge the role of ARI Timers, ***“...After the introduction of ARI Timers, my ability to diagnose and address pneumonia has improved.”***

Regarding the maintenance and safety of pulse oximeter, a paediatrician from Punjab expounded, ***“Pulse oximeter usage and maintenance is a little problematic as it is not sustainable. It's reading are misleading if not clipped properly wrong readings. Fixed pulse oximeters break. While portable ones get misplaced and lost.”***

Majority of government officials responded in affirmative when asked whether treatment protocols including EML, IMNCI, MSDP and Procurement lists for Pneumonia and Diarrhea have been updated. UNICEF's and WHO's support in this context was highlighted time and again. A respondent expressed his gratitude while saying,

“Donors are supporting us with Diarrhea Kits, ARI Kits... all commodities are also made available by WHO.”

Various respondents, also highlighted the development of DHIS2 and how it would facilitate the management of Pneumonia and diarrhea. A respondent stated with regard to DHIS2,

“This project will impact 27 million beneficiaries (Women & Children) from being affected with Pneumonia and Diarrhea morbidity & mortality.”

4.2.2 Effectiveness

Effectiveness has been assessed to determine the extent to which the project was able to attain its objectives.

EQ: To what extent the project has achieved its objectives/ outcomes and what were the major factors influencing the achievement or non-achievement of the objectives/

Finding: The project was able to update national and provincial policies and guidelines in line with GAPPD recommendations.

The project is implemented with a GAPPD approach. Project contributes to increase child survival in Pakistan, particularly by strengthening policy solutions for better management of Pneumonia and Diarrhea. Over a period of two years the project has been able to update national and provincial policies and guidelines in line with GAPPD recommendations.

Revision of IMNCI Guidelines – UNICEF, has successfully provided technical assistance to draft and endorse the revised community IMNCI training manuals on child survival from the Technical Working Groups and Government. Policymakers and professionals have concurred with the in-service IMNCI training manuals, which have been endorsed by the government. UNICEF, to help draft and endorse the pre-service and community IMNCI components by the WHO/Eastern Mediterranean Regional Office (EMRO) team, provided technical assistance. UNICEF successfully carried out a series of comprehensive consultative process with the government officials, key stakeholders and development partners to help draft the IMNCI modules, which are currently under review.

Health managers from both Sindh and Punjab informed about the success of IMNCI trainings being carried out in health facilities. A health manager from Sindh reported, “We have over 300 Basic Health Units in which staff is receiving trainings on IMNCI/WHO guidelines. A Child Survival Group member from Sindh, commented about the Revised IMNCI trainings, ***“...these trainings highlight the importance of usage of Amoxicillin DT, Zinc DT and L-ORS in management of Pneumonia and Diarrhea.”***

Procurement of quality assured commodities for Pneumonia and Diarrhea in selected provinces - According to District managers in Punjab and Sindh, procurement of medicines at the district level is being carried out through a central rate contract list, which is awarded at the provincial levels by the health departments. UNICEF continues the procurement of quality assured commodities offshore for Pneumonia and Diarrhea till end of the year, 2019. Advocacy with the government for adequate budgetary allocation to help ensure availability of these commodities at Public health facilities beyond the scope of project is being carried out.

Districts health Information System – Keeping in view, the role of Management Information system in combating mortality and morbidity due to Pneumonia and Diarrhea, UNICEF, provided technical assistance to Department of Health for updating of DHIS software and tools, and its planned linkages with LMIS. A series of consultative meetings of the CSW and TWG at national and provincial levels were conducted to build consensus on strengthening the logistics system with updated EMLs, MSDP and procurement lists. A digital logistic management information system (LMIS) software, which is linked with district health information system (DHIS), is being developed. KII’s were conducted with DHIS focal person of Sindh and Punjab. DHIS focal persons in Sindh and Punjab informed, ***“Health Information System has been upgraded in the province with technical support of partners, including UNICEF”***

UNICEF, helped in the printing and dissemination of the reporting tools in the updated DHIS with inclusion of P&D indicators. Furthermore, it was told that master trainers were trained which would train medics and paramedics on the use of these tools.

EQ: To what extent the implementation of the project approaches worked as intended, particularly after the baseline in 2016 and subsequent adjustments?

Finding: The project managed to achieve policy level planned milestones, however implementation of these policies at district and community level need to be strengthened.

In the annual progress report of Pneumonia and Diarrhea Project, UNICEF has outlined four outcomes that have to be achieved by the end of this project. First outcome being aimed is the Policy Change by updating existing national/provincial policies and guidelines in line with global recommendations WHO and GAPPD for management of Diarrhea and Pneumonia among under five children in Pakistan by the end of 2019. Second outcome is that of Policy Translation through transformation of revised and updated Pneumonia and Diarrhea treatment guidelines into relevant action plans by all provincial/areas health departments in Pakistan by the end of 2019. Then to achieve Policy Implementation, availability of essential commodities (Amoxicillin DT, Zinc DT, co-packaged ORS and Zinc, Oxygen, ARI Timers and Pulse Oximeters) should be ensured for treatment of childhood Pneumonia and Diarrhea in Pakistan, by the end of 2019. Last but not the least, Knowledge Management through translation of lessons learned from this investment to other settings/broader geographical scopes within Pakistan has to be conducted.

Time and again policy makers emphasized the incorporation of changes in national and provincial guidelines in accordance global recommendations. Updated EMLs, MSDPs, & procurement lists were also mentioned by most of the relevant stakeholders. It was revealed that provincial EMLs and procurement lists have been updated to include the new commodities and devices. In Punjab, MSDPs have also been updated, while in Sindh the MSDPs are still under revision. The procurement lists in Sindh have included the DTs whereas these lists in Punjab don't have them yet.

Despite all these policy changes, many improvements need to be made at district and community levels. Facility in-charges did complain of low supply and stock-outs of commodities, but at the same time there were participants who were satisfied with their stocks and also the DHIS reporting mechanism. A respondent mentioned,

“Medicine supply is always sent according to our demand but in case of stock out we share report with DHIS. All medicines are supplied in time to avoid any stock out.”

Regarding maintenance of adequate stock, a facility in-charge stated,

“Staff is not trained on forecasting and quantification of medicine and supplies. Proper training should be arranged for health facility staff.”

Nonetheless, the LHWs' capacity building is of immense significance to this project as they are the actual contact persons and are aware of the ground realities of the field. All stakeholders were cognizant of their importance and contribution to the healthcare system.

A provincial candidate mentioned,

“Availability and acceptability of DTs in the community can only be done by capacity building at the grass root level i.e. training LHWs.”

Mothers expressed their heartfelt gratitude to these community workers. A mother exclaimed,

“She is just like a sister, very helpful and polite and is very caring and considerate especially towards our children.”

However, mothers weren't much aware of DTs or low osmolarity ORS. They knew that LHWs provided them with ORS yet, had limited understanding whether they were provided with low or regular ORS. There were mixed views prevalent regarding availability of new commodities among the LHWs. Some groups were satisfied and believed that supply was adequate while others were not. Nonetheless all LHWs were unanimous that they were overburdened with other campaigns like polio. They admitted that regularization had made a positive impact on their performance. They also agreed that there is a dire need of trainings and refresher training, especially how to use ARI timers in treating Pneumonia.

EQ: How effectively various Federal Ministry and Provincial Health Departments and Programmes coordinated among each other?

Finding: Effective coordination mechanisms between federal and provincial levels have been established in the form of TWGs and CSGs.

Coordination as well as technical support from TWG and child survival groups were applauded by the respondents as these platforms provided opportunities to discuss challenges, share best practices and decision making for key actions. It was revealed that National RMNCAH TWG is a platform for policy dialogue and technical support for child survival activities and is a part RMNCAH&N at federal level. Policy makers agreed that meetings organized by CSG are of immense significance in uniting all relevant stakeholders on a single platform and discussing management of childhood illnesses. Federal and provincial level participants were all unanimous that the Child Survival Group should be institutionalized for a better standing in the health system. A respondent was of the view that CSG should be a similar institution as MNCH and LHW Programmes. Another participant also explained the distinction between the operation of CSGs in Punjab and Sindh,

“In Punjab CSG is only child focused, whereas in Sindh it is MNCH based.”

While discussing the achievements of CSG, a participant asserted,

“Development of curriculum for LHWs, managing issues of supplies, identification of training gaps are all examples of practical challenges solved by the CSG.”

Institutionalization of CSG is important in order to have a custodian who takes responsibility of the recommendations made by this group. During the data collection it was also observed that none of the respondents at policy level, even CSG members, were willing to take ownership of the group's actions.

As already mentioned in the baseline report, CSG recommendations need to be made mandatory and binding in order to value and to acknowledge their authority.

EQ: Whether the mechanisms available to create awareness among communities are effectively linked to the project objectives?

Finding: The project had limited scope and role in creating awareness among communities for social and behavioural change. However, LHWs were engaged over project timeline for raising awareness on exclusive breast-feeding, hand washing and immunisation.

UNICEF, in partnership with BMGF, aims to improve child survival in Pakistan. UNICEF's project “Accelerating Policy Change, Translation and Implementation for Pneumonia and Diarrhoea Commodities in Pakistan” focus on reducing under five infant deaths related to these illnesses. In order to raise positive awareness about the use of updated commodities, and transmit messages that highlight the benefit, job aids comprising of flip charts, standees, table flyers, patient counselling cards and posters were developed, printed and distributed for service providers both at community and facility levels by UNICEF.

KII's and Discussions carried out at different tiers at District and Provincial levels in both Punjab and Sindh reveal that these job aids were able to improve the knowledge of Public health care providers and community workers around the benefits of using revised commodities. However, at community levels, there has been little demand for uptake for DTs thus far. To Caregivers it seemed that the word "tablet" prompted the notion that tablets were hard to swallow by children; hence syrup was a better choice. A facility in-charge from Sindh mentioned, ***"Demand of syrup in children is comparatively more. DTs are often perceived as substance hard to swallow and hence not drug of choice."***

Caregivers, had limited understanding and knowledge about the usefulness of new formulation of ORS in comparison to regular and were not aware whether they were provided with low or regular ORS in health care facilities and by LHWs. Sustained, regular health education efforts at community level are required to overcome knowledge and information barriers and increase community awareness round these updated commodities. More efforts and investments need to be put in place in order to expand service outreach, including through the training and deployment of additional lady health workers combined with the continuation and intensification of community mobilization activities to leverage greater results for child survival.

EQ: Whether the healthcare provider and community health workers have the required knowledge and skills to proper utilization of supplies as per standards and use of reporting tools?

Finding: There is evidence to support the need for structured and regular trainings on management of P&D to enhance proper utilization of updated commodities.

In Sindh, total of 9,742 doctors were trained on use of updated commodities and 8,102 were also oriented on the use of revised DHIS tools for recording and reporting of information. Total of 22,700 LHWs were covered in these trainings from all 29 districts, covering ~100% of LHWs strength of Sindh province. In Punjab, in addition to doctors, LHWs and LHSs, Lady Health Visitors were also trained on use of updated commodities and MIS. Most participants agreed and were aware that IMNCI guidelines have been updated but there were mixed opinion on the adequacy of the trainings conducted for it. A federal participant shared,

"There has been some orientation on EML, MSDP and IMNCI guidelines."

The in-service IMNCI six-day abridged course was perceived as a positive step by the policy makers.

"The IMNCI guidelines were updated last year, finalized and approved by MOH. The IMNCI course has now been reduced from 11 days to 6 days, which has made it more effective."

Most participants were not aware of the pre-service and community IMNCI components which have also been drafted and endorsed by the WHO and awaiting finalization.

Nearly half of Medical Officers in the selected facilities had successfully received trainings in ARI, while only twenty percent of them confirmed trainings on IMNCI and breastfeeding. Around fifty percent of Women Medical Officers in the selected facilities had received trainings on ARI and micronutrient deficiency while a small percentage (25 percent) reported to have received trainings on IMCI. Less than fifteen percent of health technicians in the selected facilities had been trained on IMCI at the time of evaluation. Majority of the stakeholders at the district and community levels complained of lack of trainings and refresher trainings. At the same time, they also showed willingness to participate and keenness to learn from these trainings. GPs didn't undermine their credibility as medical specialists but wanted their knowledge to be updated through trainings. Being from the private sector, most GPs felt being left out. Thus they wanted to be involved in the training provided by donors or/and the

government. LHWs as mentioned in the other sections couldn't emphasize enough the need for their trainings and refreshers. This was also endorsed by district officers as one explained,

"They are not specialists therefore, they required to improve their health services at all the time through trainings, sessions, meetings and refreshers."

Not just general trainings, but training in the context of Pneumonia and Diarrhea and for the usage of new commodities were demanded, especially for DTs and ARI timers. A provincial respondent highlighted this need saying,

"DTs are new form of medication. Until and unless, LHWs are sure about its usage and convinced about their benefits, how will they be able to convey their knowledge to the mothers?"

EQ: Whether the monitoring and reporting mechanisms exist and effectively implemented for effective tracking and improvement in system?

Finding: Reporting tools on facility (DHIS tools) and community level data (LHW MIS tools) were updated with inclusion of indicators on P&D along with new commodities in line with GAPPD recommendations and WHO guidelines.

Management Information System plays a vital role in combating mortality and morbidity due to Pneumonia and Diarrhea. KIIs with DHIS focal persons of Sindh and Punjab were conducted to get an insight on the current status of DHIS. It was informed that in both provinces reporting tools on facility and community have been updated to reflect the new commodities in alignment with GAPPD recommendations and WHO guidelines. Provincial managers from both Sindh and Punjab, informed that the updated DHIS, includes indicators about availability of Zinc DT, Amoxicillin DT and L-ORS in the stock available at facility and community. Furthermore, Provincial manager from Sindh commented, "*Stock outs of DTs and L-ORS are reflected in the reporting tools in the DHIS.*" It was informed that by the Provincial Managers in both Punjab and Sindh that UNICEF, has provided technical support to develop Pneumonia and Diarrhea specific modules and trainings. According to them, trickle down trainings were conducted for reporting tools on community level data "*Provincial Master trainers conduct/arrange the trainings at district level and trained them for further trainings. And after this a group of district trainers trained the Hospital Staff at Facility or Taluka level.*"

EQ: How effective are the 'innovative approaches like use of ARI timers and pulse oximeter, what results can be achieved, are replicable within the Sindh/Punjab context?

Finding: ARI Timers and Pulse Oximeter are effective in timely diagnosis and severity of Pneumonia in children.

According to World Health Organization statistics, main contributors to high infant mortality rates are acute respiratory tract infections and Pneumonia. In most of the developing countries, clinical indicators, such as compromised mental state, poor intake, cyanosis and respiratory rate >60, nasal flaring, and chest indrawing are used to identify the severity of Pneumonia (Stop Pneumonia, 2016). In children, hypoxemia and rapid respiratory rate are ominous signs associated with respiratory tract infections. Hypoxia can be detected easily with a device, Pulse Oximeter. MTE has proved the benefits of using pulse oximeter, a non-invasive, readily available and cost effective way to identify hypoxemia and categorize high-risk children with respiratory tract infections. Similarly, rapid respiratory rate can be calculated using an Acute Respiratory Infection (ARI) Timer. Hence UNICEF and WHO developed ARI Timer in 1990 to support health workers in determining how long they need to count a child's breaths.

There was no denying on the usefulness of ARI timers and pulse oximeter in timely and accurate management of Pneumonia. Facility in-charges from Punjab commented on the

benefits of ARI Timers and Pulse Oximeter, ***“ If used accurately, Pulse Oximeters are cost-effective and efficient devices that can help detect cases of Pneumonia”.***

Majority of the health care facilities in both Punjab and Sindh were equipped with oxygen and pulse oximeters.

Majority of the LHWs in the intervention districts of Sindh and Punjab were equipped with ARI Timers. ARI Timers were functional and available at health houses in Jacobabad, Tharparkar and Bahawalnagar. Some of the LHWs reported non-functional ARI Timers and limitation of knowledge on their usage. LHWs from Punjab exclaimed, ***“ We have little knowledge on how to use this device accurately so it becomes quite challenging while checking for signs and symptoms of Pneumonia in a sick child.”*** Trainings relating to accurate usage of ARI Timers and its benefits can help to improve knowledge, skills and performance of these Lady Health Workers. This in turn, will result in early and accurate management of Acute Respiratory Tract Infections among infants under five years of age.

Finding: Provision of ARI timers has empowered LHWs and improved their ability to manage Pneumonia.

Female empowerment and gender equality has also been promoted through this project. LHS and LHWs role to provide curative Pneumonia and Diarrhea services in the community elevates their status as a productive member in the community. Since the project involves provision of commodities including gadgets like ARI timers to LHWs, the project is equipping the females in the community and empowering them to perform better. However, there was a consensus among these community workers on the lack of trainings and refresher trainings. One of them claimed,

“Since we aren’t specialists, we need to be trained regularly in order to perform our duties.”

ARI timer is a new piece of equipment for these community workers and they demanded being taught how to effectively use it. One LHW, insisted,

“We all don’t know how to use ARI timer, so it is quite challenging to check or deal with a Pneumonia patient.”

Maintenance and replacement of out of order ARI timers was another problematic area. An LHW from Karachi Malir explained,

“My ARI timer is out of order and not working properly, I have asked for a new one but still no response.”

LHWs agreed on the usefulness of the ARI timers and at the same time highlighted the need for trainings on their use and their maintenance.

EQ: To what extent the programme has been able to contribute to ownership and leadership of the provincial/ area DOHs?

Finding: The project has contributed in building ownership of provincial governments of its interventions.

The project has certainly made the DOH at provincial levels more conscious and involved in the management of these childhood illnesses. However, there is still room for further strengthening this ownership by the government. Members of CSG and TWG were extremely supportive and enthusiastic about improving Pneumonia and Diarrhea services, yet none seemed to take responsibility of the actions of these groups. The technical support provided by these groups was acknowledged and the need for “institutionalizing” these groups was mentioned by various stakeholders quite frequently. A policy maker while appreciating the role of UNICEF in this project expressed his concern for the project’s sustainability after UNICEF withdraws,

“There should be a detailed exit strategy planned out before UNICEF withdraws sustaining this massive project could be a big challenge for the government.”

Nonetheless, all policy level participants highlighted the government’s support and involvement in this project from decision-making and advocacy to making significant changes at the policy level.

Finding: Ownership of the government is evident through inclusion of recommended commodities in MSDP, EML and procurement lists.

The government revised and updated the child survival related policies and strategies in accordance with GAPDD and WHO recommendations. Majority of the policy level respondents were cognizant of these changes including the updating of IMNCI guidelines, which they agreed was a judicious step. Majority agreed that IMNCI trainings were taking place and some were of the opinion that the monitoring mechanism of these trainings should be strengthened for quality assurance purposes. The participants who chose to share specific insights about the updating of policies shared that in Punjab, DTs and Lo ORS have been included in provincial EML, procurement lists, and MSDP whereas in Sindh EML and MSDP have been updated, while DTs still need to be added in the procurement lists. This reflects the commitment of the federal and provincial governments in managing these childhood illnesses.

Finding: Childhood P&D has been brought into limelight through the project interventions at policy and planning level.

The project interventions have definitely put the much-needed focus on these childhood illnesses. Apart from changes in EML, MSDP and procurement lists, the structure of IMNCI trainings have been revised. Not only the training days have been reduced from eleven to six, but a separate module for Pneumonia and Diarrhea has been created in the IMNCI guidelines. However, quite a few respondents insisted that Pneumonia and Diarrhea should be incorporated in the medical curriculum in order to give them the attention they deserve. Majority of the respondents at policy level were aware of these changes and many also appreciated them as progressive efforts. Sustained advocacy and deliberations with the DOH at both federal and provincial levels has also led to the government concurrence on shifting to DHIS and implementation of LMIS in the province.

EQ: What have been the major factors influencing the achievement or non-achievement of outcomes?

Finding: There were multiple factors that influenced the project outcomes including certain hindering factors.

There were numerous obstacles that affected the outcomes of the project. There were some barriers specific to the project while some were more general ones that are embedded in the overall system.

IMNCI training component was not part of the initial design of the project however, later on it was added to strengthen the implementation process. Private sector engagement was and still remains to be a major challenge as the sector is largely unregulated while still contributing to more than 70% services. Many policy level respondents stated that it is imperative to involve the private sector, as the project cannot succeed while excluding it. Care givers/mothers’ lack of familiarity with the DTs and Lo ORS is also one of the limiting factors.

At district and community levels, there are many causes for not achieving the desired outcomes of the project. Social issues like lack of resources, rapid inflation, non-availability of clean drinking water, transport and treatment expenditures etc. were frequently mentioned in this context by the community members. Respondents, especially LHWs, believed that widespread lack of education and illiteracy in the community, especially among mothers, which interferes with their ability to understand the root cause of these diseases, their preventive as

well as their treatment measures. There is a dire need for basic consciousness for incorporating hygiene in the daily routines. To promote the concept of hygiene and cleanliness, a participant suggested, *“Every Friday sermon should necessarily include emphasis on cleanliness; Safayi nisf imaan hai.”*

Another suggestion was that public awareness on hygiene should be created by media campaigns, referring to pictorials distributed by UNICEF for this project. Shortage and high turnover of medical staff along with high burden of patients were mentioned quite recurrently at facility level. A respondent at RHC in Punjab described his workload of patients,

“We get 500 patients in OPD in a single day, hence we can’t give more than 40-50 seconds to each patient.”

LHWs complained that they were over burdened with other campaigns like dengue and polio whereas their original job description is only MNCH related work. Other stakeholders also agreed that such engagements hindered LHWs from focusing on their primary duties.

EQ: To what extent the programme learned and evolved over the 2 years?

Finding: Project has made certain adjustments mid-way to increase effective implementation.

A significant assignment undertaken for course correction is this midterm evaluation of the project. Many changes were made along the way. Supplies are now directly delivered to LHWs without involving the district managers. LHWs were satisfied with this mechanism as it ensured uninterrupted and timely supply. The DHIS2; linking of LMIS and DHIS would further facilitate in strengthening this project. Co-packaging of Zinc DT and Lo ORS, which was initially introduced, has now been discontinued due to its impracticality. However, many facility in-charges in Punjab claimed to still have and distribute these co-packaged commodities.

4.2.3 Efficiency

EQ: How well the resources, both human and financial, been managed to ensure timely, attainment of results?

Finding: The project has been able to achieve its planned milestones, activities and outcomes within stipulated timeline ensuring efficient use of resources.

In the annual progress report of Pneumonia and Diarrhea Project, UNICEF has outlined four outcomes that have to be achieved by the end of this project. First outcome, being aimed is the “Policy Change”. It has been achieved by updating existing national/provincial policies and guidelines in line with global recommendations WHO and GAPPD for management of Diarrhea and Pneumonia among under five children in Pakistan. There was an agreement among the respondents at policy level that Project objectives are highly consistent with Pakistan’s national vision and priorities for child health. UNICEF, role has been acknowledged for supporting consultations with national and provincial governments for updating and revising guidelines in line with GAPPD. The officials agreed that a strong coordination mechanism exists within the RMNCH including stakeholders from Federal & Provincial Governments, UN Agencies (WHO, UNICEF, UNFPA), PPA & the Private sectors. A respondent explained that, “They convene meetings at National level on quarterly basis.” Technical support from Groups like Technical working groups and Child Survival groups were mentioned. Child Survival groups are empowered, committed and capable of implementing the interventions.

UNICEF has been able to translate policies through transformation of revised and updated Pneumonia and Diarrhea treatment guidelines into relevant action plans by all provincial/areas health departments in Pakistan. Respondents at federal and provincial policy levels shared their insights on the service delivery of the public sector health facilities. There

was a consensus of opinion among the policy makers that national/provincial/ policies/strategies and guidelines incorporate the global recommendations for Pneumonia and Diarrhea. UNICEF, role was applauded by majority of policymakers to help draft, endorse, print the revised IMNCI manuals. It was revealed by policy makers in both Punjab and Sindh that provincial EMLs and procurement lists have been updated to include the new commodities and devices for management of Pneumonia and Diarrhea. In Punjab, MSDPs have been updated to include the new commodities while in Sindh; the MSDPs are under revision as per GAPDD recommendations.

Health Information Systems have been updated and strengthened to reflect the recommended commodities. A common opinion exists among the policy makers that UNICEF, provides support for the updating of DHIS software and tools and in the establishment of DHIS-II and its linkages with LMIS. Last but not the least, Knowledge Management through translation of lessons learned from this investment to other settings/broader geographical scopes within Pakistan have been carried out starting from the implementation phase. Policy and decision makers from two other provinces (Baluchistan and KPK) are a part of Federal RMNCAH & N technical working group and are informed on key decisions and lessons learned under this project during these meetings.

EQ: What are the implementation challenges from the perspective of both right holders and duty bearers especially for the under- five marginalized children of communities?

Finding: There are numerous factors that hindered the availability of recommended commodities for management of childhood Pneumonia and Diarrhea in open market.

The market for recommended commodities for management of childhood Pneumonia and Diarrhea is not very developed. The most salient market challenges observed for DTs are around awareness and availability. General Practitioners and Pharmacy in-charges highlighted that there was limited awareness and availability of DTs in the open market. Majority of the GPs, were not aware about the benefits of DTs and L-ORS and amongst those who were aware of L-ORS and DTs there is still a preference for regular ORS and syrups.

A common perception among the pharmacy in- charges was lack of demand for DTs by the caregivers and private health care providers. The prescribing practices of the private health care providers were not in align with the Global Health recommendations. There is a lack of understanding on the benefits and usage of DTs by the consumers leading to a general preference for Syrups. Amoxicillin and Zinc have been introduced in syrups form since long. In the presence of these substitutes in the market, demand for dispersible formulations is not very high. Presence of a substitute affects the prescribing behavior and uptake of these commodities. In general, sustained demand for commodity is considered a potent factor influencing production decision and sustainability. Low demand decreases profitability and adversely affects production decision. Unless there is a sustained and significant demand for these commodities, even companies having the capability and capacity would be reluctant to take a production decision.

EQ: What is the value added in terms of improved delivery of services for Pneumonia and Diarrhea?

Finding: The project has added value by improving management practices of healthcare providers, mainly outreach workers and primary level facilities.

The project has added value by improving management practices of healthcare providers, mainly outreach workers and primary level facilities. There was an agreement among the respondents at primary level facilities that community case management for Pneumonia and Diarrhea is updated according to the global recommendations of WHO and GAPDD. The

officials acknowledged the development of a separate module in IMNCI guidelines dedicated to Pneumonia and Diarrhea along with the introduction of new commodities for the management of these childhood illnesses. It was generally believed that these interventions contributed in improving Pneumonia and Diarrhea case management to a great extent. During key informant interviews, facility in-charges stated,

“...lot of medicines are only available in adult strength; administration of accurate dosage for children suffering from pneumonia and diarrhea is critical.”

“DT’s are more effective than syrups, exact dosage is administered through them.”

During FGDs with LHWs, they explained, “DT’s are considered the most ideal since they are manufactured in solid state and a single dose is turned into liquid at point of administration.”

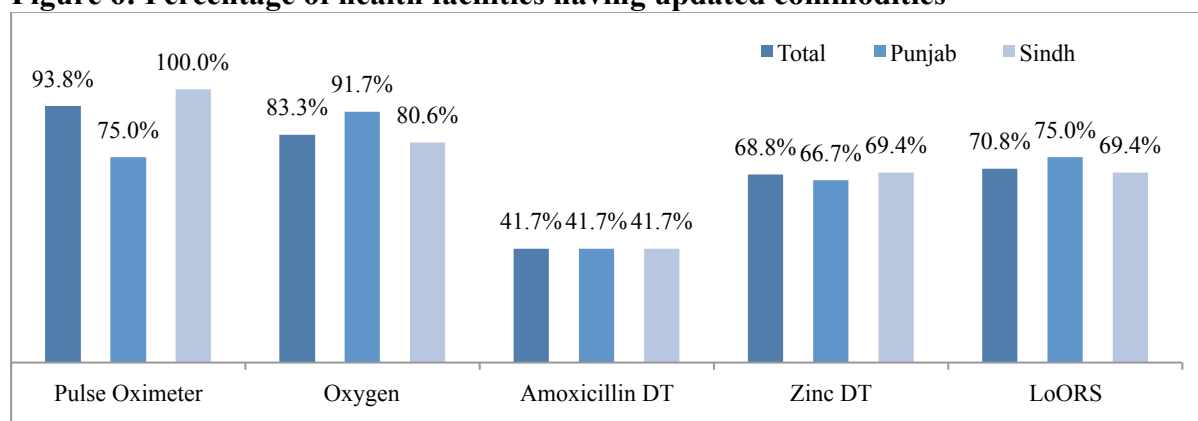
Pulse Oximeter, and Oxygen therapy introduced in primary health facilities by UNICEF, as discussed in detail in relevance section have helped to detect hypoxemia a fatal complication of Pneumonia early and accurately long with oxygen availability have helped decrease complications. Female empowerment and gender equality has also been promoted through this project. The project involves provision of commodities including gadgets like ARI timers to LHWs, the project is equipping the females in the community and empowering them to perform better.

EQ: Whether the availability of P&D supplies is adequate, timely and are prescribed?

Finding: Majority of the public health facilities had adequate supply of P&D recommended commodities but prescription behavior of care providers need improvement.

Management of Pneumonia and Diarrhea in children under 05 years of age was done through review of prescriptions of children visiting surveyed healthcare facilities. A total of 480 prescriptions were recorded with 10 from each of 48 surveyed facilities, and 473 of which turned out to be containing relevant information. Out of these prescriptions, 45.7% were Pneumonia patients and 54.3% were suffering from Diarrhea. Among these children, 52.6% were boys and 47.4% were girls, with variation more pronounced among Diarrhea patients as 54.5% male children were suffering from Diarrhea as compared to 50.5% male children suffering from Pneumonia. The ages of these children ranged from under one month to as high as 60 months. Status of the availability of updated commodities is given in the following graph. The availability of Amoxil DT was confirmed in 20 health facilities, Zinc DT in 33 while LoORS was available in 34 of all the 48 health facilities selected for evaluation.

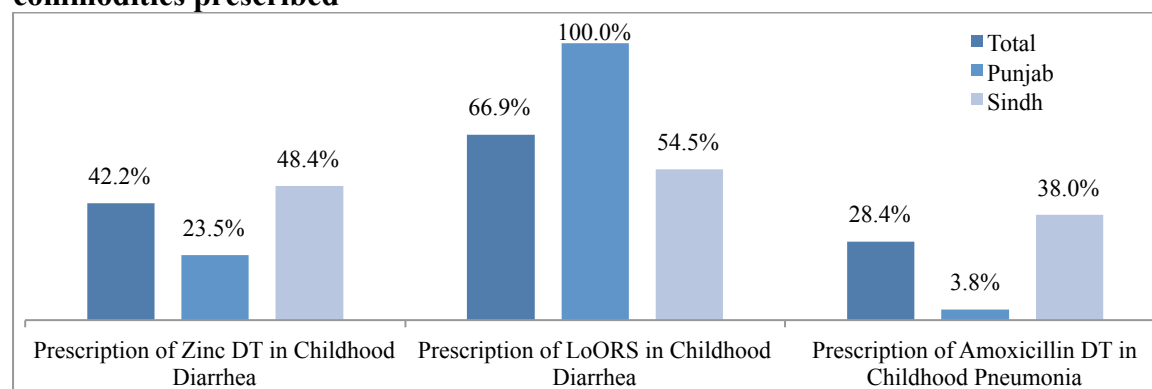
Figure 6: Percentage of health facilities having updated commodities



Findings of the assessment revealed that Zinc DT was prescribed to 42.2% of the children suffering from Diarrhea (23.5% in Punjab and 48.4% in Sindh) whereas LoORS was given to

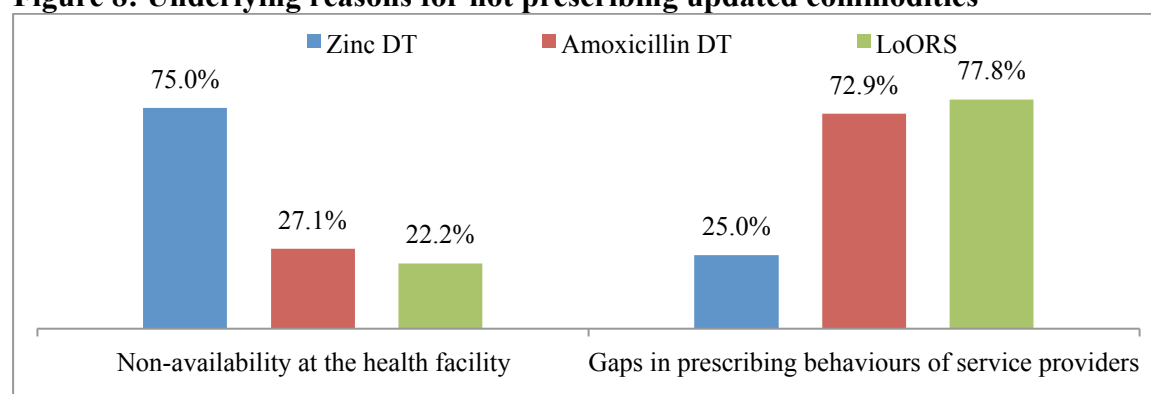
66.9% (100.0% in Punjab and 54.5% in Sindh). Use of Amoxicillin DT was reported in 28.4% of the Pneumonia prescriptions and in Punjab, only 3.8% of the prescriptions showed Amoxicillin DT while the remaining used syrup formulation. Availability of the DTs was mainly dependent on UNICEF supplies, which was not catering to 100% of the caseload. Secondly, the provincial governments were procuring syrups, therefore, the health facilities where DT was not available, syrups were being prescribed. Overall, Amoxicillin Syrup was available at 69.4% of health facilities of Sindh and 75.0% of health facilities of Punjab whereas Zinc Syrup was available at 83.3% facilities of Punjab and 80.6% health facilities of Sindh.

Figure 7: Percentage of Pneumonia and Diarrhea prescriptions having updated commodities prescribed



The reason for not prescribing Zinc DT was either its unavailability at the health facility (27.1%) or the prescribing behaviour of the healthcare provider (72.9%). The reason for not prescribing Amoxicillin DT was mainly its unavailability at the health facility (75.0%) whereas one fourth of the healthcare providers preferred the prescription of syrup despite its availability at the health facilities.

Figure 8: Underlying reasons for not prescribing updated commodities



There was a stark contrast between the awareness and availability of DTs in public and private sector. Apart from issues of stock outs and low supply, facility in-charges and LHWs were aware of DTs and their benefits. On the contrary, some GPs were not even familiar with them, leave aside their availability. Few also said that there isn't any demand for DTs as they don't get any prescriptions with them. A GP in the focus group discussion mentioned,

"If this type of provision is handed over to private sector, then the results will be better. We have heard about the DT from you for the first time. UNICEF should involve the private healthcare providers for the use of DT."

This clearly shows that GPs felt left out of the system. They also were keen on attending trainings but most of them shared that they were unfortunately not provided any. A provincial respondent also supported this notion and asserted that, *“Private sector needs to be persuaded and incentivized for the production of DTs commodities.”*

EQ: To what extent has the project achieved its goals in enhancing the health outcomes especially of under five children in the catchment communities in Sindh and Punjab target districts?

Finding: Childhood P&D management practices have improved through introduction of updated commodities.

All stakeholders were unanimous that introduction of new commodities was a great initiative to improve Pneumonia and Diarrhea management. Majority believed that this project has put the desired national focus on these critical childhood illnesses. However, respondents at community level discussed many areas for improvement.

Mothers/care givers had an obvious preference for syrup over DT. There were also mothers who were not familiar with DTs. On the contrary, ORS sachets were popular with the mothers and vouched to their efficacy in treating Diarrhea. However, most couldn't differentiate between Lo-ORS and regular ORS and shared that they used whichever sachet was provided by the LHWs. Mothers held LHWs in high regard in relation to treating these childhood illnesses and believed that they played a critical role in addressing health needs of the community.

LHWs were quite satisfied with the introduction of new commodities and their effectiveness in treating these diseases. Majority was also pleased with their supplies, both with quantity and regularity. LHWs in Punjab were quite satisfied with the new delivery system of supplies, which were either delivered at their health houses or at the facilities with their names. This supply chain didn't find favour with a few DHOs who believed that the delivery should take place through them for better monitoring. LHWs wanted more training and refreshers, especially with regards to using ARI timers. Most of them apparently considered it useful tool in diagnosing Diarrhea but were not much familiar with its usage. The ones who were aware how to use them, complained of them being out of order and not being replaced.

GPs were mostly indifferent towards the new commodities as they were unaware of them; some did not even know what DTs were. Majority was unsure about the availability of Lo ORS. Due to this they felt being excluded from the health system and showed keenness in trainings and in learning about the new commodities. Similarly, pharmacy in-charges were of the opinion that they did not get prescription of DTs and Lo ORS so they didn't keep them in stock due to no demand. Very few facility in-charges mentioned stock-outs; majority was satisfied with the supplies as well as with the effectiveness of the new commodities. Most facility in-charges believed that with DTs, dosage is more accurate. LoORS was particularly popular with facilities in both provinces. Prescribing co-packaging of zinc DT and Lo ORS was still a common practice in Punjab. In spite of admitting the efficacy of DTs, quite a few facility in-charges revealed that they did not prescribe DTs as syrup was more popular with the patients. Prescribing DTs was just not a common practice. DTs were easier to store so they would let the stock be, and prescribe syrup. An interesting suggestion came from a facility in-charge in Punjab who opined that since DT is a new commodity and its acceptability is currently low thus,

“It should have a price, even if Rs. 5-10 in order for the patient to value it. Free stuff has no value and when people are not even familiar with the usage, they just throw it away.”

District managers could not emphasize the role of LHWs enough in managing these illnesses as they considered them the backbone of the community health. Although LHWs were quite pleased with their regularization, a few managers from Punjab mentioned that they have become a little relaxed and at times rather negligent of their duties due to their permanent employment status.

Finding: The project had certain unexpected effects (both positive and negative) on the management outcomes of childhood Pneumonia and Diarrhea.

During the implementation of the project, certain unexpected effects occurred. On the positive side, it included the revision and updating of Essential Medicines List in the province of Khyber Pakhtunkhwa. The list was updated with inclusion of dispersible tablets in this province, which was not part of project intervention areas but the provincial policy makers and planners were regularly attending the federal coordination forum of Technical Working Group. Similarly, the revision of DHIS tools in the target provinces of Punjab and Sindh did not stop at the inclusion of updated Pneumonia and Diarrhea commodities but also included indicators on maternal health, immunization and breast-feeding practices. The only negative effect identified through the project was the perceptions of the mothers and caregivers developed after the introduction of dispersible tablets. Rural community did not realise the difference between a regular tablet and dispersible tablet. This led to their lack of acceptance for the use dispersible tablets in children. Although this effect was very rare but there is need to clearly explain the usage of dispersible tablets.

4.2.4 Sustainability

Sustainability has been assessed to measure whether the benefits of the project are likely to continue after donor funding has been withdrawn.

EQ: What evidences exists to see the likelihood of the project results are sustained and will be adopted by the Government to ensure that the ultimate goal of the programme is achieved?

Finding: Sustainability and replicability of the project is to be ensured through institutionalisation of key policies and guidelines through health systems strengthening approach.

Government has the decisive role in making this project sustainable by incorporating it into their policy and implementing it in true letter and spirit. Government has shown their commitment in assuming P&D project as one of its own which is reflected through various measures they have taken in this regard.

Revision of IMNCI Guidelines – IMNCI guidelines have been revised and updated allocating distinct training sessions on our diseases of interest i.e. Pneumonia and Diarrhea. This was made possible with the technical assistance from UNICEF after a series of comprehensive consultative meetings with all the stakeholders (Government, Technical Working Groups and Development Partners). In the revised version of community IMNCI training manual, not only P&D are given specific attention by assigning training sessions separately but also, the training duration of IMNCI has been made compact (shortened from eleven to six days). This will emphasize the importance of and help focus the healthcare providers these particular health problems, hence, their better management.

The trainings on revised IMNCI manuals are being carried out successfully across Sindh and Punjab.

A health manager from Sindh reported, “We have over 300 Basic Health Units in which staff is receiving trainings on IMNCI/WHO guidelines.”

A Child Survival Group member from Sindh, commented about the Revised IMNCI trainings, “...these trainings highlight the importance of usage of Amoxicillin DT, Zinc DT and L-ORS in management of Pneumonia and Diarrhea.”

Essential Medicine List and MSDP Updated - UNICEF assisted Federal and Provincial governments in Pakistan for the provision of updated six commodities for management of

Pneumonia and Diarrhea. In continuation of this policy, Essential Medicine List for the healthcare facilities was updated by the government, thereby including L-ORS and Amoxicillin, Zinc Dispersible Tablets in the EML. Provincial DHIS official informed, ***“Amoxicillin DT, Zinc DT and L-ORS have been made part of the Essential Medicine List.”***

In Punjab, updating of Minimum Service Delivery Package for the healthcare facilities has been done focusing P&D for the health and well-being of under 5 Children. Six new commodities introduced by UNICEF to improve Pneumonia and Diarrhea management and services through P&D project (Amoxicillin & Zinc DT, Lo-ORS & Zinc DT, ARI Timers, Pulse Oximeters and Oxygen for appropriate levels of service delivery) are now part and parcel of the MSDP for the healthcare facilities. Whereas in Sindh, MSDPs are still under revision/update as per GAPPD recommendations through the Healthcare Commission.

UNICEF, continues the procurement of quality assured commodities for Pneumonia and Diarrhea till end of the year, 2019. Government has included L-ORS, Amoxicillin DTs and Zinc DTs for P&D in the Procurement Lists of healthcare facilities, which will ensure the availability of these drugs at the healthcare facilities even after the year 2019.

According to District managers in Punjab and Sindh, procurement of medicines at the district level is being carried out through a central rate contract list, which is awarded at the provincial levels by the health departments. The revision of IMNCI guidelines, addition of DTs and L-ORS in EML and Procurement Lists, updating of DHIS reporting tool and inclusion of six commodities in MSDP have all contributed substantially in achieving a good part of the goal of sustainability of P&D project. These efforts on the part of Government will impact well beyond the achievement of project objectives as the future need assessments and performance analysis shall be made on the basis of these revised tools.

The above mentioned arrangements done by Government clearly depict their intent and commitment to making P&D project sustainable even when UNICEF assistance is concluded.

EQ: What internal/external factors and drivers contribute to or constrain the sustainability of the project?

Finding: Challenges faced are mitigated through enhanced government ownership and increased focus on P&D.

The commitment and resolve of the Government and Partner Organizations is of paramount importance in determining fate and shape of P&D project in future. This commitment in other words is formative of sustainability of this project. Government has taken multiple steps in making P&D project a success and also for the continuation of these activities as one of its own, even beyond 2019. The revision of IMNCI guidelines, addition of DTs and L-ORS in EML and Procurement Lists, updating of DHIS reporting tool and inclusion of six commodities in MSDP have all contributed substantially in achieving a good part of the goal of sustainability of P&D project.

All the Health Department Officials showed their zeal in the project activities and were fully aware of all the events being carried out in health departments in this regard. One of the Health Department Official told, ***“...These services regarding Diarrhea and Pneumonia are satisfactory according to our sources”***

Not only health managers displayed ownership to the project but also health facilities' staff members were equally involved in its implementation and provided positive feedback; one of the staff member from health facility told, ***“...Medicine supply provided by UNICEF is very effective”***.

Another important factor in sustainability of this project is un-interrupted budgetary allocations to buy commodities for Pneumonia and Diarrhea in sufficient quantities. Currently,

there is a mixed trend in availability of these commodities across different districts of Sindh and Punjab; most of these commodities were available but in many facilities the situation was otherwise. However sustainability demands the availability as well access to these commodities of most of if not all of the respective patients of Pneumonia and Diarrhea being reported to the public sector healthcare providers (both fixed and outreach staff). One of the health facility staff members told, “...*target is too much and medicine is very limited*”.

Trainings and refresher courses/trainings play a vital role in capacity building, developing ownership of the project and in morale boosting of the implementing staff. The more frequently these trainings and refreshers are conducted the better outcomes of the project are likely to be achieved. The information is continuously pouring in day by day and updating the staff members regularly on project objectives and their achievement modalities is crucial. The analysis revealed that there is dire need and strong demand of these trainings amongst the health facility staff members. Many staff members are still not trained on project implementation and many of the trained ones received it months ago and require refresher in order to perform better. One of the staff members explained, “...*We need more trainings about medicines of Pneumonia and Diarrhea*”.

The achievement of objectives of controlling Pneumonia and Diarrhea in Pakistan under 5 population is an uphill task and is a long and continuous process. Keeping this in mind we are rightful in saying that the current project is working good under the resolve of Government and is very much likely to be sustainable due to efforts of all the stakeholders involved. However, there are some serious challenges to be looked after carefully like budgetary allocations after the UNICEF assistance is concluded, trainings of the concerned staff members and equitable distribution of the commodities. To mitigate these challenges and cope up with them, there is viable mechanism in place, i.e., regular meetings of the federal and provincial coordination forums occur periodically and the decisions taken therein are translated into workable actions on regular basis. This process has been observed effectively during the course of the project and will be followed in future as well; many of the problems faced are already discussed and dealt with successfully through these meetings and same process will be carried on prospectively as well.

The mid-term evaluation of the P&D project clearly indicates that its achievements outweigh its constraints by far, which is evidence on project being sustainable in future.

EQ: What is required to ensure prospects of sustainability of the project outcomes and the potential for replication or scale up of good practices and/or innovative approaches?

Finding: The project envisages increased Government ownership and donor’s commitment for project sustainability.

Sustainability of P&D project depends on its achievements and performance as perceived by the stakeholders including government, partner agencies, health facility staff and patients. This means that all the services are provided to end users’ and providers’ satisfaction.

The other contributing factor is that government and organizations if manage their knowledge assets effectively will gain sustainable competitive advantage and this is why knowledge management is surety bond of sustainability in healthcare. Knowledge management and therefore sustainability depends upon coordination, transformation, and transfer of knowledge and aims to facilitating the communication of knowledge to the people that created it, as well as the people that need it. Sustainability can only be achieved if the good practices are to be simulated in the scaling up and replication of this project. The good practices in the current project such as the commitments, resolve and capacity of government and partner agencies to run the project effectively in a developing country like Pakistan with limited resources, its compromised performance issues (Pakistan ranked 120/190 in healthcare performance), high

disease load of Pneumonia and Diarrhea as well as many other communicable and non-communicable diseases and alarmingly high rates of morbidity and mortality in the children under 5 years of age. The resolve of the government and other stakeholders are evident in the form of updating of IMNCI guidelines, addition of DTs and L-ORS in EML and Procurement Lists, updating of DHIS reporting tool and inclusion of six commodities in MSDP.

Another good practice to be replicated is the coordination amongst all the stakeholders and as a result of it, the evidence based decision making by CSGs whose recommendations are conveyed and followed till the lowest operational level effectively. There are a certain innovative approaches that can be integrated into the scaled up/replicated projects: community involvement in the project at different levels such as at sub-district, district, provincial and federal level. The involvement of community is aimed to develop ownership of the project in the local community and as an accountability measure. The representation of community and its mandate can be decided after rigorous thought processes in the coordination meetings.

5. Conclusions

Policy change is a complex and painstaking process that requires clear-cut, precise and well-timed interplay of a multitude of factors. In a given context, what contributed the most to policy change is credible evidence. Building on this principle, project on Accelerating Policy Change, Translation and Implementation for Pneumonia and Diarrhea Commodities in Pakistan (P&D Project) developed, gathered, communicated and popularised the child survival as a policy agenda item on the lists of federal and provincial governments. At the time of inception, landscape analysis, baseline and evaluability studies were conducted to generate evidence, inform the project planning with ground realities and to identify drivers and barriers for policy making. The project achieved/completed/accomplished the causal chain of policy change, its translation and implementation for revised and updated commodities for management of childhood Pneumonia and Diarrhea in Pakistan during the stipulated timeframe. The evaluation has also revealed the spill over effect of the project interventions leading to achievement of broader benefits in its streams of interventions, including the updating of DHIS tools, revision of IMNCI manuals and Essential Medicines Lists. Through coordination support during these revisions, opportunities were harnessed by the relevant stakeholders in government, within Unicef and other development partners to update other areas of maternal and child health. These amendments, like EPI indicators in DHIS, were originally beyond the scope of this project but were made possible through strong stakeholders engagement during policy translation processes.

Policy change is unpredictable. No matter how calculated and premeditated the engagement and advocacy efforts are timeliness of actions is crucial to constitute an opportunity for change. The project has managed to achieve concrete milestones and outcomes on child survival during a short span of time. It was only possible because the project team availed all the available windows of opportunity and positioned their evidence to inform the policy change, its subsequent translation and implementation. Policies covering the entire continuum of care around childhood Pneumonia and Diarrhea have been translated into actions and gained results; now requiring scaling up and replication to achieve child survival goals.

The project team at all levels (M/oNHSR&C, Health Departments) critically engaged with policy makers so that opportunities are shaped for use of evidence in the policy making process. Their presence in the broader policy space and strong relationships, both in formal (steering committee, technical working groups, CSGs) and informal (networking) ways, resulted in fostering stronger association of the project with government policies and strategies on childhood Pneumonia and Diarrhea. During the evaluation, lobbying and networking of the project team for child survival was highly valued by the government counterparts and there was high buy-in of the project interventions from the federal and provincial policy makers and planners. Another example included the description of provincial Child Survival Group as a think tank – by a provincial policy maker – that has been supporting the policy makers through quality and relevant evidence for policy change. During the revisions and updating of guidelines, manuals, government yardsticks and information systems, the project established strong relationships to connect multiple stakeholders. This wide array of stakeholders that project directly engaged to achieve its planned outcomes included ministerial offices, health departments, their directorates, donors, development partners, district managers, health facility staff, professional associations, outreach staff, pharmaceutical manufacturers, regulatory bodies (Drugs Regulatory Authority of Pakistan and Drugs Testing Laboratory).

The evaluation team trailed the dimensions of OECD/DAC Criteria to gauge the success of the project. In the following section, the policy process (policy change, policy translation, policy implementation and knowledge management) for each of the interventions stream is described along with their chemistry with project assumptions. This is followed by recommendations, proposed actions and theory of change for the remainder for the project life.

At the level of **policy change**, the project had a clear analysis and expression of why, what, how and who needs to be involved and engaged to achieve policy change. Political will and commitment were evident from the stress drawn on childhood illnesses in Pakistan's National Health Vision as well as provincial health policies and sector strategies. Willingness of government institutions, federal and provincial, was evident from their ownership and buy-in of project interventions. Decentralization of health sector has created dichotomies among federal level and provinces. The project attempted to bridge this gap by supporting certain platforms to build a shared vision. Support to provincial Health Departments was provided in implementation of joint accountability framework through CSG and TWG. Both the CSG in Punjab and TWG in Sindh had representation of all relevant stakeholders. Under policy change, the receptivity of policy makers and planners was demonstrated through updating of government documents (MSDP/EPHS, Essential Medicines Lists, Provincial Procurements Lists) to include the updated commodities for management of childhood Pneumonia and Diarrhea.

Lessons Learnt – Policy Change Outcome

- Political will and commitment is essential for policy transformation and rolling out reforms agenda in Pakistan.
- Joint accountability framework assisted the project in steering the progress monitoring and tracking.
- Establishing oversight and coordination platforms at all levels are critical in building government ownership and steering the project activities.
- Decentralization of health sector has created dichotomies among federal level and provinces.

Moving to the next stage of **policy translation**, the project first built credible evidence on the effectiveness of updated commodities. Healthcare providers, at both the health facilities and within communities, acknowledged the use of updated commodities for effective management of childhood Pneumonia and Diarrhea. Inclusion of updated commodities in key government documents (MSDP/EPHS, Essential medicines list, provincial procurement lists) showed government financial commitment, whereas, the governments are already procuring alternatives to updated commodities. A key intervention for policy translation was the support to federal and provincial governments for updating IMNCI guidelines while catering to the challenges of old curriculum and training duration. However, there were still gaps in allocation of resources for conducting the trainings across the board as the district managers explained the lack of resources for conducting IMNCI trainings.

Lessons Learnt – Policy Translation Outcome

- Policy translation efforts should be strongly supported through empirical evidence for development of advocacy material.
- Acceptability of the care providers should be built for introduction of new/revised commodities..
- Without financial commitment of the governments, public procurement of updated commodities is not possible.
- Training material should cater to the local needs in terms of its contents and duration

Under **policy implementation**, the evaluation revealed that the diagnosis and treatment had improved through use of Pulse Oximeter and Oxygen at the health facilities, and ARI Timers by the LHWs. Ownership of the project interventions was built through strengthening the existing service delivery systems by improving management practices of healthcare

providers, mainly outreach workers and primary level facilities. In order to institutionalize the recording and reporting on updated commodities, DHIS tools were revised in both provinces. However, gaps in systems provisions, resources and skill sets were required to be filled through development of integrated supply chain management system for the targeted districts. The Project has initiated its support for implementation of DHIS-2 and HLMIS. The project had identified implementing partners who have the capacity and spread across the project provinces to effectively implement the project activities in their specific areas. However, a key hurdle in policy implementation was the fact that despite the resources and capacities, the local Pharma industry lacked interest in local production due to minimal demand of updated commodities in open market.

Lessons Learnt – Policy Implementation Outcome

- Evidence of improved management of childhood illnesses through updated commodities accelerated the policy implementation phase.
- LHWs were empowered through provision of modern diagnostic gadgets.
- Engagement of implementing partners pave way for enhancing the array of expertise to effectively support the policy implementation.
- Despite the resources and capacities, the local pharmaceutical manufacturing industry lacked interest in local production due to minimal demand of updated commodities in open market.

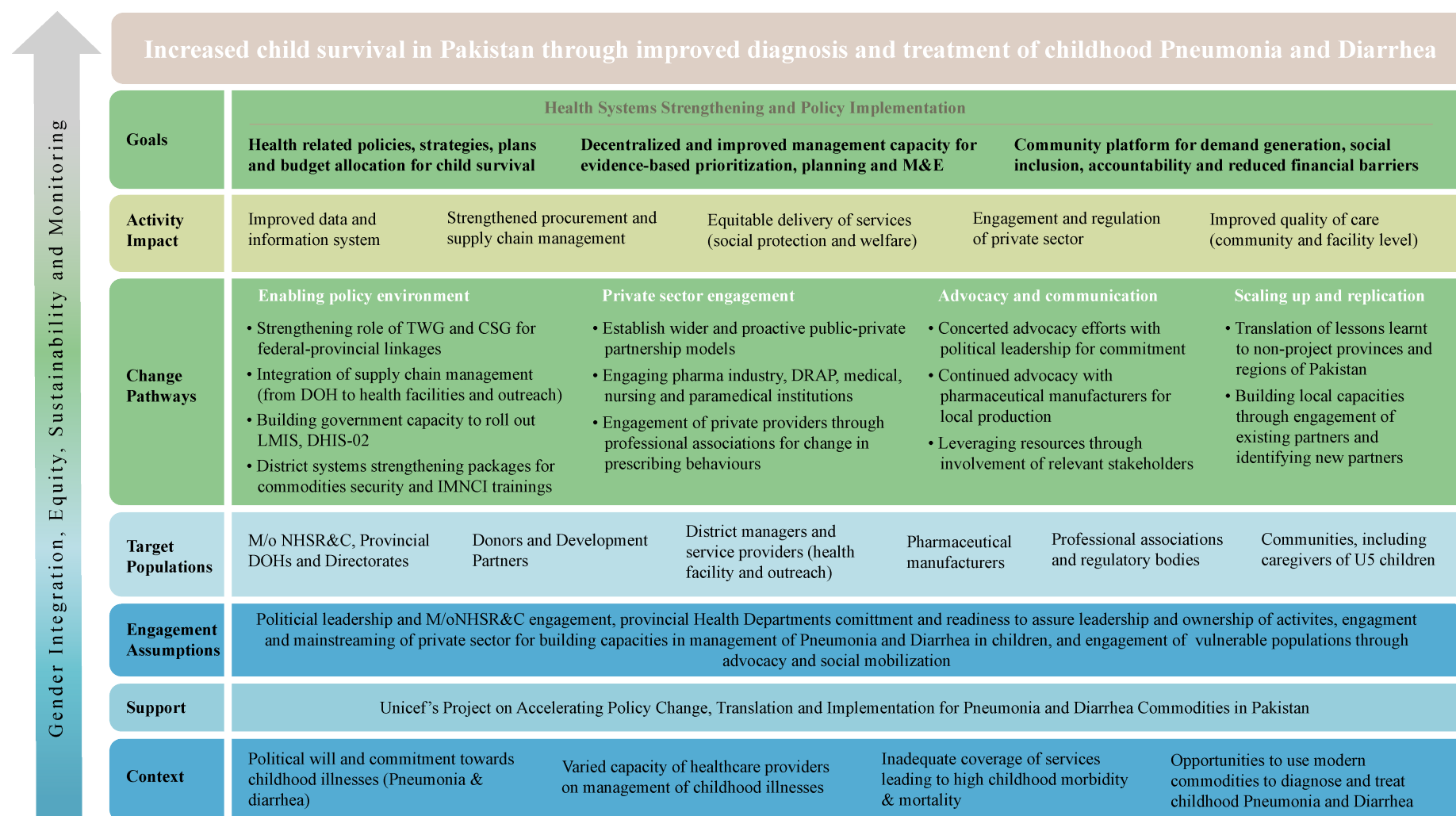
Updated federal and provincial policies and guidelines reflect the recognition of improved management as a shared goal. Now, after the project has demonstrated its effectiveness in terms of better diagnosis and treatment of childhood Pneumonia and Diarrhea, the lessons learnt from the implementation in two provinces should be replicated in other provinces and regions of Pakistan through *knowledge management*.

Lessons Learnt – Knowledge Management Outcome

- A systems strengthening approach is essential to make the project interventions well entrenched and effective for improving infant and childhood morbidity and mortality.
- Updated federal and provincial policies and guidelines reflect the recognition of improved management of Pneumonia and Diarrhea as a shared goal.
- Sustainability of P&D project depends on achievement of results as perceived by the stakeholders including government, partner agencies, health facility staff and patients.

5.1 Proposed Theory of Change

While assessing the current ToC, the evaluation team found the given TOC not adequately developed and unable to convey the pathways of change as described by the project management team. A valuable result of the discussion around the given TOC of the project was the inference shared by the management team that certain issues and challenges that need addressing for resolving bottlenecks in project implementation to ensure achievement of results needed different pathways of change. Based on the consultations with all stakeholders, the current TOC was thoroughly revised to achieve the desired outcomes of supporting child survival in Pakistan through improved diagnosis and treatment of childhood P&D. Proposed TOC (Figure 9) was further connected with the proposed recommendations put forward by the evaluation team as described in Annex 7. It was reviewed and endorsed by the counterparts at Federal Ministry, Provincial Health Departments and UNICEF for course correction and ensuring the best use of the remaining project duration.

Figure 9: Proposed Theory of Change

6. Recommendations

The following recommendations were elicited through a consultative and participatory process among all key stakeholders mentioned above. Guides used for key informant interviews and focus group discussion included sub-sections on the respondents' perspective on bringing improvement in the project and its interventions while moving forward. In addition to these respondents, members of the Reference Group at federal level and project implementation team at UNICEF all provided valuable inputs in identifying the recommendations. During the meetings with these stakeholders, these recommendations were discussed at length and finalized in the participatory manner. The recommendations are further divided into short, medium and long-term actions according to the stipulated timelines for their implementation. While the Project will focus on the short to medium term recommendations, there are certain long-term recommendations that will be achievable beyond the project life. Below are the key, strategic recommendations followed by a matrix for easy understanding and action:

In the causal pathway of policy transformation and reforms, the stage of policy implementation is the most critical. This is the point where rightly developed policies fail due to lack of systems' capacity to sustain and implement this change on its own. In order to make the P&D Project interventions institutionalized in the existing system, the strengthening of health sector at individual, organization and systems' level is of utmost importance. A **systems-strengthening approach** is recommended while moving forward so that the interventions under P&D Project are well entrenched in existing system for sustainability beyond the project life. This systems strengthening approach will fill the gaps in each building block of the existing health system to ensure child survival in Pakistan. This includes building an equitable health system, improving quality of care at health facilities and outreach, development of robust information systems, establishing integrated supply chain management system and most importantly, engagement of private sector, which is providing services to nearly 70% during childhood illness in Pakistan.

Enabling policy environment - Sustenance of federal-provincial-district linkages and coordination should be strengthened and continued through existing platforms at federal level (RMNCAH Group), provincial level (CSG and TWG), and district level (District Health & Population Management Teams). Government capacities at all level should be built and a district systems strengthening package should be implemented in the project areas to ensure commodities security and training of care providers on IMNCI and information systems. Details of the recommended actions along with timeline and roles are given in Table 7, whereas proposed short-term immediate actions are as follows:

- Build a systems strengthening framework with package of system strengthening interventions at district and provincial levels.
- Policy dialogues and roundtables on improving management of Pneumonia and Diarrhea through updated commodities.

Private sector engagement – Being a service provider of more than two-thirds of population, the project should establish wider and proactive public-private partnership models. This includes engagement with pharma industry along with Drugs Regulatory Authority of Pakistan, institutions regulating medical, nursing and paramedical education. Private care providers should be reached out through their professional associations. Details of the recommended actions for private sector engagement along with timeline and roles and responsibilities are explained in Table 7, whereas proposed short-term immediate actions are as follows:

- Initiate dialogues on opportunities for pharma industry and drugs distribution networks and DRAP
- Build stronger collaborations with private sector professional associations

Advocacy and Communication - Concerted advocacy efforts with political leadership for sustained will and commitment is important in policy transformation and reforms. Side by side, continued networking and lobbying with pharmaceutical manufacturers for local production important. The project should engage relevant partners for leveraging resources and coordinated actions. Details of the recommended actions for advocacy and communication along with timeline and roles and responsibilities are explained in Table 7, whereas proposed, short-term immediate actions are as follows:

- Develop and implement a mass-media engagement plan focusing on childhood Pneumonia and Diarrhea
- Develop an advocacy toolkit for policy makers and planners

Scaling up and replication – Despite the devolution of health sector, the project is encouraged to scale up the interventions to cover entire Pakistan. Now, after having experience implementation in two of *the* big provinces of Pakistan, the provincial level interventions will not take much effort for replication. Details of the recommended actions for scaling up and replication along with timeline and roles and responsibilities are explained in Table 7, whereas proposed, short-term immediate actions are as follows:

- Cross-sharing of achievements, challenges and lessons learnt among uncovered regions and provinces of Pakistan

Detailed recommendations are described in the following table with proposed actions and timeframe indicating responsibility for each actor.

Table 7: Recommendations’ Matrix				
Recommendations	Proposed Actions	Timelines	Primary & Secondary Responsibility	
ENABLING POLICY ENVIRONMENT				
Fostering CSG/TWG for harnessing its optimal potential and taking measure for sustenance beyond the project life	Institutionalization of CSG/TWG through inclusion in PC-1 documents of relevant health programs (IRMNCH, MNCH and LHWs)	Long term	CSG and TWG	-
	Setting-up a small secretarial unit for the CSG and TWG	Short term	CSG and TWG	UNICEF Project Management Team
Strengthening of supply chain management (from DOH to health facilities and outreach)	Capacity building of DOH on supply chain management	Medium term	SCM implementing partner	Federal and Provincial Governments
Strengthen federal, provincial linkages and coordination to implement National Health Vision and provincial policies/strategies priorities f	Encourage participation of federal representative in CSG/TWG	Short term	M/o NHR&C and Health Departments	UNICEF Project Management Team
	Cross-sharing of challenges, lessons learnt among the provinces	Short term	UNICEF Project Management Team	M/o NHR&C and Health Departments
Revitalization of district level decision-making and coordination platform for enhancing ownership	Building district capacities on forecasting, procurement, distribution, maintenance and warehousing	Long term	Implementing partners	District Health Department
	Linkage of DHIS with LMIS	Medium term	District Health & Population Management	UNICEF Project Management Team

			Teams	
Strengthening government capacity to roll out LMIS and DHIS-02	Training of provincial and district level managers on enhancing analytical skills, data interpretation skills and use of information Experience sharing through visits of areas/countries where DHIS 02 is operational for federal and provincial managers	Long term	UNICEF Project Management Team and	SCM partners
PRIVATE SECTOR ENGAGEMENT				
Establish wider and proactive public-private partnership models through engaging pharma industry along with DRAP, medical, nursing and paramedical institutions	Dialogues on opportunities for pharma industry and drugs distribution network	Short term	UNICEF Project Management Team	CSG and TWG
Enhancing private sector engagement for updating their prescribing behaviours	Active involvement of professional associations in advocacy for private practitioners	Medium term	Professional Associations	UNICEF Project Management Team
	Continued capacity building program for service providers from both public and private providers	Long term	UNICEF Project Management Team	Professional Associations
Engagement of private providers through professional associations for change in prescribing behaviours	Government initiating pre-qualification of potential pharmaceutical manufacturers	Medium term	UNICEF Project Management Team	Professional Associations
ADVOCACY AND COMMUNICATION				
Continued advocacy with pharmaceutical manufacturers for local production	Engaging with private pharma and distribution networks by sharing areas of opportunities and informing them about benefits of treatment	Short term	UNICEF Project Management Team	Pharma industry and DRAP
	Building stronger collaborations with DRAP and private sector			
	Direct Demand Creation: Convincing provincial health authorities to include updated commodities in provincial procurement lists	Medium term	Provincial Health Department	Pharmaceutical industry through relevant associations
Concerted advocacy efforts with political leadership for sustained will and commitment under the GAPPD for childhood Pneumonia and Diarrhea management	Policy dialogues and roundtables on improving management of Pneumonia and Diarrhea through updated commodities	Short term	UNICEF Project Management Team	M/o NHR&C and Health Departments
	Mass-media engagement: Opinion Editorials in leading national newspapers Televised debates	Medium term	UNICEF Project Management Team	Mass-media agencies

Advocacy on the benefits of updated commodities for replacing existing finances for alternative procurement	Support to provincial health departments in sustainable implementation of integrated HLMIS through development of PC-1	Medium term	UNICEF Project Management Team	Provincial Health Departments
Leveraging resources through advocacy and involvement of relevant stakeholders	Strengthen district level action planning with inclusion of trainings with required resources Engagement of donors and development partners Engagement of Nursing Council, Midwifery Associations , PPP, Healthcare Commissions relevant DoH / Health inst. for curricula revision and trainings of HCPs and CHWs	Long term	UNICEF Project Management Team	Provincial Health Departments and Districts
Two-pronged approach with concurrent advocacy efforts to mitigate inherent barriers in local production and public procurement	Advocate with government and pharma industry for procurement of revised commodities Private sector engagement and proactive collaboration with DRAP	Short term	UNICEF Project Management Team	Provincial Health Departments
SCALING UP AND REPLICATION				
Translation of lessons learnt to other geographical settings	Evidence-based policy advocacy - enhancement of geographic coverage of the interventions Rolling-out IMNCI in target provinces and advocacy for replication in other provinces and regions of Pakistan	Long term	M/o NHSR&C and Health Departments	UNICEF Project Management Team
Scale up and replication of these interventions to non-project provinces and regions of Pakistan	Sharing of findings of MTE with Health Departments and stakeholders of other provinces and regions of Pakistan	Short term	UNICEF Project Management Team	Provincial Health Departments
	Resource allocation - donor identification and earmarking of funds	Medium term	Provincial Health Departments	CSG and TWG
	Evidence-based policy advocacy - enhancement of geographic coverage of the interventions	Long term	UNICEF Project Management Team	Provincial Health Departments

7. Dissemination

The report has been accepted by all the key counterparts through a validation workshop that was conducted on 24th December as part of the IRMNACH Technical Committee meeting. Another dissemination is planned by UNICEF where evaluation team will present the key findings and recommendations along with case studies (Annex 4) and a policy brief to disseminate the evaluation to a wider audience.

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10. Annexes

Annex 1: Evaluation Matrix

Criteria	Evaluation Questions	Indicators	Data Sources	Data Collection Methods	Data Collection Tools
Relevance	How relevant and meaningful are the project objectives and activities in addressing the needs and priorities of the marginalized and vulnerable children in the project areas?	Empirical evidence supporting use of updated commodities for management of Pneumonia and Diarrhea in girls and boys	Secondary	Desk review of empirical evidence	Review guidelines
		Acceptance of communities for use of recommended commodities	Primary	- KII with facility in-charges (BHUs and RHCs) - FGD with service providers (GPs) - FGD with LHWs - FGD with Beneficiaries	- KII Guides - FGD Guides
	What is the relevancy of the project with the Federal and provincial DoH as well as the services being provided by the private sector?	Level of priority of childhood Pneumonia and Diarrhea in federal and provincial policies and strategic frameworks	Primary & Secondary	- Desk review - KIIs with M/oNHSR&C representative - KIIs with provincial government representatives - Representatives of pediatric associations and CSG	- Review guidelines - KIIs Guides
	To what extent the objectives of the project are consistent with the existing national/provincial policies and guidelines in line with global recommendations (WHO/ GAPPD) for management of Diarrhea and Pneumonia among under-five children in Pakistan and are sustainable?	Objectives of the project are aligned to the recommendation of WHO/GAPPD for management of Pneumonia and Diarrhea	Primary & Secondary	- Desk review - KIIs with M/oNHSR&C representative - KIIs with provincial government representatives - Representatives of pediatric associations and CSG	- Review guidelines - KIIs Guides
	Are the strategies or approaches appropriate and	Appropriateness and adequacy	Primary &	Desk review	Review

Criteria	Evaluation Questions	Indicators	Data Sources	Data Collection Methods	Data Collection Tools
	adequate to achieve results?	of interventions under Pneumonia and Diarrhea Project for child survival, both girls and boys	Secondary	• KIIs with M/oNHSR&C representative • KIIs with provincial government representatives • Representatives of CSG	guidelines • KIIs Guides •
Effectiveness	To what extent the project has achieved its objectives/ outcomes and what were the major factors influencing the achievement or non-achievement of the objectives/ outcomes?	• Updated national and provincial policies and guidelines, in line with GAPPD recommendations, available • Joint accountability framework endorsed and notified by the competent authority at federal and provincial level • Local manufacturers advocated for production of recommended P&D commodities in Pakistan	Primary & Secondary	• Desk review • KIIs with M/oNHSR&C representative • KIIs with provincial government representatives	• Review guidelines • KIIs Guides •
	To what extent the implementation of the project approaches worked as intended, particularly after the baseline in 2016 and subsequent adjustments?	• Project milestones identified at the inception of the project • Achievement of project millstones against planned	Primary & Secondary	• Desk review of project documents • KIIs with UNICEF project staff	• Review guidelines • KIIs Guides
	Whether the mechanisms available to create awareness among communities are effectively linked to the project objectives?	• Mass media campaign conducted for social and behavioural change in target communities, particularly focusing on girl child	Primary & Secondary	• Desk review of project documents • KIIs with government representatives at federal, provincial and district levels • KIIs with UNICEF project staff • FGDs with LHWs and beneficiaries	• Review guidelines • KIIs and FGDs Guides

Criteria	Evaluation Questions	Indicators	Data Sources	Data Collection Methods	Data Collection Tools
	Whether the healthcare provider and community health workers have the required knowledge and skills to proper utilization of supplies as per standards?	- Revised National P&D/IMNCI guidelines utilized to improve knowledge and capacity building of Healthcare Workers - Capacities of health care providers built on management of P&D on updated commodities	Primary & Secondary	- Desk review of project documents - KIIs with government representatives at federal, provincial and district levels - KIIs with UNICEF project staff - FGDs with LHWs and beneficiaries	- Review guidelines - KIIs and FGDs Guides
	Whether the monitoring and reporting mechanisms exist and effectively implemented for effective tracking and improvement in system?	- Joint accountability framework implemented at federal and provincial levels - Reporting tools on facility level data (DHIS Tools) updated with revised indicators on recommended commodities	Primary & Secondary	- Desk review of project documents - Desk review of revised DHIS tools - KIIs with government representatives at federal, provincial and district levels - KIIs with UNICEF project staff - FGDs with LHWs and beneficiaries	- Review guidelines - KIIs and FGDs Guides
	How effective are the 'innovative approaches like use of ARI timers and pulse oximeter, what results can be achieved, are replicable within the Sindh/Punjab context?	- Effect of use of ARI Timers by LHWs on management of Pneumonia and Diarrhea in girls and boys - Effect of use of updated commodities by healthcare providers (public and private) on Pneumonia and Diarrhea management in girls and boys	Primary & Secondary	- Global best practices on recommended commodities - Case study analysis - KIIs with Federal, Provincial and District stakeholders - KII with facility in-charges - FGDs with LHWs and beneficiaries	- Case study analysis guidelines - KIIs and FGDs Guides - Facility checklists

Criteria	Evaluation Questions	Indicators	Data Sources	Data Collection Methods	Data Collection Tools
	To what extent the programme has been able to contribute to ownership and leadership of the provincial/ area DOHs?	- Level of participation of provincial government officials in child survival groups - Provincial Minimum Service Delivery Package (MSDP) containing recommended commodities - Revision of EML and procurement lists	Primary & Secondary	- Review of MSDP, EML and procurement lists - KIIs with federal, provincial and district levels - KIIs with UNICEF project staff	- Review guidelines - KIIs and FGDs Guides
	What have been the major factors influencing the achievement or non-achievement of outcomes?	Factors identified that influenced achievements or non-achievement of project outcomes	Primary & Secondary	- Desk review of project documents - KIIs with federal, provincial and district stakeholders - KII with facility in-charges - FGDs with service providers and beneficiaries	- Review guidelines - KIIs and FGDs Guides
	To what extent the programme learned and evolved over the 2 years and whether there were other alternatives, more cost-effective strategies available to reach intended results?	Project adjustments made for behind schedule or under target outcome/output	Primary & Secondary	- Desk review of project documents - KII with project staff	- Review guidelines and Tool - KIIs Guides

Criteria	Evaluation Questions	Indicators	Data Sources	Data Collection Methods	Data Collection Tools
Efficiency	How well the resources, both human and financial, been managed to ensure timely attainment of results?	Timeliness of the achievement of project activities	Primary & Secondary	- Desk review of project documents - KIIs with UNICEF project staff	- Review Guidelines - KIIs Guides
	To what extent planning, budgeting, monitoring and evaluation, supervision, coordination, logistics and financial management systems are functioning in support of the project objectives?	Project management functions performed over time	Primary & Secondary	- Desk review of project documents - KIIs with UNICEF project staff	- Review Guidelines - KIIs Guides
	What are the implementation challenges from the perspective of both right holders and duty bearers especially for the under-five marginalized children of communities?	- Availability of recommended commodities in the open market - Uninterrupted supply of P&D recommended commodities at health facilities	Primary	- KIIs with Federal, Provincial and District stakeholders - KII with facility in-charges - FGDs with LHWs, GPs, pharmacists and beneficiaries - KII with representative of pharmaceutical manufacturers	- KIIs and FGDs Guides - Facility checklist
	To what extent has the project leveraged additional resources to address identified gaps?	- Additional resources leveraged through support of government at federal and provincial levels - Additional resources leveraged through advocacy with other donors and development partners	Primary & Secondary	- Review of project documents - KIIs with government representatives at federal and provincial levels - KIIs with UNICEF project staff	- Review Guidelines - KIIs Guides
	What is the value added in terms of service delivery for Pneumonia and Diarrhea?	- Supply of P&D recommended commodities at public facilities - Availability of recommended commodities in open market - Effect on prescribing behaviours of healthcare providers and outreach workers	Primary	- KII with provincial and district levels - KII with facility in-charges - Prescription review analysis - FGDs with LHWs, GPs, pharmacists and beneficiaries	- Prescription review tool - FGD and KII guides

Criteria	Evaluation Questions	Indicators	Data Sources	Data Collection Methods	Data Collection Tools
	What actions including innovations are needed to improve the coverage?	- Revised national IMNCI guidelines for healthcare providers with shorter duration - Replicability potential of the project to other provinces and regions of the country - Supply chain management for commodities security at public facilities	Primary & Secondary	- Review of IMNCI guidelines - Consultative meetings in uncovered provinces - KII with provincial and district levels - KII with facility in-charges	- Review Guidelines - Tool for consultative meetings
	Whether the availability of P&D supplies is adequate, timely and cost effective?	- Stock-out of recommended of recommended commodities at public facilities - Availability of recommended commodities at medical stores and pharmacies	Primary & Secondary	- Desk review of project documents - KIIs with federal, provincial and district stakeholders - KII with facility in-charges - FGDs with LHWs, GPs, pharmacists and beneficiaries	- Review guidelines - KIIs and FGDs Guides
	To what extent has the project achieved its goals in enhancing the health outcomes especially of under five children in the catchment communities in Sindh and Punjab target districts?	Effect of use of recommended commodities on Pneumonia and Diarrhea management among girls and boys in project areas	Primary	- Prescription review analysis - KII with provincial and district levels - KII with facility in-charges - FGDs with LHWs, GPs and beneficiaries	- Prescription review tool - FGD and KII guides
Sustainability	What evidences exists to see the likelihood of the project results are sustained and will be adopted by the Government to ensure that the ultimate goal of the programme is achieved?	Policies and guidelines updated through revision of IMNCI guidelines, MSDP, EML, procurement list and DHIS tools	Primary & Secondary	- Review of federal and provincial policy documents and guidelines - KIIs with Federal, Provincial, and District stakeholders	- Review guidelines and Tool - KII Guides
	What internal/external factors and drivers contribute to or constrain the sustainability of the project?	- Regularity of meetings of the federal and provincial coordination forums - Actions taken on decisions of CSG and TWG meetings	Primary & Secondary	- CSG minutes of meetings - KIIs with provincial stakeholders	- Review Guidelines - KIIs Guides

Criteria	Evaluation Questions	Indicators	Data Sources	Data Collection Methods	Data Collection Tools
	What is being planned to sustain the process with government support beyond 2019?	Availability of project sustainability plan with roles and responsibilities for government support at federal and provincial levels	Primary & Secondary	- Review of project documents - KIIs with Federal, Provincial, and District stakeholders	- Review Guidelines - KIIs Guides
	What is required to ensure prospects of sustainability of the project outcomes and the potential for replication or scale up of good practices and/or innovative approaches?	Level of government ownership and donors commitment for project sustainability	Primary	- KIIs with Federal, Provincial, and District stakeholders - KIIs with UNICEF staff	KIIs Guides

Annex 2: Evaluation Team Composition

Evaluation Team Composition	
BACKSTOPPING AND ADVISORY TEAM	
Health Specialist	Dr. Naeem uddin Mian
Health Systems Experts	Professor Dr. Ashraf Majrooh
TECHNICAL TEAM	
Evaluation Expert	Dr. Muhammad Adeel Alvi
Research Expert	Ms. Mariam Z. Malik
Public Health Specialists	Dr. Farooq Umer Dr. Rabia Suljuk
Qualitative Research Expert	Professor Dr. Rubeena Zakar
Medical Anthropologist	Ms. Hira Hasan
Bio-Statistician	Mr. Abdul Hamid
Field Supervisor	Mr. Saleem Shehzad
FIELD TEAM	
Enumerators	8 females and 4 males
Sociologists	5 female moderators and 5 female note-takers

Annex 3: Ethical Review Board Certificate**Research Ethics Approval**

13 September 2019

Dr. Muhammad Adeel Alvi
Principal Investigator
c/o UNICEF MENARO
Aman, Jordan

RE: Ethics Review Board findings for: *Midterm Evaluation of UNICEF's Pneumonia and Diarrhea Project in Pakistan*

Dear Dr. Alvi,

Protocols for the protection of human subjects in the above study were assessed through a research ethics review by HML Institutional Review Board on 26 August – 12 September 2019.

This study's human subjects' protection protocols, as stated in the materials submitted, received **ethics review approval**. Please notify this IRB of any changes in this study's design, risks, consent, or other human subject protection protocols.

Sincerely,

D. Michael Anderson, Ph.D., MPH
Chair & Human Subjects Protections Director, HML IRB

cc: Shamshad Begum, Penelope Lantz, JD

HML Institutional Review Board
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US Department of Health & Human Services, Office of Human Research Protections IRB #00001211

Annex 4: Case Studies

Midterm Evaluation of Project for Accelerating Policy Change, Translation and Implementation for Pneumonia and Diarrhea Commodities in Pakistan (2016-2020): Case Study

This is one of two case studies carried out as part of Midterm Evaluation of Unicef's Pneumonia and Diarrhea Project in Pakistan. The evaluation aimed to ascertain that the progress is made in the right direction to meet the project's ultimate results, documenting its overall achievement and lessons learned. The evidence generated is envisaged to navigate the scaling up and replication of these interventions to curtail childhood morbidity and mortality in the country.

1 Case Study 1: Addressing diagnostic challenges for Pneumonia through use of ARI Timers by Lady Health Workers (LHWs)

PROBLEM STATEMENT

Pakistan has one of the highest mortality rates globally among children under 5 years of age, at 74 deaths per 1,000 live births (PDHS, 2017-18). Pneumonia is considered one of the main causes of childhood mortality, attributed to delayed diagnosis and inaccurate identification of symptoms or risk factors. Early and accurate diagnosis of childhood pneumonia, its timely referral and treatment decreases the childhood morbidity and mortality (Bhutta et al., 2013).

Traditionally, medical practitioners – based on clinical signs and symptoms – use various diagnostic tests to determine illness etiology and severity; however, childhood pneumonia still faces diagnostic challenges. Front line workers at primary healthcare facilities and community health workers face difficulties in assessing the severity of illness in children suffering from Pneumonia due to the absence of standard diagnostic tools. Situation is further aggravated with their limited capacities on IMNCI, lack of proper case definitions and delayed or absence of referral. Misclassification of an observed respiratory rate is common, leading to incorrect diagnosis. Further, many diagnostic methods are initially validated in adult populations, making interpretation in the pediatric setting more difficult.

APPROACH

In order to improve the child survival in Pakistan, UNICEF initiated "Accelerating Policy Change, Translation and Implementation for P&D Commodities in Pakistan," Project (2016-2020), aiming to link the most disadvantaged children and women with an integrated package of high impact and quality commodities to ensure a continuum of healthcare. The project promotes the use of evidence-based and cost-effective commodities for accurate and early diagnosis for improved management of childhood Pneumonia & Diarrhea (P&D). Counting the respiratory

ARI Timer given distributed by UNICEF to LHWs in Pakistan



Source: MTE Field Team

rate in children with cough or difficulty in breathing is a useful sign for diagnosis of pneumonia and assessing its severity. In order to equip healthcare providers at health facilities and in the communities with tool for accurately counting the respiratory rate, UNICEF and WHO introduced ARI Timer in 1990 to support health workers in developing countries for determining how long they need to count a child's breathing to aide in correct and timely diagnosis. ARI Timer is a very simple device, which was provided to each LHW in the project districts. All the LHWs were briefly oriented on its usage to accurately count the respiratory rate.

Lady Health Workers (LHWs), as the first level healthcare providers, with unrestricted access to households, close rapport with local community and admitted acceptability, made them an ideal entry point for such interventions. The program, thus utilized the channel of LHWs in introducing ARI Timers for early diagnosis and timely referral of pneumonia at community level.

In alignment with Global Action Plan for Pneumonia & Diarrhea (GAPD) and WHO updated recommendations, UNICEF provided continuous technical support to federal and provincial governments for inclusion of ARI Timers in the procurement lists. UNICEF initially procured these devices for the early detection of pneumonia in children under 5 years at primary health care levels, following UNICEF's stringent supply procedures. These ARI Timers were distributed to target districts of Punjab and Sindh from 2017 to 2019 followed by capacity building of LHWs on their use. The benefit and impact of these commodities on child survival was established through their usage and UNICEF advocated their inclusion in essential medicines list of the Health Departments. Midterm Evaluation of the project further provided substantial evidence on the use of these commodities for an improved management of both the childhood diseases.

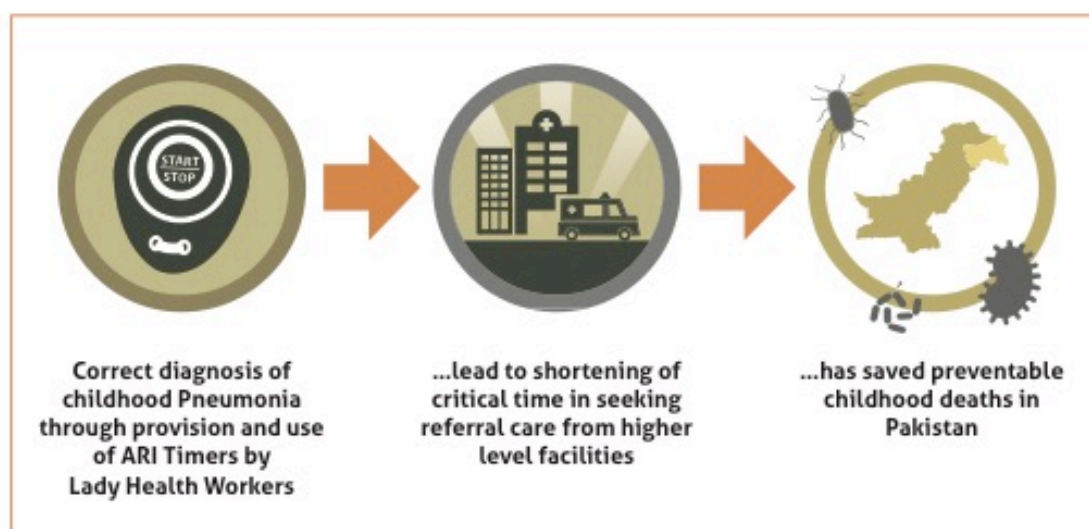
OUTCOME

Prevention of pneumonia, its early diagnosis and treatment in children is an essential component to reduce child mortality and morbidity. Analysis revealed that UNICEF distributed ARI Timers to 29,788 LHWs including 22,700 from Sindh and 7,088 from Punjab for early detection of pneumonia/ARI to ensure timely provision of healthcare services. Analysis demonstrated that ARI Timers addressed the challenges faced by front line workers related to unnecessary delay and inaccurate diagnosis of pneumonia, leading to severity

of illness, complication, or even to deaths among children.

Provision of ARI Timers has empowered LHWs and improved their ability to manage ARI and timely refer childhood Pneumonia. Both facility in-charges and LHWs acknowledged the contribution of UNICEF for child survival in Pakistan through provision of requisite and modern commodities. They were also of the view that ARI Timers, not only enhanced their efficiency and reliance in early detection of pneumonia, but also helped them in saving lives of a number of children.

LHWs acknowledged the usefulness of the ARI Timers and highlighted the importance of trainings imparted on their proper usage. Facility in-charges from Punjab commented on the benefits of ARI Timers, "If used accurately, ARI Timers are cost-effective and efficient devices that can help detect cases of Pneumonia at community level". The same is also evident from the quantitative analysis of two recent waves of Pakistan Demographics and Health Surveys (PDHS), which highlighted that the incidence of ARIs has declined over a period of time among children under age 5 from 16% in 2012-13 to 14% in 2017-18. Most significantly, the behavior of community regarding seeking treatment or advice from a healthcare facilities and providers, including LHWs has also improved, showing an increase from 64% to 84% of children with ARI symptoms being brought to health facilities over time.



CONCLUSION

Better diagnosis and management of childhood Pneumonia is beneficial in reducing morbidity and mortality in the developing world, including Pakistan. Use of ARI Timers is simple, cost-effective and efficient tool to improve the diagnosis and management of Pneumonia. Evaluation findings are strongly suggestive of the scale up of the use of ARI Timers to uncovered areas of Punjab as well as other provinces and regions of Pakistan.

Bhutta, Z., Das, J., Hafeez, A., Rizvi, A., Ali, N., Khan, A., & Ahmad, F. et. al. (2013). Reproductive, maternal, newborn, and child health in Pakistan: challenges and opportunities. *The Lancet*, 381(9884), 2207-2218. Doi: 10.1016/s0140-6736(12)61999-0

National Institute of Population Studies (NIPS) [Pakistan] and ICF. 2019. Pakistan Demographic and Health Survey 2017-18. Islamabad, Pakistan, and Rockville, Maryland, USA: NIPS and ICF.

Midterm Evaluation of Project for Accelerating Policy Change, Translation and Implementation for Pneumonia and Diarrhea Commodities in Pakistan (2016-2020): Case Study

This is one of two case studies carried out as part of Midterm Evaluation of Unicef's Pneumonia and Diarrhea Project in Pakistan. The evaluation aimed to ascertain that the progress is made in the right direction to meet the project's ultimate results, documenting its overall achievement and lessons learned. The evidence generated is envisaged to navigate the scaling up and replication of these interventions to curtail childhood morbidity and mortality in the country.

Case Study 2: Use of Dispersible Tablets for improved management of Pneumonia and Diarrhea

PROBLEM STATEMENT

Pneumonia and Diarrhea are the two leading causes of death among children under 05 years of age. Each year, in Pakistan, about 91,000 children die from Pneumonia and 53,300 from Diarrhea (UNICEF, 2019). Before the introduction of UNICEF's intervention, healthcare providers at both public and private sector healthcare facilities relied on use of syrups and adult dosage antibiotic to treat childhood Pneumonia and Diarrhea (P&D). Findings of midterm evaluation revealed that caregivers often crushed, compressed and mixed adult tablets into liquid of varying concentrations in attempts to prepare pediatric formulation.

Zinc, an important micronutrient, is known to play a vital role in overall health and development of infants and young children. Its usage during Diarrheal episodes is well-established and both WHO and UNICEF recommend a daily dosage of 20 mg of zinc supplements for 10 days for children suffering with Diarrhea and 10 mg for infants under six months. Use of antibiotics is essential in treating childhood Pneumonia as WHO recommends a 250 mg Amoxicillin DT, to be taken twice daily for 3-5 days for its treatment.

APPROACH

UNICEF, through Bill and Malinda Gates Foundation (BMGF) Fund, introduced "Accelerating Policy Change, Translation and Implementation for Pneumonia and

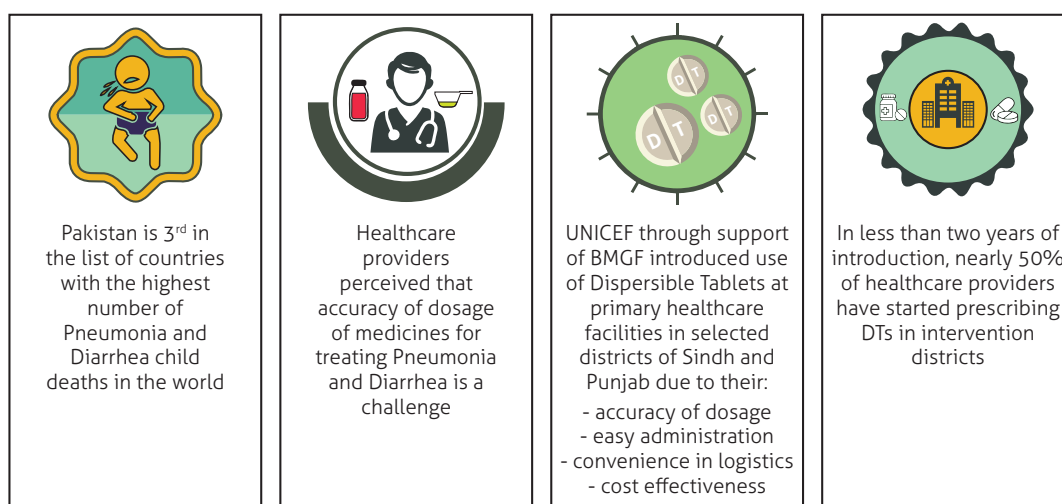
Diarrhea Commodities in Pakistan" program (2016-2020), to increase child survival through improved diagnosis and treatment of P&D in children under five years of age. Realising the critical role of Zinc and Amoxicillin in management of childhood P&D, the project promoted evidence based, high impact, low cost and lifesaving commodities, including Amoxicillin and Zinc Dispersible Tablets. Given the fact that DTs are more cost effective and provide accurate dosage of child friendly formulations, the project procured and distributed Zinc and Amoxicillin DTs to all the healthcare facilities in the project area.

DT can be easily dispersed in 5-10 ml of liquid preferably clean drinking water, thereby simplifying treatment administration. DTs are known to facilitate and simplify dose accuracy compared to liquid formulations, which have to be measured and mixed manually. DTs offer cost effective logistical and supply chain advantages in terms of volume and weight. Caregivers in rural community settings in low-resource countries have preferred DTs to syrups and tablets due to easy administration to treat P&D in children (UNICEF, n.d.). Findings of the evaluation revealed that care providers were of the view that dosing of liquids can be uncertain due to inaccuracy of spoon measurement and size, while on the contrary, DT dosage is more accurate as each dose requires the complete intake of one or half tablet. World Health Organization and Global Action Plan for Pneumonia and Diarrhea (GAPPD) strongly recommend Amoxicillin and Zinc DTs as effective first line treatment for under-five P&D compared to existing alternate treatments in countries with high caseloads of these illnesses.

OUTCOME

Based on the situation analysis and learning from best practices, UNICEF provided technical support to federal and provincial governments to procure and distribute DTs to ensure availability at primary level healthcare facilities and community level in target districts of Punjab and Sindh. Provincial manager informed, "Amoxicillin DT and Zinc DT have been made part of the Essential Medicine List." Amoxicillin & Zinc DT are now part and parcel of Minimum Service Delivery Package (MSDP) for the healthcare facilities in Punjab. Analysis revealed that UNICEF procured and supplied approximately 169,744 Amoxicillin DTs and Co-packaged Zinc DTs at ~2,458 primary healthcare facilities in Sindh and ~ 450 health facilities in Punjab. At community level, Zinc DTs were made available to LHWs in both Punjab and Sindh. Through the efforts of this project, Government has included Amoxicillin DTs and Zinc DTs for P&D in the procurement lists of healthcare facilities, ensuring the sustainability of these drugs at the healthcare facilities beyond the life of the project. The facility in-charges in both Punjab and Sindh discussed DTs, in terms of their availability, acceptability and practicality as a formulation in comparison with syrup. Dispersible Tablets have gained

popularity and Zinc DTs were present in majority of the facilities in Sindh and Punjab, and facility in-charges were aware of their benefits. Facility in-charge from Sindh shared their views on DTs benefits, "These are more effective than syrup, exact dosage is administered through them which is essential for recovery." DTs were perceived to not only provide appropriate management but also had additional benefits of accurate dosage and easy administration for caregivers with low literacy rates. Further, in support of evaluation findings, quantitative analysis of two recent Pakistan Demographic and Health Surveys reveal that there has been significant improvement regarding seeking treatment or advice from healthcare providers and front line workers. Statistics show that the practice of seeking advice or treatment for both Pneumonia and Diarrhea has significantly improved (61% in 2012-13 to 71% in 2017-18 for Diarrhea) and (64% in 2012-13 to 84% for ARI). Evidence also suggest that during the last five years, use of zinc for treatment of Diarrhea has increased significantly, i.e. 2% in 2012-13 to 13% in 2017-18. Thus, improvement in management of P&D is evident, which can be taken as an indication of positive impact of UNICEF' support and advocacy to warrant early and accurate treatment.



CONCLUSION

It is established fact that DTs are more efficient than syrup in terms of their preparation, similar plasma concentration, dosage, palatability, packaging, storage, handling, logistics and cost. The inclusion of DTs in essential medicines lists, procurement lists and Minimum Service Delivery Packages (MSDPs) are all contributing substantially in achieving the goal of healthy children and their survival in Pakistan. UNICEF's approach to promote and advocate for use of DTs in management of Pneumonia and Diarrhea are eventually expected to facilitate decline in childhood morbidity and mortality in Pakistan. The evaluation results recommend that in order to maximise this positive impact of DTs in better P&D management, UNICEF should take up two-pronged approach involving concurrent advocacy efforts to mitigate inherent barriers in local production and public procurement, and engagement of private providers through professional associations for change in prescribing behaviors.

UNICEF Pakistan (2019). Retrieved on August 15, 2019, from <https://www.unicef.org/pakistan/health>

UNICEF. (n.d). Dispersible tablets. Retrieved on August 21, 2019 from https://www.UNICEF.org/supply/files/Dispersible_Tablets.pdf

Annex 5: Guides for Interviews and Focus Group Discussions

Guide for Ministry of Health Service Regulation and Coordination and Provincial DoH

Before starting interview, remember to:

- ⇒ Introduce yourself and explain purpose of visit
- ⇒ Get formal consent for the interview and provide brief of study objectives
- ⇒ Ask for permission for audio recording of the interview

1. Do the national/provincial/ policies/strategies and guidelines incorporate the global recommendations for pneumonia and diarrhea?
2. Who are key stakeholders supporting the Government in P&D management?
3. Have the policies on community case management for pneumonia and diarrhea updated according to the global recommendations (World Health Organization (WHO)/Global Action Plan for Pneumonia and Diarrhea (GAPPD)?
 - a. Who has supported the Government in updating these guidelines (development partners, e.g. UNICEF)?
4. Has UNICEF provided any technical support to national and provincial governments for updating/revising the following in alignment with GAPPD:
 - a. Child survival related policies/strategies in terms of updating of management guidelines (IMNCI)
 - b. Essential medicines lists with inclusion of Amoxicillin DT, Zinc DT, LORS and Zinc
 - c. Minimum Service Delivery Package (MSDP)
 - i. Have these been endorsed by TWG, Child Survival and the Government through consensus?
 - ii. Was there any orientation carried out?
 - iii. Is it being implemented?
5. Has UNICEF's P&D project facilitated the Government in strengthening of logistics system and in updating the procurement lists (MSDP and PC-1s) including ARI Timers, Pulse Oximeters and Oxygen for appropriate levels of service delivery?
 - a. Have these been endorsed by TWG, Child Survival and the Government through consensus?
 - b. Was there any orientation carried out?
 - c. Is it being implemented? Has the supply chain management system updated according to the required commodities?
 - d. How is the Government ensuring supply of pneumonia & diarrhea commodities from DoH and district health offices downstream to facility levels?
6. Has UNICEF P&D project provided any support to Government in strengthening of comprehensive supply chain management system including forecasting, procurement, distribution, real time stock maintenance, and warehousing involving different programmes and their linkage with DHIS?

7. Has UNICEF P&D project provided any technical assistance to DoH/DHIS cell for updating of DHIS software and tools and implementing recommendations for its linking with LMIS and for:
 - a. Printing and dissemination of revised DHIS tools
 - b. Orientation of master trainers (provincial level) and relevant health facility staff (selected district level) on revised DHIS tools
8. What coordination mechanism exist for management of diarrhea and pneumonia among under five children in Pakistan at national level, e.g. Child Survival Groups, Technical Working Group
9. Are the community IMNCI training manuals revised? Who was responsible for their revision? Has UNICEF provided technical assistance for the revision?
 - a. Have these been endorsed by TWG, Child Survival and the Government through consensus?
 - b. Was there any orientation carried out to endorse the revised community IMNCI Training Modules?
10. Have the treatment protocols for pneumonia and diarrhea updated according to new recommendations (EML, IMNCI, MSDP and Procurement lists etc.)?
 - a. Are job-aids, reporting tools and training material for healthcare providers revised accordingly?
 - b. What plans are there for refresher training of health workers (facility and community) that will be an opportunity to introduce new treatment protocols?
 - c. Has UNICEF facilitated the Government in the process?
11. What are the structural nonstructural factors that affect policy change, its translation and subsequent implementation with reference to pneumonia and diarrhea commodities?
12. Have the policy changes been implemented in consideration of Gender and HRB approaches? Any example
13. How will Government ensure the sustainability of these policy changes?
14. How can the policy translation be achieved effectively?
15. Who are potential key stakeholders that can institute a policy change?
16. What are some of the barriers that are hindering the policy level processes?
17. What are some of the facilitators that act as a catalyst for the desired change?

Any suggestions for improvement in UNICEF P&D Project

Guide for LHW Coordinator

Before starting interview, remember to:

- ⇒ Introduce yourself and explain purpose of visit
- ⇒ Get formal consent for the interview and provide brief of study objectives
- ⇒ Ask for permission for audio recording of the interview

1. First I'd like to ask you about the role of LHWs and LHSs in reducing morbidity and mortality associated with Pneumonia and Diarrhea in children.
 - What are the provincial policies and guidelines for LHW Programme to deliver child health services?
 - In your opinion, how adequate are these policies/guidelines to ensure the availability of services, particularly for the management of Pneumonia and Diarrhea?
2. To what extent the programme is sustainable after regularization of LHWs and provincialization of the programme?
 - Probe about financial sustainability of the programme in relation to further expansion and improvement of existing quality of services
3. What is the status/coverage of LHWs and LHSs in your province?
 - Is required number of LHWs present to cover your entire province? If not, what are the reasons for non-availability of LHWs?
 - Is the required number of LHSs available to supervise the LHWs?
 - In your opinion, how availability of LHSs and LHWs can be improved in your province?
4. What do you think about the competence of LHSs and LHWs to manage diarrhea and pneumonia in children at household level and for referral of complicated cases?
 - What is routine mechanism for in-service training and capacity building of LHSs and LHWs in your province?
 - Is there any specific training for prevention and treatment of Pneumonia and Diarrhea, like IMCI or any other training?
 - If you don't have in built mechanism of in-service training in your province, what else have you done for this purpose?
5. What is the current status of availability of commodities and equipment for child health services, particularly for diagnosis and treatment of Pneumonia and Diarrhea?
 - Are sufficient amount of antipyretics, antibiotics (particularly Amoxicillin), L-ORS and Zinc are provided to the LHSs and LHWs?
 - In what formulation are these commodities usually provided? Probe for DT?
 - What are the common reasons for stock-outs of these commodities?
 - Are any specific equipment, like ARI timers/pulse oximeter, provided to the LHWs for better managing the cases of Pneumonia and Diarrhea?
 - In your opinion, how can the uninterrupted supply of these commodities be ensured for LHSs and LHWs?
6. Can you put some light on LHW-MIS?
 - What do you think about the sufficiency of LHW-MIS tools for recording and reporting regarding management of Pneumonia and Diarrhea?
 - Do these reports provide adequate cover and indicate the shortage of commodities?
 - Have you made any decisions regarding management of Pneumonia and Diarrhea on the basis of the reports generated at provincial level? Kindly give some examples.
 - How can the recording and reporting of LHSs and LHWs' services be improved?

7. What are your in built mechanisms for monitoring and supervision of LHWSs and LHWs?
 - What is the current supervision policy? At which level responsibility of supervision is fixed?
 - Is the supervision integrated or service specific e.g. for EPI, CMWs, LHWs, etc.?
 - What is the role of Lady Health Supervisor regarding supervision and ensuring uninterrupted supply of commodities for LHWs?
 - Is there any notified schedule for supervisory visits to the community-based workers?
 - What are your suggestions to improve the monitoring and supervision of child health services in your province?
8. What do you think about the integration of the LHWs with other health programmes, local community, key persons, Community Based Organizations (CBOs) or elected representatives?
 - What is the level integration of LHWs, LHS and CMWs in management of Pneumonia and Diarrhea in children?
 - What role do they play for acceptance of health services provided by LHWs? Kindly give some examples about the support they support.
 - What is usefulness of local community liaison?
9. What is your overall impression about the scope and scale of Pneumonia and Diarrhea case management by the LHSs and LHWs (diagnosis, treatment and referral services) in your province?
 - If you are asked to take measures for improvement of Pneumonia and Diarrhea case management services by LHSs and LHWs, what measures will you suggest at managerial, facility and community level?

Guide for Unicef Team Interview

Before starting interview, remember to:

- ⇒ Introduce yourself and explain purpose of visit
- ⇒ Get formal consent for the interview and provide brief of study objectives
- ⇒ Ask for permission for audio recording of the interview

1-What measures have been taken to ensure that the existing national/provincial policies and guidelines are updated in line with global recommendations (World Health Organization (WHO)/Global Action Plan for Pneumonia and Diarrhea (GAPPD)) for management of diarrhea and pneumonia among under five children in Pakistan?

a-Is there a Joint accountability framework and workplan with timelines under the stewardship of national and provincial Child Survival Groups?

- Is there a Coordination mechanism in the selected province and at national/provincial levels?
- What measures have you taken to ensure appropriate levels of service delivery for P&D commodities?
- Is there any kind of technical support to national and provincial governments I aligning theses to GAPPD & in line with recommendations of landscape analysis?

b- Do National, provincial, and district essential medicines and procurement lists include Amoxicillin DT, co-packaged LoORS & Zinc DT, ARI Timers, Pulse Oximeters and Oxygen?

c- Is there any kind of technical support to national and provincial governments for updating

management guidelines (IMNCI), essential medicines lists, Minimum Service Delivery Package (MSDP)?

- Did you develop/print evidence-based briefing papers/policy notes on benefits of Amoxicillin DT, Zinc DT, co-packaged ORS and Zinc, Pulse Oximeters, ARI timer and oxygen in management of Pneumonia and Diarrhea?
- Do you have National & provincial consultations with child survival groups and other relevant stakeholders to review and endorse briefing papers/policy notes to help build consensus on the recommendations for strengthening of logistics system and updated EMLs & procurement lists?

2-Has there been a translation of revised and updated pneumonia and diarrhea treatment guidelines into relevant action plans by all provincial/areas health departments in Pakistan? If yes:

a-To what extent did the utilization of revised National Pneumonia & Diarrhea management/IMNCI guidelines help improve knowledge, skills, and capacities of Healthcare workers at national/provincial levels?

- Technical assistance to draft the revised community IMNCI training manuals?
- Dissemination of IMNCI Guidelines and training modules with the public and private sectors?
- Has there been any kind of orientation on updated EMLs, MSDPs, and procurement lists

b-Are concerned national, provincial, and district level policy and decision makers sensitized on necessary resource allocation for management of childhood pneumonia and diarrhea?

c-

3-In order to ensure availability of essential commodities (Amoxicillin DT, Zinc DT, co-packaged ORS and Zinc, Oxygen, ARI Timers and Pulse Oximeters) for treatment of childhood pneumonia and diarrhea in Pakistan, can you please explain:

- Logistics, procurement, and supply chain management systems updated and strengthened to include recommended commodities for pneumonia and diarrhea management?
- Phased procurement and supply of pneumonia and diarrhea commodities for selected provinces/districts

4-what measures are required to translate lessons learned from this project to other geographical scopes within Pakistan?

- What actions are necessary to support and strengthen the DHIS2 and link it with LMIS?
- What measures are necessary to improve advocacy for local production and manufacturing of these commodities?
- How can Unicef, advocate with the government for adequate budgetary allocation, while increasing procurement of newly recommended commodities?
- How can the capacity of health service providers on health information, supply chain management and use of data collection tools be built/enhanced?

Guide for Provincial DHIS Official

Before starting interview, remember to:

- ⇒ Introduce yourself and explain purpose of visit
- ⇒ Get formal consent for the interview and provide brief of study objectives
- ⇒ Ask for permission for audio recording of the interview

1 Can share something about routine MIS? What is the current status of DHIS?

- *What do you think about the sufficiency of DHIS tools for recording and reporting regarding management of Pneumonia and Diarrhea? If yes,*
- *Do these reports provide adequate cover and indicate the shortage of commodities?*
- *If no, Do you have any other system?*

2 Have you made any decisions regarding management of Pneumonia and Diarrhea on the basis of the reports generated from the district level? If yes Kindly give some examples.?

- *How can the recording and reporting of services regarding management of Pneumonia and Diarrhea be improved?*

3 Now, we would like to know about the process of forecasting and quantification of medicines and supplies at your facility?

Probes:

- *Is it based on burden of disease from the catchment area or based on data from DHIS?*
- *Are the supplies provided according to the demands submitted/ need based? If no, kindly explain reasons for under-supply.*

4 Kindly comment on whether the supply of new commodities like Zinc DT and Low Osmolarity ORS is according to the generated demand?

- *Have you or other staff member received any formal training on forecasting and quantification of medicines and supplies?*

5 In your opinion, what is the use of DHIS in context of management of children suffering from Pneumonia and Diarrhea?

Probe:

- *Are you satisfied with the quality (accuracy of information, completeness and timeliness) of DHIS recording and reporting? If not, how can we improve it?*
- *Does DHIS help keeping the record of referrals made in complicated cases?*

6 Do you think that current DHIS indicators sufficiently cover the data requirements for managing Pneumonia and Diarrhea? Do you suggest to include some indicators?

- *Are any indicators related to new commodities included in the DHIS? If yes, does the facility staff felt any difficulty in using the tools after these changes?*
- *Can you comment on the capacities of staff on using DHIS tools? If there are any gaps, how can these be addressed?*

7 Has UNICEF P&D project provided any support to Government in strengthening of comprehensive supply chain management system including forecasting, procurement, distribution, real time stock maintenance, and warehousing involving different programmes and their linkage with DHIS?

8 Has DHIS been updated to include any relevant pneumonia and diarrhea treatment /commodity indicators and linked with LMIS?

9. Has revision of LHWs MIS tools for reporting on updated commodities for management of pneumonia and diarrhea taken place in line with GAPPD as per project requirement?

10. Has the revision, updation and printing of DHIS tools and manuals and software in Sindh taken place?

11. Kindly provide any suggestions for improving DHIS that would facilitate the management of pneumonia and diarrhea?

CSG Member

Before starting interview, remember to:

- ⇒ Introduce yourself and explain purpose of visit
- ⇒ Get formal consent for the interview and provide brief of evaluation objectives
- ⇒ Ask for permission for audio recording of the interview

During the initial discussion to explain the evaluation, the interviewer and participant should have had the opportunity to introduce themselves and to make small talk to break the ice. You can begin by saying these words to the group of participants:

'As you know Pneumonia and Diarrhea are common causes of deaths among children under age 5. We are conducting a study to learn about of services being provided to treat and manage these diseases among children and would like to know your views in this regards. We

will start when you are ready, will listen to your view points and ask few questions to clarify some of the things you would talk about.' May we begin?

The guideline contains multiple questions and each question is followed by a set of probes. Please let participants initiate the discussion and use probes only when certain concerns are not addressed by the group

1. What is your role in CSG? What is the role of CSG?
2. What has CSG quarterly meetings contributed in improving Under-5 childhood mortality and morbidity? any specific contribution in improving pneumonia and diarrhea services?
3. What has the overall role of CSG been in improving overall Under-5 childhood mortality and morbidity? any specific role in making pneumonia and diarrhea services better?
4. Do you believe that CSG has been meeting its objectives? What have been the challenges in this achieving these objectives?
5. Does Child Survival Group play any sort of coordination mechanism role for management of diarrhea and pneumonia among under five children in Pakistan at national level?
6. Have any interventions taken place under this Child Survival Group?
7. In your opinion, how can CSG contribute in a more effective manner in improving pneumonia and diarrhea services?
8. What support would CSG require in making this contribution from the Government?
9. What support would CSG require in making this contribution from the development partners and donors? Has the support from Unicef been satisfactory?
10. In your opinion what should be done to improve the quality of pneumonia and diarrhea services at various levels in the health system (policy, district and community levels)?

Guides for Implementing Partners

Before starting interview, remember to:

- ⇒ Introduce yourself and explain purpose of visit
- ⇒ Get formal consent for the interview and provide brief of evaluation objectives
- ⇒ Ask for permission for audio recording of the interview

1 Do the national/provincial/ policies/strategies and guidelines incorporate the global recommendations for pneumonia and diarrhea? Are these guidelines being implemented at your facilities?

2 Have the policies on community case management for pneumonia and diarrhea updated according to the global recommendations (World Health Organization (WHO)/Global Action Plan for Pneumonia and Diarrhea (GAPPD)?

3 Has UNICEF provided any technical support to national and provincial governments for updating/revising the following in alignment with GAPPD:

- a. Child survival related policies/strategies in terms of updating of management guidelines (IMNCI)
- b. Essential medicines lists with inclusion of Amoxicillin DT, Zinc DT, LORS and Zinc
- c. Minimum Service Delivery Package (MSDP)
 - i. Have these been endorsed by TWG, Child Survival and the Government through consensus?
 - ii. Was there any orientation carried out?
 - iii. Is it being implemented?

4 Has UNICEF's P&D project facilitated the Government in strengthening of logistics system and in updating the procurement lists (MSDP and PC-1s) including ARI Timers, Pulse Oximeters and Oxygen for appropriate levels of service delivery?

- d. Have these been endorsed by TWG, Child Survival and the Government through consensus?
- e. Was there any orientation carried out?
- f. Is it being implemented? Has the supply chain management system updated according to the required commodities?
- g. How is the Government ensuring supply of pneumonia & diarrhea commodities from DoH and district health offices downstream to facility levels?

5 Has UNICEF P&D project provided any support in strengthening of comprehensive supply chain management system including forecasting, procurement, distribution, real time stock maintenance, and warehousing involving different programmes and their linkage with DHIS?

6 UNICEF P&D project provided any technical assistance to DoH/DHIS cell for updating of DHIS software and tools and implementing recommendations for its linking with LMIS and for:

- h. Printing and dissemination of revised DHIS tools
- i. Orientation of master trainers (provincial level) and relevant health facility staff (selected district level) on revised DHIS tools

7 What coordination mechanism exist for management of diarrhea and pneumonia among under five children in Pakistan at national level, e.g. Child Survival Groups, Technical Working Group?

8. Are the community IMNCI training manuals revised? Who was responsible for their revision? Has UNICEF provided technical assistance for the revision?

- c. Have these been endorsed by Technical Working Group, Child Survival Group or/and the Government through consensus?
- d. Was there any orientation carried out to endorse the revised community IMNCI Training Modules?

9 Have the treatment protocols for pneumonia and diarrhea updated according to new recommendations (EML, IMNCI, MSDP and Procurement lists etc.)?

- d. Are job-aids, reporting tools and training material for healthcare providers revised accordingly?
- e. What plans are there for refresher training of health workers (facility and community) that will be an opportunity to introduce new treatment protocols?
- f. Has UNICEF facilitated the Government in the process?

10. What are the structural and non-structural factors that affect policy change, its translation and subsequent implementation with reference to pneumonia and diarrhea commodities?

11. How can the policy translation be achieved effectively?

12. What are some of the barriers that are hindering the policy level processes?

13. What are some of the facilitators that act as a catalyst for the desired change?

14. Any suggestions for improvement in UNICEF P&D Project?

PPA Member

Before starting interview, remember to:

- ⇒ Introduce yourself and explain purpose of visit
- ⇒ Get formal consent for the interview and provide brief of evaluation objectives
- ⇒ Ask for permission for audio recording of the interview

During the initial discussion to explain the evaluation, the interviewer and participant should have had the opportunity to introduce themselves and to make small talk to break the ice. You can begin by saying these words to the group of participants:

'As you know Pneumonia and Diarrhea are common causes of deaths among children under age 5. We are conducting a study to learn about of services being provided to treat and manage these diseases among children and would like to know your views in this regards. We will start when you are ready, will listen to your view points and ask few questions to clarify some of the things you would talk about.' May we begin?

The guideline contains multiple questions and each question is followed by a set of probes. Please let participants initiate the discussion and use probes only when certain concerns are not addressed by the group

Q1 a) What is your role in PPA? What has PPA's role been in improving Under-5 childhood mortality and morbidity? Any particular improvement in treating pneumonia and diarrhea?

b) Has PPA conducted or contributed to any study related to Under-5 child mortality and morbidity? Any particular research on pneumonia and diarrhea?

c) In your opinion how can PPA make a significant contribution in improving pneumonia and diarrhea services?

d) What support would PPA require in making this contribution from the Government?

e) What support would PPA require in making this contribution from the development partners and donors?

2-I'd like to know about main childhood related illnesses diagnosed in your clinic/hospital. Can you please comment on the caseload of Pneumonia and Diarrhea?

Probes:

- *About what proportion of children (less than 5 years) is brought to your clinic/hospital that suffer from pneumonia and diarrhea?*
- *What is your view about the major causes of spread of these diseases and how can this be prevented?*
- *Do you think male and female children equally seek treatment for these illnesses? Do you think any disparity exists when parents seek treatment for their sons and daughters?*

3- I would like to know about your skills/competence for managing childhood illnesses? Can you please tell us about any training(s) received since you have been practicing for dealing such diseases? In your view, what can be done to improve quality of training?

4- Now, I would like to discuss your awareness about protocols and guidelines (like IMNCI guidelines) for managing pneumonia and diarrhea in children at your clinic/hospital. Are these protocols and guidelines followed?

Probes:

- *Can you please explain their usefulness in diagnosing and treating childhood illness?*
- *If no such specific protocols/guidelines are available and being implemented, can you please comment on the reasons?*

5- Now, we will like to know about the diagnosis of pneumonia in your clinic/hospital? Please tell us how do you diagnose children suffering from Pneumonia?

Probes:

- *Do you think any particular equipment (ARI Timers, Oxygen cylinders and Pulse Oximeters) is essential for the diagnosis of pneumonia? If there is any deficiency, please explain reasons for shortage of equipment.*
- *How commonly antibiotics are used for the treatment of pneumonia?*
- *Do you think there is any role and advantage of using oxygen in treating pneumonia? Is it being used at your clinic/hospital? If not, can you explain the reasons?*
- *In your opinion, how deficiencies in equipment can be address?*

6- Can you please tell us how do you manage children suffering from Diarrhea at your clinic/hospital?

Probes:

- *How do you diagnose childhood diarrhea on the basis of severity of illness and do you differentiate it from dysentery?*
- *What options do you have to treat childhood diarrhea with oral medications?*
- *Is there any role of oral Zinc and low Osmolarity ORS in its treatment? If so, what type of Zinc formulation do you usually prescribe?*
- *Do you think there is any advantage of using Low Osmolarity ORS compared to regular ORS in treating Diarrhea?*
- *What do you think is the reason for their acceptability/ non-acceptability?*
- *Do you have new commodities like Zinc DT and Low Osmolarity ORS in your clinic/hospital?*
- *If yes, do you prefer prescribing new commodities like Zinc DT and Low Osmolarity ORS in your clinic/hospital over the other ones?*
- *Do you think there is a difference in results in recovery process using new commodities as compared to using syrup and regular ORS?*
- *In your opinion, what is the beneficiaries' reaction to using dispersible tablets and LO ORS? Do you think they are acceptable?*

7-Now, I would like you to comment on the cost of treating Pneumonia and Diarrhea?

- *In your opinion, does cost play a significant influence over the choice of medicines prescribed and service provider?*
- *If you are prescribing new commodities, do you think there has been any change in cost of treatment per patient?*
- *How commonly do you prescribe medicines related to the treatment of Pneumonia and Diarrhea to be bought from market in case of its stock-out?*

8- In your opinion what should be done to improve the quality of pneumonia and diarrhea services in the entire health system?

INTERVIEW GUIDE FOR DISTRICT MANAGERS

CEO-DHA/EDOH/DHO

Before starting interview, remember to:

- ⇒ **Introduce yourself and explain purpose of visit**
- ⇒ **Get formal consent for the interview and provide brief of evaluation objectives**
- ⇒ **Ask for permission for audio recording of the interview**

1. First I'd like to ask you about the provincial guidelines/policies to address the challenges related to child health services.
 - What autonomy district health department/authority has got to make its own policies?
2. Do you have your district strategic or operational health plans? Do these plans include district

targets for child health services, particularly management of Pneumonia and Diarrhea?

- What is the situation of services for Pneumonia and Diarrhea in your district?
 - What initiatives district has taken for improvement of these services?
3. What is the status of human resource related to Pneumonia and Diarrhea, in primary level health facilities in your district?
 - What are care providers related issues for delivery of these services?
 - What initiatives government has taken to retain the staff on health facilities?
 - How do you address the absenteeism?
 - If not, what are the reasons for unavailability of staff?
 - In your opinion, how can situation of HR improve at public sector health facilities?
 4. We are interested to learn about community-based healthcare providers (LHW/CMW).
 - What is their role in provision of Pneumonia and Diarrhea management services, regarding diagnosis, treatment and referral?
 - What is the quality of Pneumonia and Diarrhea management services provided by them?
 - What are your suggestions to further improve their services?
 5. What do you think about the competence level of staff at primary level facilities and community based staff to manage diarrhea and pneumonia in children?
 - What is your routine mechanism for in-service training and capacity building of these staff in your district?
 - Do you have any in-service training programme for staff at primary level health facilities and community?
 - Any specific training for Pneumonia and Diarrhea care provider, like IMNCI or any other training?
 - If you don't have in built mechanism of in-service training in your district what else you do for this purpose?
 6. What is the current status of facility resources (building, equipment and supplies) for child health services, particularly Pneumonia and Diarrhea?
 - Are sufficient amount of antibiotics (particularly Amoxicillin, L-ORS and Zinc) provided to the primary health facilities and community-based service providers?
 - What are the common reasons for stock-outs of these medicines and supplies?
 - Are any specific equipment, like ARI timers and pulse oximeters, provided to the service providers for better managing cases of Pneumonia and Diarrhea?
 - Are all your primary level health facilities provided with oxygen supply for managing childhood Pneumonia?
 - In your opinion, how can the availability of medicines, supplies and equipment be ensured at the primary health facilities and the community-based workers?
 7. Can you throw some light on your routine MIS? What is the current status of DHIS?
 - What do you think about the sufficiency of DHIS tools for recording and reporting regarding management of Pneumonia and Diarrhea?
 - Do these reports provide adequate cover and indicate the shortage of commodities?
 - Have you made any decisions regarding management of Pneumonia and Diarrhea on the basis of the reports generated at district level? Kindly give some examples.
 - How can the recording and reporting of services regarding management of Pneumonia and Diarrhea be improved?

8. What are your inbuilt mechanisms for monitoring and supervision of facility and community based staff?
 - What is the current supervision policy? At which level, is responsibility of supervision fixed?
 - Is the supervision integrated or service specific e.g. for EPI, CMWs, LHWs, etc.?
 - Is there any notified district schedule for supervisory visits to the primary level facilities and community based workers?
 - What are your suggestions to improve the monitoring and supervision of child health services in your district?
9. How frequently do you conduct evaluation to monitor the achievement of your facilities?
 - Have you generated or published any report/s of your evaluations. Kindly give some examples.
10. Do you have liaison with the local community, key persons, Community Based Organizations (CBOs) or elected representatives of the district?
 - What role do they play for acceptance of health services provided by you? Kindly give some examples about the support they support.
 - What is usefulness of local community liaison?
11. What is your user-charges/fee policy for services, particularly services of pneumonia and diarrhea?
 - What are your suggestions to minimize the cost as a barrier for underutilization of these services?
12. What is your impression of current utilization of Pneumonia and Diarrhea management services at primary level health facilities and in the community? How do you rate it as under, normal or over utilization?
 - If you are asked to take measures at various levels for improvement of Pneumonia and Diarrhea management services, what measures will you suggest at managerial, facility and community level?

INTERVIEW GUIDES OF DISTRICT IRMNCH COORDINATOR

District IRMNCH Coordinator

Before starting interview, remember to :

- ⇒ **Introduce yourself and explain purpose of visit**
- ⇒ **Get formal consent for the interview and provide brief of evaluation objectives**
- ⇒ **Ask for permission for audio recording of the interview**

10. First I'd like to ask you about the role of LHWs in reducing morbidity and mortality associated with Pneumonia and Diarrhea in children.
 - What are the policies and guidelines for LHW Programme to deliver child health services?
 - In your opinion, how adequate are these policies/guidelines to ensure the availability of services, particularly for the management of Pneumonia and Diarrhea?
11. To what extent the programme is sustainable after regularization of LHWs and provincilization of the programme
 - Probe about financial sustainability of the programme in relation to further expansion and improvement of existing quality of services
12. What is the status/coverage of LHWs in your district?
 - Is required number of LHWs present to cover your entire district? If not, what are the reasons for unavailability of LHWs?
 - In your opinion, how availability of LHWs can be improved in your district?
13. What do you think about the competence of LHWs to manage diarrhea and pneumonia in children at household level and for referral of complicated cases?
 - What is routine mechanism for in-service training and capacity building of LHWs in your district?
 - Is there any specific training for prevention and treatment of Pneumonia and Diarrhea, like IMCI or any other training?
 - If you don't have in built mechanism of in-service training in your district, what else have you done for this purpose?
14. What is the current status of availability of commodities and equipment for child health services, particularly for diagnosis and treatment of Pneumonia and Diarrhea?
 - Are sufficient amount of antipyretics, antibiotics (particularly Amoxicillin), L-ORS and Zinc are provided to the LHWs?
 - In what formulation are these commodities usually provided? Probe for DT?
 - What are the common reasons for stock-outs of these commodities?
 - Are any specific equipment, like ARI timers/pulse oximeter, provided to the LHWs for better managing the cases of Pneumonia and Diarrhea?
 - In your opinion, how can the uninterrupted supply of these commodities be ensured for LHWs?
15. Can you throw some light on LHW-MIS?
 - What do you think about the sufficiency of LHW-MIS tools for recording and reporting regarding management of Pneumonia and Diarrhea?
 - Do these reports provide adequate cover and indicate the shortage of commodities?
 - Have you made any decisions regarding management of Pneumonia and Diarrhea on the

basis of the reports generated at district level? Kindly give some examples.

- How can the recording and reporting of LHWs' services be improved?

16. What are your in built mechanisms for monitoring and supervision of LHWs?

- What is the current supervision policy? At which level responsibility of supervision is fixed?
- Is the supervision integrated or service specific e.g. for EPI, CMWs, LHWs, etc.?
- What is the role of Lady Health Supervisor regarding supervision and ensuring uninterrupted supply of commodities for LHWs?
- Is there any notified district schedule for supervisory visits to the community-based workers?
- What are your suggestions to improve the monitoring and supervision of child health services in your district?

17. What do you think about the integration of the LHWs with other health programmes, local community, key persons, Community Based Organizations (CBOs) or elected representatives?

- What is the level integration of LHWs, LHS and CMWs in management of Pneumonia and Diarrhea in children?
- What role do they play for acceptance of health services provided by LHWs? Kindly give some examples about the support they support.
- What is usefulness of local community liaison?

18. What is your overall impression about the scope and scale of Pneumonia and Diarrhea case management by the LHWs (diagnosis, treatment and referral services) in your district?

- If you are asked to take measures for improvement of Pneumonia and Diarrhea case management services by LHWs, what measures will you suggest at managerial, facility and community level?

INTERVIEW GUIDE FOR FACILITY IN-CHARGE (BHU & RHC)

CHECKLIST FOR AVAILABILITY OF COMMODITIES FOR MANAGEMENT OF < 5 PNEUMONIA & DIARRHEA CASES

A	Trainings/Refreshers for Management of Pneumonia & Diarrhea (Write the number of staff who received these trainings)						
	Staff trained during past 24 months	No of Staff	IM CI	ARI	CD D	Breast Feeding/ Micronutrient Deficiency	EPI/ Cold Chain Management
A1	Consultant						
A2	SMO/MO						
A3	WMO						
A4	Staff Nurse						
A5	Health Technician						
IMCI (Integrated Management of Childhood Diseases) ARI (Acute Respiratory Infection) CDD (Control of Diarrhoeal Diseases) EPI (Extended Programme On Immunization)							
B	Standard Guidelines for Diagnosis & Management of Pneumonia & Diarrhea (Mark ✓ for available and not available)						

	Guidelines	Available	Not Available
B1	Guidelines for Pneumonia/ARI case management		
B2	Guidelines for Diarrhea/Dysentery Case Management		
B3	Guidelines on Storage of Drugs		
B4	Guidelines for Cold Chain Management		
B5	EPI Guidelines		

C	Essentials for Pneumonia & Diarrhea Case Management (Mark ✓ for yes or No)				
	Commodities	Available		Functional	
		Yes	No	Yes	No
C1	Oxygen cylinder				
C2	Oxygen flow meter				
C3	Pulse oximeter				
C4	ARI timer				
C5	Ambo bag (For Child)				
C6	Nebulizer				
C7	ILR to maintain cold chain				
C8	Supplies to mix and dispense Low-Osmolarity ORS (e.g. cup)				
C9	Source of clean drinking water				
C10	Ambulance				
• ILR (Ice Lined Refrigerator) • LORS (Low-Osmolarity Oral Rehydration Solution)					

D	Drugs for Pneumonia and Diarrhea Treatment (Mark ✓ for available and not available status of drugs but for stock outs write the number of months for which medicine was not available)			
	Drugs for Pneumonia and Diarrhea Treatment	Current status		Stock Outs * (July 2018 - June 2019)
		Available	Not Available/Expired	
D1	Syrup Amoxicillin			
D2	Amoxicillin Dispersible Tablets (DT)			
D3	Injection Amoxicillin			

D	Drugs for Pneumonia and Diarrhea Treatment (Mark ✓ for available and not available status of drugs but for stock outs write the number of months for which medicine was not available)			
	Drugs for Pneumonia and Diarrhea Treatment	Current status		Stock Outs * (July 2018 - June 2019)
		Available	Not Available/Expired	
D4	Pneumococcal Vaccine			
D5	Syrup Zinc			
D6	Zinc Dispersible Tablet (DT*)			
D7	LORS packets			
D8	LORS-Zinc Co-packaging			
D9	Ringer lactate Solution			
D10	0.45% Normal Saline/ Dextrose Saline			
• DT (Dispersible tablet) • Note: Please check the Stock register (medicines/supplies) for stock outs and write number of months for mentioned period				

E	Record keeping and Reporting (DHIS) (Mark ✓ for yes or No)				
	Material	Available		Updated	
		Yes	No	Yes	No
E1	OPD Register				
E2	In-patients Register (RHC)				
E3	Stock Register (Medicine/Supplies)				
E4	Stock Register (Equipment/Furniture)				
E5	PHC Facility Monthly Report Form				
E6	Procedures Manual for DHIS/MIS				
• OPD (Out-patient Department) • RHC (Rural Health Centre) • PHC (Primary Health Care) • DHIS (District Health Information System)					

1. I'd like to know about main childhood related illnesses in the catchment area of your facility. Can you please comment on the caseload of Pneumonia and Diarrhea?
Probes:

- i. *About what proportion of children (less than 5 years) is brought to this facility that suffers from Pneumonia and Diarrhea?*
 - ii. *What type of treatment approach or professional help is commonly preferred in the community for these diseases (e.g. self-treatment, traditional healer, homeopathic, allopathic)?*
2. I would like to know about the skill/competence of the staff at your facility for managing childhood illnesses? Can you please tell us about any training(s) received during the service for dealing such diseases?

Probes:

- i. *What in your view is the importance and benefits of in-service trainings for enhancing skill/competence to manage childhood illness, particularly Pneumonia and Diarrhea?*
 - ii. *When was the last time you or any of your staff member received such training? If no, what could be the reason for not arranging such trainings?*
 - iii. *Who usually arrange such trainings (like Health Department, development partners e.g. UNICEF or pharmaceutical companies)?*
 - iv. *Are these trainings conducted at the facility or some outside venue? What is your preference for the venue of such trainings and why?*
 - v. *What deficiencies did you notice in these trainings? In your view, how can these be improved?*
 - vi. *If you have received training, did any refresher follow it? If no, what could be the reason for not providing refresher training?*
3. Now, I would like to know about the use of protocols and guidelines (like IMCI guidelines) for managing pneumonia and diarrhea in children at your facility. What are these protocols and guidelines, kindly explain?

Probes:

- i. *Can you please explain their usefulness in diagnosing and treating childhood illness?*
 - ii. *If no such specific protocols/guidelines are available, can you please comment on the reasons?*
4. Now, we will like to know about the diagnosis and management of Pneumonia at your facility? Please tell us how do you diagnose children suffering from Pneumonia?

Probes:

- i. *How do you diagnose and classify Pneumonia on the basis of its severity and does its treatment also varies accordingly? Kindly explain.*
 - ii. *What is your prescription practice?*
 - iii. *Do you think any particular equipment is essential for the diagnosis of pneumonia? If there is any deficiency, please explain reasons for shortage of equipment.*
 - iv. *How commonly antibiotics are used for treatment of Pneumonia?*
 - v. *Have you ever heard of any dispersible (soluble in water) antibiotic tablet? If so, do you perceive any advantage of dispersible tablets over syrups?*
 - vi. *In your opinion, what is the perception of the community regarding use of dispersible tablets?*
 - vii. *Do you think there is any role and advantage of using oxygen in treating pneumonia? Is it being used at your facility? If not, can you explain the reasons?*
 - viii. *In your opinion, how deficiencies in equipment and medicines can be addressed?*

- ix. *In case of complicated Pneumonia cases, what referral mechanisms exist for transferring these children to higher-level health facilities?*
5. Can you please tell us how do you manage children suffering from Diarrhea at your facility?
Probes:
- i. *How do you diagnose childhood diarrhea and do you differentiate it from dysentery?*
 - ii. *Now focusing on diarrhea, do you classify patients on the basis of severity of dehydration? If so, how?*
 - iii. *What options do you have to treat childhood diarrhea with oral medications? What is your prescription practice?*
 - iv. *Is there any role of oral Zinc in its treatment? If so, what type of Zinc formulation is usually provided at your facility?*
 - v. *Have you ever heard of any dispersible (soluble in water) tablets of Zinc? If so, do you perceive any advantage of dispersible tablets over syrups?*
 - vi. *In your opinion, how would community take the idea of using dispersible tablets?*
 - vii. *Do you think there is any advantage of using Low Osmolarity ORS compared to regular ORS in treating Diarrhea?*
 - viii. *Is your facility provided with Low Osmolarity ORS? If no, please state reasons?*
 - ix. *In case of complications, what referral mechanisms exist for transferring these children to higher-level health facilities?*
6. Please tell us about any stock-out of aforementioned medicines that are used for treating Pneumonia and Diarrhea?
Probes:
- i. *How common are these stock-outs? Kindly comment on the medicines that suffer from stock-out more frequently.*
 - ii. *How do you treat these children, when there is stock-out of medicines at the facility?*
 - iii. *In your opinion, what are the reasons for these stock-outs?*
 - iv. *Do you have suggestions for ensuring un-interrupted supply of these medicines and commodities?*
7. Now, we would like to know about the process of forecasting and quantification of medicines and supplies at your facility?
Probes:
- i. *Is it based on burden of disease in your catchment area or based on data from DHIS?*
 - ii. *Are the supplies provided according to the demands submitted? If no, kindly explain reasons for under-supply.*
 - iii. *Have you or other staff member received any formal training on forecasting and quantification of medicines and supplies?*
 - iv. *What do you suggest for improving the process of forecasting and quantification to overcome stock-outs?*
8. In your opinion, what is the use of DHIS in context of management of children suffering from Pneumonia and Diarrhea?
Probe:
- i. *Are you satisfied with the quality (accuracy of information, completeness and timeliness) of DHIS recording and reporting? If not, how can we improve it?*
 - ii. *Do you find any duplication of reporting (like Disease Surveillance System and DHIS) that may affect the decision-making?*

- iii. *Does DHIS help keeping the record of referrals made in complicated cases?*
 - iv. *Do you think that current DHIS indicators sufficiently cover the data requirements for managing Pneumonia and Diarrhea? Do you suggest to include some indicators?*
 - v. *Can you comment on the capacities of your staff on using DHIS tools? If there are any gaps, how can these be addressed?*
9. Now, coming to the role of community health workers in management of Pneumonia and Diarrhea, particularly Lady Health Workers (LHWs) and Community Midwives (CMWs), what is their importance in reducing morbidity and mortality associated with these diseases?
- Probes:
- i. *Is there any network of these workers associated with your facility?*
 - ii. *Are these workers fully equipped and have been provided with sufficient supplies to perform their role in management of these illness?*
 - iii. *How their role can be further strengthened?*
 - iv. *Do you find any gap in linkages between these workers and public/private health facilities? What are your suggestions to improve this situation?*
10. Now, I would like you to comment on the cost of treating Pneumonia and Diarrhea?
- Probes:
- i. *In your opinion, does cost play a significant influence over the choice of medicines prescribed and service provider?*
 - ii. *How commonly do you prescribe a medicine to be bought from market in case of its stock-out?*
 - iii. *What do you do in case of a non-affording patient? Do you prescribe inexpensive medicines?*
11. I want to know, if you have established any liaison with any or more public health facilities for referral? How does health facility staff generally behave with your referred cases of severe pneumonia and diarrhea?
- i. *Do you have linkages with higher level facilities? (e.g. RHC, THQH, DHQH & tertiary care hospital)*
 - ii. *What is reported experience of patients referred to these facilities? Do they generally receive supportive and considerate treatment? Give some examples.*
 - iii. *How are patients transported to referral facilities and who pays for transport charges? Do they avail facility or community ambulance services?*
 - iv. *Do you keep record of referrals and feedbacks?*
12. What is the usual monitoring mechanism of outreach services?
- i. *Describe working relationship between LHWs and facilities*
 - ii. *Do officials (e.g. DDOH, DOH or EDOH) usually make informed or surprised monitoring visits?*
 - iii. *Do they provide supportive or traditional supervision? Do you get appreciation for good work?*
13. I'd now like to ask you to summarize your role in the provision of pneumonia and diarrhea services? What else do you think you need to improve the quality of pneumonia and diarrhea services at community level?
- i. *To what extent are you satisfied with your services? If not, why not?*
 - ii. *What must be done to improve the quality of pneumonia and diarrhea services at community level?*

GUIDE FOR FOCUS GROUP DISCUSSION (FGDs) WITH COMMUNITY HEALTH WORKERS (LHWs, LHSs)

Name of District	
Name of Health Facility	
Number of Participants (LHWs and LHSs)	
Date	

During the initial discussion to explain the evaluation, the interviewer and participant should have had the opportunity to introduce themselves and to make small talk to break the ice. You can begin by saying these words to the group of participants:

‘As you know Pneumonia and Diarrhea are common causes of deaths among children under age 5. We are conducting a study to learn about of services being provided LHWs and LHSs to prevent and control these diseases among children and would like to know your views in this regards. We will start when you are ready, will listen to your view points and ask few questions to clarify some of the things you would talk about.’ May we begin?

The guideline contains multiple questions and each question is followed by a set of probes. Please let participants initiate the discussion and use probes only when certain concerns are not addressed by the group.

Disease burden or cases of pneumonia and diarrhea

1. I’d like to ask you a little bit about your own perceptions about estimated number of cases of pneumonia and diarrhea.

Probes:

- i. *Average cases of pneumonia and diarrhea LHWs dealt daily/weekly/monthly.*
- ii. *Do you get sufficient time to deal with pneumonia & diarrhea cases, in addition to your other responsibilities?*

Trainings and skills to handle pneumonia and diarrhea

2. Do you have sufficient knowledge & skill to handle cases of pneumonia and diarrhea?

Probes:

- i. *Have you received induction & on job trainings (e.g. IMCI) regarding pneumonia and diarrhea?*
- ii. *How much time has passed since you received last training/refreshers?*
- iii. *Did you find trainings and refreshers useful? If not- why not? Any suggestion?*

Availability of treatment protocols/guidelines

3. Do you have WHO treatment and referral guideline for pneumonia and diarrhea in the form charts, posters or booklets?

Probe:

- i. *Did you receive any audio-visual and pictorial material related to pneumonia and diarrhea?*
- ii. *Do you use these materials for awareness raising of the community, especially mothers of children under 5 years?*

Availability of sufficient supplies for management of pneumonia and diarrhea

4. Do you have sufficient medicines and supplies for management of pneumonia and diarrhea?

Probe:

- i. *E.g. amoxicillin, antipyretic, zinc, LORS, ARI timer, thermometer, dispersible tablets*

Case management of pneumonia and diarrhea

5. How comfortable do you feel in handling a case of pneumonia or diarrhea at your health house or during home visits?

Probe:

- i. *Can you detect danger signs for referral?*
- ii. *Did you receive any training to detect danger signs of pneumonia and diarrhea?*
- iii. *Do you know how to classify pneumonia and diarrhea by using WHO guidelines? How frequently do you use these guidelines*
- iv. *Can you describe home remedies commonly used?*

Challenges in case handling

6. What challenges do you face while managing cases of pneumonia & diarrhea in the community?

Probe:

- i. *Do you face any challenge with respect to your skill and knowledge, and feel the need for more training?*
- ii. *Do you have sufficient supplies and equipment according to the need?*
- iii. *Do you suffer from lack of self- motivation? If yes- why?*

Health facility liaison and referral support

7. Have you established any liaison with any or more public health facilities for referral? How does health facility staff generally behave with your referred cases of severe pneumonia and diarrhea?

Probe:

- v. *Do you have linkages with higher level facilities? (e.g. RHC, THQH, DHQH & tertiary care hospital)*
- vi. *What is reported experience of patients referred to these facilities? Do they generally receive supportive and considerate treatment? Give some examples.*
- vii. *How are patients transported to referral facilities and who pays for transport charges? Do they avail facility or community ambulance services?*
- viii. *Do you keep record of referrals and feedbacks?*

Liaison with other Community Health Workers

8. Do you have any formal or informal liaison with other community health workers (CHWs) in your catchment area?

Probe:

- i. *Do they belong to NGO or public sector organization?*

- ii. *Do you consider their role supportive/productive in the management of pneumonia and diarrhea?*

Coordination and Monitoring Mechanisms

9. What is the usual coordination and monitoring mechanism of outreach services?

Probe:

- iv. *Describe the coordination mechanisms and working relationship between LHWs and LHSs and other departments?*
- v. *Do officials (e.g. DDOH, DOH or EDOH) usually make informed or surprised monitoring visits?*
- vi. *Do they provide supportive or traditional supervision? Do you get appreciation for good work?*

Case reporting and documentation

10. How pneumonia and diarrhea cases are routinely recorded and reported?

Probe:

- i. *What is the frequency of reporting?*
- ii. *Where these reports are sent? Are they entered in DHIS?*
- iii. *Do you receive feedback from health authorities?*
- iv. *Do health authorities use your data for managing supplies and improving quality of services?*

Community behavior and response

11. What key challenges and constraints do you face regarding community behaviour?

Probe:

- i. *Do majority of clients like receiving drugs or prefer home remedies? Does the beneficiaries prefer DTs over syrups or vice versa for treatment of pneumonia and diarrhoea?*
- ii. *Do you face any resistance or displeasure during your home visits?*
- iii. *Do you feel any security threat while working?*
- iv. *Do the community respect you for your role and services?*

Social beliefs and taboos

12. What is the general level of awareness about health issues among the local community?

Probe:

- i. *Do you feel cultural beliefs and practices could be one reason for not seeking proper treatment for pneumonia and diarrhea? If yes, please explain.*
- ii. *Are there taboos regarding diet for children suffering from diarrhea and pneumonia? If yes, give examples.*
- iii. *Do you think, raising awareness level of the community can improve utilization of services for pneumonia and diarrhea at community and facility level*

Suggestions for improvement of services

13. I'd now like to ask you to summarize your role in the provision of pneumonia and diarrhea services? What else do you think you need to improve the quality of pneumonia and diarrhea services at community level?

Probe:

- iii. *Do you agree with your role as agent of change in provision of health services at community? Please describe your experience in this regard.*
- iv. *To what extent are you satisfied with your services? If not much satisfied, why not?*
- v. *What must be done to improve the quality of pneumonia and diarrhea services at community level?*

FGDs GUIDE FOR GENERAL PRACTITIONERS

Name of District	
Number of Participants	
Date	

During the initial discussion to explain the evaluation, the interviewer and participant should have had the opportunity to introduce themselves and to make small talk to break the ice. You can begin by saying these words to the group of participants:

‘As you know Pneumonia and Diarrhea are common causes of deaths among children under age 5. We are conducting a study to learn about of services being provided to treat and manage these diseases among children and would like to know your views in this regards. We will start when you are ready, will listen to your view points and ask few questions to clarify some of the things you would talk about.’ May we begin?

The guideline contains multiple questions and each question is followed by a set of probes. Please let participants initiate the discussion and use probes only when certain concerns are not addressed by the group.

CHECKLIST FOR AVAILABILITY OF COMMODITIES FOR MANAGEMENT OF < 5 PNEUMONIA & DIARRHEA CASES

A	Trainings/Refreshers for Management of Pneumonia & Diarrhea					
	No of trainings received during past 24 months	IM CI	ARI	CD D	Breast Feeding/ Micronutrient Deficiency	EPI/ Cold Chain Management
A1						
IMCI (Integrated Management of Childhood Diseases) ARI (Acute Respiratory Infection) CDD (Control of Diarrhoeal Diseases) EPI (Extended Programme on Immunization)						

B	Standard Guidelines for Diagnosis & Management of Pneumonia & Diarrhea (Mark ✓ for available and not available)		
	Guidelines	Available	Not Available
B1	Guidelines for Pneumonia/ARI case management		
B2	Guidelines for Diarrhea/Dysentery Case Management		
B3	Guidelines on Storage of Drugs		
B4	Guidelines for Cold Chain Management		
B5	EPI Guidelines		

C	Essentials for Pneumonia & Diarrhea Case Management (Mark ✓ for yes or No)
---	---

	Commodities	Available		Functional	
		Yes	No	Yes	No
C1	Oxygen cylinder				
C2	Oxygen flow meter				
C3	Pulse oximeter				
C4	ARI timer				
C5	Ambo bag (For Child)				
C6	Nebulizer				
C7	ILR to maintain cold chain				
C8	Others				
C9	Supplies to mix and dispense ORS (e.g. cup)				
C10	Source of clean drinking water				
C11	Ambulance				
• ILR (Ice Lined Refrigerator) • ORS (Oral Rehydration Solution)					

D	Drugs for Pneumonia and Diarrhea Treatment (Mark ✓ for available and not available status of drugs but for stock outs write the number of months for which medicine was not available)			
	Drugs for Pneumonia and Diarrhea Treatment	Current status		Stock Outs * (July 2018 - June 2019)
		Available	Not Available/Expired	
D1	Syrup Amoxicillin			
D2	Amoxicillin Dispersible Tablets (DT)			
D3	Injection Amoxicillin			
D4	Pneumococcal Vaccine			
D5	Syrup Zinc			
D6	Zinc Dispersible Tablet (DT*)			
D7	Low osmolarity ORS packets			
D8	LORS-Zinc Co-packaging			
D9	Ringer lactate Solution			
D10	0.45% Normal Saline/ Dextrose Saline			
• DT (Dispersible tablet) • Note: Please check the Stock register (medicines/supplies) for stock outs and write number of months for mentioned period				

E	Record keeping and Reporting (Mark ✓ for yes or No)				
	Material	Available		Updated	
		Yes	No	Yes	No
E1	OPD Register				
E2	In-patients Register				
E3	Stock Register (Medicine/Supplies)				
E4	Stock Register (Equipment/Furniture)				
E5	Facility Monthly Report Form				
• OPD (Out-patient Department)					

1. I'd like to know about main childhood related illnesses in your catchment area. Can you please comment on the caseload of Pneumonia and Diarrhea?

Probes:

- About what proportion of children (less than 5 years) is brought to your clinic that suffer from Pneumonia and Diarrhea?*
- What is your view about the major causes of spread of these diseases in your community and how can this be prevented?*
- What type of treatment approach or professional help is commonly preferred in the community for these diseases (e.g. self-treatment, traditional healer, homeopathic, allopathic – public or private)?*

2. I would like to know about your skills/competence for managing childhood illnesses? Can you please tell us about any training(s) received since you have been practicing for dealing such diseases?

Probes:

- What in your view is the importance and benefits of in-service trainings for enhancing skill/competence to manage childhood illness, particularly Pneumonia and Diarrhea?*
- When was the last time you received such training? If no, what could be the reason for not arranging such trainings?*
- Who usually arrange such trainings (like Health Department, development partners or pharmaceutical companies)?*
- What is your preference for the venue of such trainings and why?*
- What deficiencies did you notice in these trainings? In your view, how can these be improved?*
- If you have received training, did any refresher follow it? If no, what could be the reason for not providing refresher training?*

3. Now, I would like to know about the use of protocols and guidelines (like IMCI guidelines) for managing pneumonia and diarrhea in children at your clinic. What are these protocols and guidelines, kindly explain?

Probes:

- i. *Can you please explain their usefulness in diagnosing and treating childhood illness?*
 - ii. *If no such specific protocols/guidelines are available, can you please comment on the reasons?*
 4. Now, we will like to know about the diagnosis and management of Pneumonia at your clinic? Please tell us how do you diagnose children suffering from Pneumonia?
Probes:
 - i. *How do you diagnose and classify Pneumonia on the basis of its severity and does its treatment also varies accordingly? Kindly explain.*
 - ii. *Do you think any particular equipment is essential for the diagnosis of pneumonia? If there is any deficiency, please explain reasons for shortage of equipment.*
 - iii. *How commonly antibiotics are used for treatment of Pneumonia?*
 - iv. *Have you ever heard of any dispersible (soluble in water) antibiotic tablet? If so, do you perceive any advantage of dispersible tablets over syrups? What do you routinely prescribe?*
 - v. *In your opinion, what is the perception of the community regarding use of dispersible tablets?*
 - vi. *Do you think there is any role and advantage of using oxygen in treating pneumonia? Is it being used at your clinic? If not, can you explain the reasons?*
 - vii. *In case of complicated Pneumonia cases, what referral mechanisms exist for transferring these children to higher-level health facilities?*
5. Can you please tell us how do you manage children suffering from Diarrhea at your clinic?
Probes:
 - i. *How do you diagnose childhood diarrhea and do you differentiate it from dysentery?*
 - ii. *Now focusing on diarrhea, do you classify patients on the basis of severity of dehydration? If so, how?*
 - iii. *What options do you have to treat childhood diarrhea with oral medications?*
 - iv. *Is there any role of oral Zinc in its treatment? If so, what type of Zinc formulation is usually provided at your clinic?*
 - v. *Have you ever heard of any dispersible (soluble in water) tablets of Zinc? If so, do you perceive any advantage of dispersible tablets over syrups?*
 - vi. *In your opinion, how would community take the idea of using dispersible tablets?*
 - vii. *Do you think there is any advantage of using Low Osmolarity ORS compared to regular ORS in treating Diarrhea? What do you routinely prescribe?*
 - viii. *In case of complications, what referral mechanisms exist for transferring these children to higher-level health facilities?*
6. Now, I would like you to comment on the cost of treating Pneumonia and Diarrhea?
Probes:
 - i. *In your opinion, does cost play a significant influence over the choice of medicines prescribed and service provider?*
 - ii. *How commonly do you prescribe a medicine to be bought from market in case of its stock-out?*
 - iii. *What do you do in case of a non-affording patient? Do you prescribe inexpensive medicines?*
7. What must be done to improve the quality of pneumonia and diarrhea services at community level?

GUIDE FOR FOCUS GROUP DISCUSSION (FGDs) WITH BENEFICIARIES (MOTHERS OF CHILDREN UNDER 5 YEARS)

Name of District	
Number of Participants	
Date	

During the initial discussion to explain the evaluation, the interviewer and participant should have had the opportunity to introduce themselves and to make small talk to break the ice. You can begin by saying these words to the group of participants:

‘As you know Pneumonia and Diarrhea are common causes of deaths among children under age 5. We are conducting a study to learn about of health-seeking behaviors regarding these diseases among children and would like to know your views in this regards. We will start when you are ready, will listen to your view points and ask few questions to clarify some of the things you would talk about.’ May we begin?

The guideline contains multiple questions and each question is followed by a set of probes. Please let participants initiate the discussion and use probes only when certain concerns are not addressed by the group.

1. What causes diarrhoea and pneumonia and what are the symptoms?
 - *In your opinion, which preventive measures and/or habits can prevent spread of pneumonia or diarrhoeal diseases?*
 - *From which source have you received this information?*
 - *Do you undertake these actions for your child/children?*
 - *Where does your family get water from? Do you drink it directly from the source?*
 - *Where does your family (adults and children) go to the toilet?*
 - *Does your family (adults and children) wash their hands? With water only?*
2. Where do most people prefer to go to seek treatment of children? What degree of influence do you think the cultural and family norms and beliefs have over the choice of treatment?
 - *Do you use a traditional doctor / spiritual healer for child illness?*
 - *What cultural beliefs influence child illness and treatment seeking in this community?*
 - *Do you go to the health centre for child illness?*
 - *What is your opinion about the quality of the services provided by the health centre?*
 - *Do you prefer Dispersible tablets or syrup for treatment of diarrhea and pneumonia?*
 - *Do you directly go to the chemist for your child illness?*
 - *Why do you use the chemist (instead of the health centre?)*
3. Where have you learnt from about child illness?
 - *What child survival information, education and communication activities are targeted at mothers/fathers?*
 - *What measures should be taken to improve the community’s knowledge about child illness?*
4. What activities do health workers (LHWs) undertake in the community regarding diarrhea and pneumonia?
 - *Are they providing sufficient care for diarrhea and pneumonia?*

- *What challenges do health workers face doing their work in the community?*
 - *How can their services be improved?*
5. What are the main challenges you face in going to the health centre or accessing treatment for child illness?
 - *What are the reasons that some families not take their child for treatment if they are ill?*
 - *Does the cost of accessing treatment sometimes prevent some families from taking the child?*
 - *What do you say about the overall quality of services? Would you like to visit this facility again for child care?*
 - *Do you have any suggestion to improve the services?*
 6. In your view, what have been the biggest obstacle in combating childhood diarrhea and pneumonia? What are the solutions to these challenges / barriers?
 7. What can be done to improve the health of children in this area?

FGDs GUIDE WITH IN-CHARGES OF PHARMACIES/MEDICAL STORE

1. Which antibiotic is most commonly prescribed by clinicians (specialists/ General Practitioners) for the treatment of respiratory tract infections in children in your catchment area?

Probe:

- i. *Do you find any difference in prescribing pattern of Specialists and GPs? If yes, please elaborate.*
- ii. *In your opinion, what are the prime considerations of clinician to prescribe any drug?*

2. Which medicine is most commonly prescribed by clinicians (specialists/ General Practitioners) for the treatment of diarrhea in children in your catchment area?

Probe:

- i. *Do you receive prescription of Zinc for children suffering from diarrhea? If yes how frequently and by whom? GP or specialist?*
- ii. *If not, probe if it was ever prescribed in the past. If so, what is probable reason for non-continuation?*
- iii. *Do you find any difference in prescribing pattern of Specialists and GPs? If yes, please elaborate.*
- iv. *In your opinion, what are the prime considerations of clinician to prescribe any drug?*

3. What factor/s do you routinely consider while placing order for any new drug not acquired and sold before?
4. If any medicine is available in syrup as well as dispersible tablet form, which one, in your experience, client/patient would prefer and why so?

5. Do you have amoxicillin and zinc in dispersible tablets (DT) formulation in stock for sale?
If not, why not?

Probe:

- i. Will you like to keep Amoxicillin and Zinc (DT) in your pharmacy?
- ii. Is the supply of these medicines regular or intermittent? What are the reasons for delay, if any?
- iii. In your opinion, which marketing strategy is more likely to promote demand and sale of these drugs?

6. Do you receive prescriptions from the public sector health facilities as well (e.g. BHUs, RHC, THQ, DHQ, and MCH)? If so, about what percentage do you get such prescriptions?
If not, why not?

Probe:

- i. Is the client/patient specifically referred to your pharmacy from public sector facility or is the client given free choice to select any?

Tool for Prescription Review

FACILITY IDENTIFICATION	
Name of the district	
Name of the facility	
Address	
Facility Code (HID No)	
PATIENT IDENTIFICATION	
Age of Child	Gender
Provisional Diagnosis	

PRESCRIPTION				
Drug	Formulation*	Dose	Frequency	Duration
Amoxicillin				
Zinc Sulfate				
ORS (LO or regular)				
* Formulations include Syrup, Dispersible Tablet, Sachet or Tablet.				

Annex 6: Risk Register

1	EXTERNAL RISKS				
	Project risks	Assessment of impact of risks	Assessment of probability of risks	Possible mitigation	Residual risks that cannot be managed
1.1	Political/Terrorism Situation				
1.1.1	Protests	High	Medium	Contingency operational planning of field operations	Country-wide impact
1.1.2	Military operation	High	Medium	Security measures to be adopted and local personnel/organizations to be involved	Security threat in high risk areas/no-go areas
1.1.3	Unstable political situation	High	Medium	Support from local political parties and government	Change of government
1.2	Health Priorities and staff transitions				
1.2.1	Field activities and operations	High	Medium	Days to be identified beforehand and planning done accordingly	Emergency outbreaks/epidemic/accidents causing disruption in operationalization
				Set plan for operations	
1.2.2	Posting/Transfer of any potential key officer (in Government/ programme)	Medium	Medium	Rapport building sessions with Programme /Government officials	Key stakeholders officials transferred and vacant positions
1.3	Security Situation				
1.3.1	Law & order situation	High	High	Only local field and operational teams to be involved/hired	Security threats in high risk areas/no-go areas
1.3.2	Movement in hard to reach areas	High	High	Security measures to be adopted and local personnel to be involved	
				Security planning to be done beforehand	
1.4	Natural Disasters or Disease Outbreaks				

1.4.1	Extreme weather conditions	Medium	Medium	Variation in functioning hours in winters and summer according to the terrain/location	Natural calamities
1.4.2	Natural Calamities and Disasters	Medium	Low	Forecasting, precautionary measures for disaster management	Crisis and disasters
1.5	Capacity issues				
1.5.1	Inexperienced and un-qualified field teams	Medium	Low	Recruitment on the basis of previous experience within similar and local context and refresher trainings periodically	Local staff with requisite skills and qualification difficult to identify
1.6	Internal dynamics				
1.6.1	Potential drop-outs of field staff at any stage	High	Low	Local context specific reputable and experienced team/consultants would be involved for operations	Unforeseen drop-outs during operationalization
1.7	Inflation and economic situation				
1.7.1	Implication of new budgetary cycle	Low	Low	Agreement on rates beforehand and forecasting keeping flexibility done	Additional cost of transport or personnel in lieu of hard to reach areas
1.8	Approval for Inception report				
1.8.1	Delay in seeking approval on Inception report	High	High	Regular meetings with UNICEF team for seeking early approval on report	Unusual delay in getting feedback for inception report

2 INTERNAL RISKS					
	Programme risks	Assessment of impact of risks	Assessment of probability of risks	Possible mitigation	Residual risks (that cannot be managed)
2.1	Security situation				
2.1.1	Mobility issues of field teams	High	High	Local staff/teams will be engaged. Moreover, Instructions given in Security manual to be adhered in letter and spirit	Hard terrain and poor road network
2.2	Government/ Programme Support				
2.2.1	Inactive engagement of programme in supporting the	Medium	Low	Rapport building meetings with Technical team to gauge support	Key stakeholders/programme officials transferred and vacant

	vision, goals, objectives and operations				positions
2.2.2	Lack of coordination and rapport with governments	Medium	Low	Placement of a reliable and transparent process to resolve disagreements and form coordination	Governments' policy and own interests
2.2.3	Lack of support from programme for getting operational or financial data	Medium	Low	Rapport building meetings with Technical team to gauge support from Programme	Key stakeholders/programme officials transferred and vacant positions
2.3	Staff transitions				
2.3.1	Posting/change of project technical staff	High	Low	Pool of alternate and equally qualified professionals maintained A team of experts shall be working together so that in case of any change, transition remain smooth	Key positions become unavailable/vacant due to any unforeseen and unavoidable reason
2.4	Tight implementation timelines				
2.4.1	Delay in completion of linked planned activities due to interdependent nature	High	Low	Workplan to be adhered as much as possible	Project coming to an abrupt end or change in project activities

Annex 7: Description of Proposed Change Pathways

Assumption	Link with Project Outcomes	Findings of MTE based on DAC Criteria	DAC Linkage	Recommendation	Proposed Action	Timeline	Priority	Prime Responsibility	Supporting Role
1. Proposed commodities (ones replacing existing) are more effective for the treatment of Pneumonia and Diarrhea.	Policy Translation	Empirical evidence on effectiveness of updated commodities	Relevance and Effectiveness	Scaling up in existing provinces and replicability in rest of Pakistan	Evidence-based policy advocacy - enhancement of geographic coverage of the interventions	Short term	+++	UNICEF Project Management Team	Federal and Provincial Governments
		Use of updated commodities is more effective in terms of better management through early diagnosis (ARI Timers) and accuracy of dosage (use of DTs)			Resource allocation - donor identification and earmarking of funds	Medium term	++	Provincial Health Departments	CSG and TWG
2. Ensuring better availability of these commodities at the public health facilities will ensure better treatment of Pneumonia and Diarrhea patients.	Policy Implementation	Better diagnosis and treatment through use of Pulse Oximeters and Oxygen at health facilities and ARI Timers for Lady Health Workers	Efficiency and Effectiveness	Strengthening of supply chain management (from DOH to health facilities and outreach)	Capacity building of DOH on supply chain management	Medium term	+++	SCM implementing partner	Federal and Provincial Governments
		Varied availability of DT at public sector facilities and no availability of DTs in open market of the revised commodities		Establish wider and proactive public-private partnership models through engaging pharma industry along with DRAP, medical, nursing and paramedical institutions	Dialogues on opportunities for pharma industry and drugs distribution network	Short term	+++	UNICEF Project Management Team	CSG and TWG

Assumption	Link with Project Outcomes	Findings of MTE based on DAC Criteria	DAC Linkage	Recommendation	Proposed Action	Timeline	Priority	Prime Responsibility	Supporting Role
3. Government institutions are willing to participate in the project.	Policy Change	Willingness of federal and provincial government institutions is evident from their concurrence on project interventions. The project is supporting provincial health departments in implementation of joint accountability framework through CSG/TWG.	Relevance	Continuance of government support for health system strengthening and policy implementation	Sharing MTE results with stakeholders with special focus on role of CSG/TWG	Short term	+	UNICEF Project Management Team	Federal and Provincial Governments
4. Effectiveness of the promoted methods and commodities are well acknowledged at the local level.	Policy Translation	Healthcare providers and outreach workers at local levels acknowledged the use of updated commodities for effective management.	Effectiveness	Enhancing private sector engagement for updating their prescribing behaviours	Active involvement of professional associations in advocacy for private practitioners	Medium term	++	Professional Associations	UNICEF Project Management Team
		On the contrary, private sector providers lacked familiarity with the updated commodities due to their limited engagement in the project activities.			Continued capacity building program for service providers from both public and private providers	Long term	+++	UNICEF Project Management Team	Professional Associations
5. All stakeholders recognize improved diagnosis and treatment of Pneumonia and Diarrhea as a shared	Knowledge Management	Updated federal and provincial policies and guidelines reflect the recognition of improved management as a shared goal	Sustainability	Translation of lessons learnt (scale up and replication) to other geographical settings	Evidence-based policy advocacy - enhancement of geographic coverage of the interventions Rolling-out IMNCI in target provinces and advocacy for	Long term	++	M/o NHR&C and Health Departments	UNICEF Project Management Team

Assumption	Link with Project Outcomes	Findings of MTE based on DAC Criteria	DAC Linkage	Recommendation	Proposed Action	Timeline	Priority	Prime Responsibility	Supporting Role
goal.					replication in other provinces and regions of Pakistan				
6. Policy makers, health institutions and pharmaceutical industry are as receptive to new ideas and demands.	Policy Change	Evident receptivity of policy makers, health institutions from updated government documents (MSDP/EPHS, Essential medicines list, provincial procurement lists)	Relevance & Effectiveness	Continued advocacy with pharmaceutical manufacturers for local production	Engaging with private pharma and distribution networks by sharing areas of opportunities and informing them about benefits of treatment	Short term	++	UNICEF Project Management Team	Pharma industry and DRAP
		Limited interest of pharmaceutical manufacturers in local production			Building stronger collaborations with DRAP and private sector				
					Direct Demand Creation: Convincing provincial health authorities to include updated commodities in provincial procurement lists	Medium term	++	Provincial Health Department	Pharmaceutical industry through relevant associations
7. Sufficient political will is available to control Pneumonia and Diarrhea.	Policy Change	Political will and commitment is evident from inclusion of Pneumonia and Diarrhea in Pakistan's national health vision and priorities for child health as well as the provincial health sector strategies	Relevance and Sustainability	Concerted advocacy efforts with political leadership for sustained will and commitment under the GAPPD for childhood Pneumonia and Diarrhea management	Policy dialogues and roundtables on improving management of Pneumonia and Diarrhea through updated commodities	Short term	+++	UNICEF Project Management Team	M/o NHR&C and Health Departments

Assumption	Link with Project Outcomes	Findings of MTE based on DAC Criteria	DAC Linkage	Recommendation	Proposed Action	Timeline	Priority	Prime Responsibility	Supporting Role
		Project interventions are aligned with provincial health sector strategies of Sindh and Punjab			Mass-media engagement: Opinion Editorials in leading national newspapers Televised debates	Medium term	++	UNICEF Project Management Team	Mass-media agencies
8. Child survival group represent all relevant stakeholders and produce ownership of its recommendation.	Policy Change	Child Survival Group of Punjab and Technical Working Group of Sindh have representation of all relevant stakeholders.	Effectiveness	Fostering CSG/TWG for harnessing its optimal potential and taking measure for sustenance beyond the project life	Institutionalization of CSG/TWG through inclusion in PC-1 documents of relevant health programs (IRMNCH, MNCH and LHWs)	Long term	+	CSG and TWG	-
		Ownership of CSG/TWG is evident from the decision-making and its follow-up by the members during CSG/TWG meetings.							
		Recommendations of CSG/TWG have been critical in policy translation and implementation.			Setting-up a small secretarial unit for the CSG and TWG	Short term	++	CSG and TWG	UNICEF Project Management Team
9. Credible and well-organized implementing partners.	Policy Implementation	Project has identified implementing partners who have the capacity and spread across the country to effectively implement the project activities in their specific areas.	Efficiency	Building local capacities through engagement of existing partners and identifying new partners	Prioritize areas for local partnership based on the findings of MTE	Short term	+++	UNICEF Project Management Team	Implementing partners

Assumption	Link with Project Outcomes	Findings of MTE based on DAC Criteria	DAC Linkage	Recommendation	Proposed Action	Timeline	Priority	Prime Responsibility	Supporting Role
10. Province and federal level stakeholders share the same vision for improved diagnosis and treatment of Pneumonia and Diarrhea.	Policy Change	Decentralization of health sector has created dichotomies among federal level and provinces. The project attempts to bridge this gap by supporting certain platforms to build a shared vision.	Relevance	Strengthen federal, provincial linkages and coordination to implement National Health Vision and provincial policies/strategies priorities focusing on health system strengthening for improved management P&D	Encourage participation of federal representative in CSG/TWG	Short term	+++	M/o NHR&C and Health Departments	UNICEF Project Management Team
					Cross-sharing of challenges, lessons learnt among the provinces	Short term	++	UNICEF Project Management Team	M/o NHR&C and Health Departments
11. Ownership of the initiatives at district level health management.	Policy Implementation	Ownership of the project interventions is being built through strengthening the existing service delivery by improving management practices of healthcare providers, mainly outreach workers and primary level facilities	Effectiveness	Revitalization of district level decision-making and coordination platform for enhancing ownership, side by side focusing on district specific gaps in provision of updated commodities and capacity building (DHIS-2, LMIS, IMNCI) implementation commitment and sustainability	Building district capacities on forecasting, procurement, distribution, maintenance and warehousing	Long term	+++	Implementing partners	District Health Department
					Linkage of DHIS with LMIS	Medium term	+	District Health & Population Management Teams	UNICEF Project Management Team

Assumption	Link with Project Outcomes	Findings of MTE based on DAC Criteria	DAC Linkage	Recommendation	Proposed Action	Timeline	Priority	Prime Responsibility	Supporting Role
12. Government has sufficient resources to ensure sustainable commodities supply.	Policy Translation	Inclusion of updated commodities in key government documents (MSDP/EPHS, Essential medicines list, provincial procurement lists) shows government financial commitment..	Efficiency and Effectiveness	Advocacy on the benefits of updated commodities for replacing existing finances for alternative procurement	Support to provincial health departments in sustainable implementation of integrated HLMIS through development of PC-1	Medium term	++	UNICEF Project Management Team	Provincial Health Departments
13. Relevant institutions prioritize capacity building of service providing staff and allocate suitable time and financial resources to the same.	Policy Translation	Federal and provincial governments have updated IMNCI guidelines while catering to the challenges of previous curriculum and training duration. However, there are still gaps in allocation of resources for conducting the trainings across the board. District does not have resources for conducting IMNCI trainings	Efficiency and Effectiveness	Leveraging resources through advocacy and involvement of relevant stakeholders	Strengthen district level action planning with inclusion of trainings with required resources Engagement of donors and development partners Engagement of Nursing Council, Midwifery Associations , PPP, Healthcare Commissions relevant DoH / Health inst. for curricula revision and trainings of HCPs and CHWs	Long term	+++	UNICEF Project Management Team	Provincial Health Departments and Districts

Assumption	Link with Project Outcomes	Findings of MTE based on DAC Criteria	DAC Linkage	Recommendation	Proposed Action	Timeline	Priority	Prime Responsibility	Supporting Role
14. Relevant stakeholders agree to update essential medicine list and minimum service delivery standards to include new areas for Pneumonia and Diarrhea control.	Policy Change	Federal and provincial governments have updated essential medicine list and minimum service delivery standards to include the updated commodities for Pneumonia and Diarrhea through the support of the project.	Effectiveness	Scale up and replication of these interventions to non-project provinces and regions of Pakistan	Evidence-based policy advocacy - enhancement of geographic coverage of the interventions	Long term	++	UNICEF Project Management Team	Provincial Health Departments
					Sharing of findings of MTE with Health Departments and stakeholders of other provinces and regions of Pakistan	Short term	+++	UNICEF Project Management Team	Provincial Health Departments
15. System provisions, resources and adequate skill sets exist to facilitate inclusion of EML in logistics management information system.	Policy Implementation	Gaps in systems provisions, resources and skill sets are filled through development of integrated supply chain management system for the targeted districts	Impact and Sustainability	Strengthening government capacity to roll out LMIS and DHIS-02	Training of provincial and district level managers on enhancing analytical skills, data interpretation skills and use of information Experience sharing through visits of areas/countries where DHIS 02 is operational for federal and provincial managers	Long term	++	UNICEF Project Management Team and	SCM partners
16. Local drug providers have the resources, capacity and interest in	Policy Implementation	Despite the resources and capacities, the local Pharma industry lacked interest in local production due to	Sustainability	Two-pronged approach with concurrent advocacy efforts to mitigate inherent barriers in	Advocate with government and pharma industry for procurement of revised commodities	Short term	+++	UNICEF Project Management Team	Provincial Health Departments

Assumption	Link with Project Outcomes	Findings of MTE based on DAC Criteria	DAC Linkage	Recommendation	Proposed Action	Timeline	Priority	Prime Responsibility	Supporting Role
producing required Pneumonia and Diarrhea commodities.		minimal demand of updated commodities in open market		local production and public procurement	Private sector engagement and proactive collaboration with DRAP				
		Public procurement of updated commodities has yet to be initiated.		Engagement of private providers through professional associations for change in prescribing behaviours	Government initiating pre-qualification of potential pharmaceutical manufacturers	Medium term	+++	UNICEF Project Management Team	Professional Associations
		Demand for updated commodities increased through engagement with general practitioners and their professional associations.							
17. Existing information systems include information on updated commodities	Policy Implementation	DHIS tools are revised to record and report on updated commodities in Sindh	Sustainability	Revision of DHIS tools for updated commodities in Punjab	Technical assistance to relevant stakeholders for revision and support in printing of revised tools	Short term	+	UNICEF Project Management Team	Provincial Health Departments
				Ensuring implementation of DHIS-2 and LMIS					
				Improving quality of DHIS data	Training of facility and district level staff relevant to recording and reporting on data quality	Long term	+++	UNICEF Project Management Team	District Health Department

